

Deep Learning for Metal Artifact Reduction in Computed Tomography

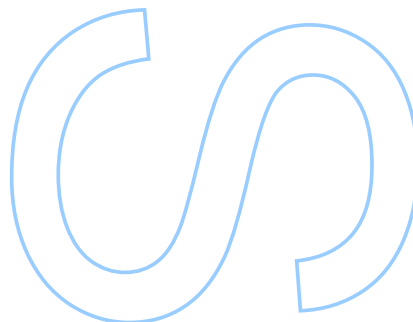
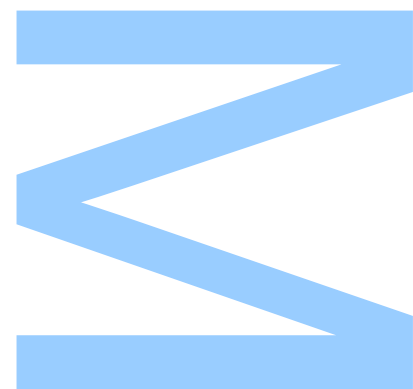
Ludmila Montorsi Queiroz

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MASTERS THESIS

Deep Learning for Metal Artifact Reduction in Computed Tomography

Author:

Ludmila MONTORSI QUEIROZ

Supervisor:

Hélder Filipe PINTO DE OLIVEIRA

Co-supervisor:

Eduardo MATOS RODRIGUES and

Tânia PEREIRA

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"The Universe is under no obligation to make sense to you."

Neil deGrasse Tyson, *Astrophysics for People in a Hurry* (2017)

"Nothing in life is to be feared; it is only to be understood."

Marie Curie, *Madame Curie: A Biography* (1937)

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I'm deeply grateful to my husband for his unwavering support, patience, and encouragement throughout this journey. His love and understanding have been a constant source of strength, and I am truly thankful for his presence in my life.

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I need to express my sincere gratitude to my mother, who, even though not physically present, has been a guiding light in my life. Her love, wisdom, and values continue to inspire me every day, and I carry her memory with me as I face the challenges and triumphs of this journey. I hope to honor her legacy by doing good to all who cross my path. Without her love and presence, I would not have made it this far.

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Abstract

Faculdade de Ciências da Universidade do Porto

Departamento de Ciência de Computadores

Departamento de Física e Astronomia

INESC TEC

MSc. Medical Physics

Deep Learning for Metal Artifact Reduction in Computed Tomography

by [Ludmila MONTORSI QUEIROZ](#)

Metal artifacts remain a common and clinically significant source of image degradation in computed tomography (CT), affecting an estimated 21% of scans and potentially obscuring or mimicking pathology. This thesis develops and rigorously evaluates a reproducible deep learning pipeline for image-domain metal artifact reduction, comparing a conventional U-Net against a dual-supervision variant that adds an auxiliary artifact-mask prediction head to a shared encoder-decoder. Using the AAPM CT-MAR benchmark dataset (14,000 image pairs derived from DeepLesion and UCLH Stroke EIT), a patient-level traceability procedure was implemented to recover exam and patient identity absent from the original metadata, enabling a leakage-free patient-level split (187/41/41 patients) across three deterministic training seeds per architecture. The primary, pre-specified hypothesis, that reconstruction improves structural similarity (SSIM) specifically in the region directly affected by the artifact while leaving distant tissue comparatively unchanged, was confirmed on the internal test set (41 patients, 1,478 slices) for both architectures: artifact-region SSIM improved by 0.328 for the single-head model (95% CI [0.296; 0.355], Holm-adjusted $p = 2.7 \times 10^{-12}$) and by an equivalent margin for the dual-head model, against a much smaller change of about 0.06 in distant tissue. As secondary, complementary evidence, both architectures also significantly improved global PSNR (27.4 to ~ 35.1 dB), SSIM (0.836 to ~ 0.944) and MAE (36.6 to ~ 17.5 HU) relative to the corrupted input (patient-level sign and t -tests, $p < 0.001$). A direct paired comparison showed a small but statistically significant global advantage for the single-head model and a small but significant distant-region preservation advantage for the dual-head model, indicating a correction-versus-preservation trade-off rather than a clear superiority of either design. Based on this pre-specified artifact-region criterion, the single-head model (seed 2024) was selected, frozen, and

characterized on an independent external set of real exams from The Cancer Imaging Archive, where, in the absence of a clean reference, only the magnitude of input-to-output change could be quantified. A blinded, paired multi-reader evaluation by physicians was designed to assess perceived anatomical preservation and safety on these external cases; data collection was still ongoing at the time of writing, so this thesis reports the technical evidence supporting internal validity and reproducibility, while clinical benefit and external safety remain to be established by that evaluation.

Keywords: Computed Tomography, Metal Artifact Reduction, Deep Learning, U-Net, Multi-task Learning, Reproducibility, Patient-level Data Partitioning, External Validation, Image Quality Assessment

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Resumo

Faculdade de Ciências da Universidade do Porto

Departamento de Ciência de Computadores

Departamento de Física e Astronomia

INESC TEC

MSc. Medical Physics

Aprendizado Profundo para Redução de Artefatos de Metal em Tomografia Computadorizada

por [Ludmila MONTORSI QUEIROZ](#)

Artefatos metálicos permanecem uma causa comum e clinicamente relevante de degradação de imagem em tomografia computadorizada (CT), estimados em cerca de 21% dos exames, podendo esconder ou simular patologias. Esta dissertação desenvolve e avalia rigorosamente um *pipeline* reproduzível de aprendizagem profunda para redução de artefatos metálicos no domínio da imagem, comparando uma U-Net convencional com uma variante de dupla supervisão que acrescenta uma saída auxiliar de predição da máscara de artefacto a um *encoder-decoder* partilhado. Utilizando o *dataset* de referência CT-MAR da AAPM (14.000 pares de imagens derivados do DeepLesion e do UCLH Stroke EIT), foi implementado um procedimento de rastreabilidade ao nível do paciente para recuperar a identidade do exame e do paciente, ausente nos metadados originais, permitindo um particionamento por paciente sem vazamento de dados (187/41/41 pacientes) ao longo de três sementes de treino determinísticas por arquitetura. A hipótese primária, definida à partida, de que a reconstrução melhora a similaridade estrutural (SSIM) especificamente na região diretamente afetada pelo artefacto, mantendo o tecido distante comparativamente inalterado, foi confirmada no conjunto de teste interno (41 pacientes, 1.478 fatias) para ambas as arquiteturas: o SSIM na região do artefacto melhorou 0,328 no modelo de supervisão única (IC 95% [0,296; 0,355], $p = 2,7 \times 10^{-12}$ ajustado por Holm) e por uma margem equivalente no modelo de dupla supervisão, contra uma alteração muito menor, de cerca de 0,06, no tecido distante. Como evidência secundária e complementar, ambas as arquiteturas também melhoraram significativamente o PSNR global (27,4 para $\sim 35,1$ dB), o SSIM (0,836 para $\sim 0,944$) e o MAE (36,6 para $\sim 17,5$ HU) face à entrada corrompida (testes de sinais e t ao nível do paciente, $p < 0,001$). Uma comparação pareada direta

mostrou uma pequena mas estatisticamente significativa vantagem global do modelo de supervisão única e uma pequena mas significativa vantagem de preservação na região distante do modelo de dupla supervisão, indicando um compromisso entre correção e preservação, e não uma superioridade clara de nenhuma das arquiteturas. Com base neste critério pré-definido de SSIM na região do artefacto, o modelo de supervisão única (semente 2024) foi selecionado, congelado e caracterizado num conjunto externo independente de exames reais provenientes do The Cancer Imaging Archive, onde, na ausência de referência limpa, apenas foi possível quantificar a magnitude da alteração entrada-saída. Uma avaliação cega e pareada por múltiplos médicos foi desenhada para avaliar a preservação anatômica percebida e a segurança nesses casos externos; a recolha de dados ainda estava em curso à data de redação, pelo que esta dissertação relata a evidência técnica que sustenta a validade interna e a reprodutibilidade, ficando o benefício clínico e a segurança externa por confirmar por essa avaliação.

Palavras-chave: Tomografia Computadorizada, Redução de Artefatos Metálicos, Aprendizagem Profunda, U-Net, Aprendizagem Multitarefa, Reprodutibilidade, Particionamento por Paciente, Validação Externa, Avaliação da Qualidade de Imagem

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This thesis is dedicated to my beloved late mother, who would have been filled with pride and genuine happiness with each step of this achievement.

Chapter 1

Introduction

When Wilhelm Conrad Röntgen, University of Würzburg's physics teacher, in Germany, decided to repeat the experiment done by Philipp Lenard with a modified Crookes tube to study cathode rays, no one could foresee the impact of his discoveries on medicine.

It didn't take long for his experiments to present surprising results that demonstrated that the observed behavior was not compatible with cathode rays, but rather with a kind of more penetrating ray that did not suffer deflections in the presence of a magnetic field. During the handling of the pieces used in his experiments, he noticed that the outline of his bones was being imprinted on the fluorescent screen he was using.

Like that, on December 22, 1895, he invited his wife, Anna Bertha, to make a 15-minute exposure to these rays, obtaining what would be the first radiography of the human body. In this same year, Röntgen published the article "On a new kind of rays", in which he named the responsible agent as "X-rays". Already in January 1896, newspapers announced on the front page the discovery as a revolutionary way to observe the interior of the human body without the need for surgical incisions, pointing out its potential use in medical diagnosis [1, 2].

On February 1896, in Massachusetts (USA), Edwin Brant Frost produced a radiograph of a Colles fracture of a patient for his brother, a local doctor, marking the beginning of the incorporation of X-rays into medical practice [1]. Thus, less than a year after the announcement of Röntgen's discovery, X-rays quickly moved from laboratories to extensive use in diagnosis and therapy [1]. Since that, the techniques have undergone successive modernizations and diversified into multiple modalities and devices, consolidating themselves as an indispensable tool in modern healthcare [3, 4].

From these successive modernizations emerged, in 1971, the first computerized tomography scanner. This technology is based on the principle of x-rays attenuation, in which the amount of radiation absorbed or scattered by the material is related to its density [5, 6].

Computed Tomography (CT) was the first medical imaging modality to provide quantitative information about tissue attenuation, something that was not possible with conventional radiography [7]. Furthermore, allowing sectional visualization of the human body, it eliminated the overlap of anatomical structures inherent to two-dimensional images, enabling precise localization and assessment of the extent of lesions [7].

However, these advances brought a significant increase in technological complexity. The acquisition of multiple projections, combined with computational reconstruction and digital image processing, introduced new forms of degradation that were nonexistent or of little relevance in conventional radiography. Thus, computed tomography became particularly susceptible to the presence of artifacts, which can compromise the diagnostic quality and clinical reliability of images [7].

Various types of artifacts can be observed in computed tomography, including noise, beam hardening, motion, ring artifacts, and metal artifacts, which can obscure or simulate pathologies [8]. The causes of these artifacts are varied, ranging from detector failures or miscalibration, statistical limitations associated with low photon counts, to physical phenomena such as beam hardening and scattering, which produce dark streaks between high-attenuation objects such as metal, bone, or iodinated contrast media [7].

Specifically, streak artifacts caused by metallic implants are observed in approximately 21% of computed tomography scans, being considered extremely common in clinical practice [7]. The high attenuation or even total blockage of X-rays by these implants prevents the radiation from adequately reaching the detectors, resulting in low-quality projection data and compromising image reconstruction [9].

The increase use of metallic implants, driven by the evolution of healthcare, has intensified concerns regarding the negative impact of these artifacts on image quality and the reliability of diagnostic analyses [9]. Traditionally, various **Metal Artifact Reduction (MAR)** algorithms have been proposed, including image inpainting techniques, sinogram inpainting, and **Model-Based Iterative Reconstruction (MBIR)** or combinations of these approaches [10].

Although analytical methods such as **Filtered Back Projection (FBP)** have dominated tomographic reconstruction for decades due to their computational efficiency and conceptual simplicity, their limitations become evident under non-ideal clinical conditions, requiring higher radiation doses to ensure image quality [11]. **MBIR** methods emerged as an alternative to overcome these limitations, at the cost of higher computational demand and the introduction of artificial textures in the reconstructed image [11].

Recently, deep learning-based approaches have been widely investigated for artifact correction in computed tomography, demonstrating superior performance and greater clinical applicability compared to conventional methods [12]. In particular, [Convolutional Neural Networks \(CNNs\)](#) have been proposed for metal artifact reduction, enabling noise suppression and preservation of anatomical and soft tissue details [12].

1.1 Clinical and Technical Context

Due to the fact that artifacts in [CT](#) can hide or simulate pathologies, its presence can compromise the diagnostic quality and clinical reliability of images. A diagnostic mistake can have serious consequences for the patient, including treatment delays, inadequations or unnecessary, and even life-threatening risks. Additionally, the presence of artifacts can lead to increased radiation exposure, as additional scans may be required to obtain a clear image, which can increase the risk of radiation-induced complications. The constant technological evolution has made the use of metallic implants increasingly common in the medical context, which significantly increases the presence of this type of artifact and also the associated diagnostic mistakes.

Therefore, the reduction of artifacts in [CT](#) is a research area with significant clinical importance, and the development of effective methods to reduce artifacts gains high value in medical practice.

1.2 Work Motivation

Although studies using deep learning for metal artifact reduction are not new, a lot of the work published to date does not present a rigorous and detailed evaluation of the performance of the proposed models. In particular, the evaluation of the performance of models on external datasets (which were not used during model training) or evaluation of images by qualified professionals is often lacking. Moreover, the training of supervised models depends on reference images, which are difficult to obtain in a clinical context, creating a dependency on synthetic or simulated images. This lacking turns out to be still more relevant when considering that the performance of models trained on synthetic images may not be representative of their performance on real images, which can lead to overestimation of their effectiveness and limit their clinical applicability.

In this sense, the present study proposes a more rigorous and detailed evaluation of the performance of deep learning models for this purpose.

Chapter 2

Objectives and Investigation Question

2.1 General Objective

The general objective of this dissertation is to develop, evaluate and rigourously compare a [CNN](#) model for metal artifact reduction in [CT](#), prioritizing anatomical preservation, robustness to domain variation and interpretability of the correction process. Additionally, the study aims to investigate the generalization of the best performing model to real and external data, with a focus on clinical applicability. Healthcare professionals will be involved in the evaluation of the model's performance on real data, providing a comprehensive assessment of its effectiveness and potential for clinical use.

2.2 Specific Objectives

The specific objectives of this study include:

- Develop a reproducible pipeline for training and evaluating metal artifact reduction models in [CT](#) in the image domain, with rigorous separation between training, validation, and test data.
- Evaluate the performance of a [U-shaped Network \(U-Net\)](#) to image reconstruction for metal artifact reduction and compare it with a version of itself with an additional supervision of artifact mask.
- Evaluate the performance of the two versions using quantitative metrics like: [Peak Signal-to-Noise Ratio \(PSNR\)](#), [Structural Similarity Index Measure \(SSIM\)](#), [Mean Absolute Error \(MAE\)](#), [Binary Cross-Entropy \(BCE\)](#) and [Dice Similarity Coefficient \(DICE\)](#). Besides, performing error maps, statistical and qualitative analysis to assess the quality of the reconstructed images.

- Investigate the generalization capability of the best performing version in the internal evaluation stage on real and external data from the [The Cancer Imaging Archive \(TCIA\)](#).

2.3 Hypothesis and Outcomes

The primary hypothesis of this study is that, in the internal test partition, the correction produced by each architecture makes a real, meaningful difference to the structural quality of the image compared to the artifact input. This is measured with [SSIM](#), calculated separately in the [Region of Interest \(ROI\)](#) where the metal artifact was inserted and in the clean [ROI](#), without an artifact, which should stay about the same or change only a little. Each patient's slices were averaged into a single value per patient ($n = 41$), and the comparison asked whether there was typically no difference at all between the input and the model.

The secondary hypotheses are: (i) that [PSNR](#) and [MAE](#), in HU, also improve compared to the input, using the same per-patient averaging; (ii) that the same three metrics ([SSIM](#), [PSNR](#) and [MAE](#)) also improve when calculated over the whole image; and (iii) a direct comparison between the Single Model and the Dual Model on these same three metrics, addressing the architecture-comparison objective.

As a way to judge anatomical preservation, the change introduced in the Clean Zone was expected to be clearly smaller than the correction observed in the Artifact Zone, and the overall shift in average HU values, checked with a Bland–Altman plot at the patient level, was expected to be small compared to the width of the [HU](#) window used. No fixed numeric limit was set in advance for how small the Distant-Zone change had to be, so preservation is described rather than formally tested.

Finally, since the external [TCIA](#) set has no clean reference image to compare against, the generalization hypothesis is necessarily more limited: the selected architecture is expected to produce changes of a plausible size in the external cases, without simply reproducing the input unchanged, but only the doctors' blinded evaluation ([section 4.6](#)), not a reference-based metric, can confirm whether artifact removal is accurate and clinically safe.

Chapter 3

Related Works

3.1 Metallic Artifacts in CT

The traditional methods of artifact reduction in CT are predominantly based on physical and mathematical principles associated with image formation. Barrett and Keat[13] provide a detailed description of the origin of the main artifacts in CT, including beam hardening, photon starvation, scattering, and geometric effects, as well as classical strategies for their mitigation.

Reconstruction techniques such as FBP were widely used due to their computational efficiency, but they are highly sensitive to noise and metallic artifacts. More advanced methods, such as Iterative Reconstruction (IR) and MBIR, incorporate statistical models of the acquisition system and noise, allowing for partial artifact reduction. However, these approaches often introduce artificial textures and have limitations in complex clinical scenarios [11].

Recent revisions indicate that commercial algorithms for MAR, such as Orthopedic Metal Artifact Reduction (O-MAR), Iterative Metal Artifact Reduction (iMAR), and Single Energy Metal Artifact Reduction (SEMAR), although effective in specific situations, rely on proprietary parameters and exhibit inconsistent performance in cases of dense or multiple implants [9, 14]. These limitations have driven the investigation of deep learning-based approaches.

3.2 Artifact Reduction based on Deep Learning

The First applications of Deep Learning (DL) para MAR focused on the image domain, exploring convolutional networks to map degraded images to corrected versions. Gjestebj et al. [15] investigated metal artifact reduction in the image domain based on CNN and demonstrated significant quantitative improvements compared to conventional techniques.

Zhang e Yu [16] also purposed a supervised model based on Convolutional Neural Network (CNN) for metal artifact reduction in CT, achieving consistent gains in PSNR and Root Mean

Square Error (RMSE). These works helped consolidate the use of convolutional networks for **MAR** in the image domain. In particular, Zhang and Yu used a shallow fully convolutional **CNN** for fusing information from multiple pre-corrected images. Subsequent works increasingly adopted encoder-decoder architectures, especially those based on **U-shaped Network (U-Net)**, for correction/restoration tasks. **U-Net**-type networks and their variants are capable of capturing global contextual information through the contraction path, and recovering fine spatial details through the expansion path and *Skip Connections*[17].

Recent Systematic reviews have highlighted these advances, emphasizing that most supervised methods in the image domain exhibit competitive performance, especially when trained with high-quality paired data [9, 11].

In addition to image domain, some studies explored the sinogram domain, with linear interpolation and **Normalized Metal Artifact Reduction (NMAR)** methods. These approaches operate directly on the projection data and act directly on the physical origin of metallic artifacts[18]. However, access to raw projection data is often restricted in commercial **CT** systems, limiting the applicability of such approaches in clinical settings [15].

Beyond that, hybrid models have been purposed to leverage the advantages of each approach. The method purposed by Zhang and Yu [16] seeks to merge **CNN** in the image domain and classical **MAR** in the sinogram domain. In the adopted approach, the initial refinement occurs in the image domain, but the complete method uses the image generated by the **CNN** as a prior for replacement of affected regions in the sinogram. This characterizes a hybrid approach, but still does not employ **CNN** in both domains. More recent models, such as DuDoNet[19], integrate deep networks in both domains, demonstrating high performance and characterizing what we call Dual Domain. However, these methods present greater computational complexity and dependence on access to raw sinogram, which motivates the continuous exploration of robust solutions in the image domain.

To address the scarcity of anatomically paired data with metal artifact/metal artifact-free, self-supervised learning-based approaches have emerged. Yu et al. [20] proposed a method capable of reducing metal artifacts without the need for fully clean reference images, expanding the clinical applicability of these techniques.

Others works explored the use of **Generative Adversarial Network (GAN)** for **MAR**. Huang et al. [21] used multimodal fusion and **GANs** to improve texture preservation and reduce excessive smoothing. Despite good visual results, **GAN**-based can introduce training instability and a higher risk of structural hallucination[22], raising clinical concerns.

3.3 Reproducible Evaluation and Generalization

The evaluation of DL models for CT is typically performed through quantitative metrics such as PSNR, RMSE, and structural similarity-based indices. Traditional error-based metrics, such as RMSE and PSNR, quantify pixel-wise discrepancies but do not adequately capture structural information relevant to visual perception and clinical interpretation. In this context, Wang et al. [23] introduced the SSIM as a perceptual metric that explicitly models luminance, contrast, and structure. Subsequently, the Multi-Scale Structural Similarity Index Measure (MS-SSIM) was proposed to incorporate scale-dependent structural variations, extending SSIM to multiple resolution levels [24]. Horé and Ziou [25] demonstrated that both PSNR and SSIM have limitations depending on the type of distortion considered, suggesting that the choice of metric may depend on the application context. Given that metallic artifacts manifest primarily as structural distortions affecting anatomical boundaries and long-range streak patterns, metrics sensitive to structural preservation, such as SSIM and MS-SSIM, are particularly suitable for evaluating the performance of MAR methods, as they explicitly consider structural components at different spatial scales [23, 24].

Additionally, a critical challenge identified in the literature is the models generalization. Zech et al. [26] demonstrated that models with high performance on internal datasets can fail significantly when applied to external data, highlighting the impact of Domain Shift. Recent reviews reinforce the need for rigorous external and clinical validation to ensure safety, robustness and real applicability [27, 28].

The reviewed literature demonstrates significant advancements in artifact reduction in CT through DL, but also evidences persistent gaps: lack of experimental standardization, limited external validation, and risk of anatomical hallucination.

Chapter 4

Materials and Methods

4.1 Data, curation and traceability

4.1.1 CT-MAR Dataset and Reference Sources

The main dataset used in this work comes from the [CT Metal Artifact Reduction Challenge \(CT-MAR\)](#) [29, 30] promoted by the [American Association of Physicists in Medicine \(AAPM\)](#) and consists of 14,000 training pairs containing [CT](#) images with simulated metal artifacts and their original artifact-free versions. The data was generated through a hybrid simulation framework [31], which incorporates publicly available clinical images and simulated metallic objects with the CatSim simulator, integrated into the open-source XCIST toolkit [32]. The dataset is distributed into two anatomical categories:

- **Head:** 1.626 (11.61%) samples derived from the *UCLH Stroke EIT Data* [33, 34];
- **Body:** 12.374 (88.38%) samples derived from the *NIH DeepLesion Data* [35].

Where the folders in the *body* category contained trunk [CTs](#), covering axial, thoracic, and abdominal slices. The data is organized into subfolders, each containing up to 1,000 samples. Only the following files were used for training and validation:

1. **Metal-free images (reference):** Files contained in the *Target* folder of size 512×512×1 pixels;
2. **Images with metal artifacts (corrupted):** Files contained in the *Baseline* folder of size 512×512×1 pixels;

All image files (.raw) in the dataset are presented in binary format without a header, with float32 precision in little-endian order. The images were already converted from linear attenuation coefficients (in mm^{-1} units) to [Unidade Hounsfield \(HU\)](#), using the normalization value of

0.02 mm^{-1} for water. There was not additional information about which exam each slice belonged to, if they were from the same patient, or which specific anatomical regions were represented. The only available information was the organization by folder, indicating the anatomical category (Head or Body). The lack of detailed metadata imposed the need for a careful pre-processing approach to ensure that the data was properly organized and used during model training and validation. Therefore, a significant effort was dedicated to tracking the images to the original datasets (UCLH Stroke EIT Data and NIH DeepLesion Data) to obtain additional information about the exams and patients, allowing for an appropriate partitioning of the data.

4.1.2 Partitioning at patient level

The data partitioning was stratified by patients, ensuring that slices from the same patient were not distributed across training, validation, and test sets. This approach prevents information leakage and ensures a more realistic assessment of the models generalization capability.

Due to the lack of detailed clinical metadata in the [CT-MAR](#) dataset, an algorithmic screening of the slices to the original datasets (UCLH Stroke EIT and NIH DeepLesion Data) was necessary to enable the association of the provided metadata with the [CT-MAR](#) slices. This process allowed for the identification of which exam and patient from DeepLesion/UCLH Stroke EIT each [CT-MAR](#) image belonged to. This information was subsequently used to perform patient-level partitioning of the data into training, validation, and test sets, reducing the risk of data leakage between experimental sets.

The correct separation of the data is particularly important in medical imaging studies based on [DL](#), once slices belonging to the same exam often present high anatomical similarity. Thus, the simultaneous presence of images from the same patient in training and testing can artificially inflate the model's performance and compromise the assessment of its generalization capability [36].

The Deep Lesion images [35] are stored in 16-bit PNG format, while the [CT-MAR](#) images are provided in .raw format directly in Hounsfield units (HU). The UCLH Stroke EIT [33] provides the exams in DICOM format. Therefore, regardless of the original format, all images were converted to HU before comparison, ensuring uniformity across the different datasets.

After the conversion, all images was represented on the same intensity scale to allow direct comparison.

The screening algorithm was based on fact that [CT](#) images preserve quantitative information in [HU](#), which directly reflects the tissue attenuation properties. Since anatomical structures

exhibit high spatial continuity between consecutive slices of the same acquisition, images corresponding to the same anatomical slice tend to present similar HU distributions and highly consistent global structural patterns [23, 37].

That way, the method was not designed to verify absolute pixel-to-pixel equality between images, but rather to determine whether the overall structural organization of the anatomy remains extremely similar across different representations of the same slice. This behavior is particularly important because small local variations in intensity, numerical discretization, and linear transformations associated with storage format conversion tend to preserve the global anatomical structure of the image even when small local intensity variations occur [23, 37].

Thus, the adopted methodology seeks to identify correspondences between images based on global structural similarity, allowing for a margin of local variation that does not compromise the correct identification of the exam and patient of origin. Initially, each image was reduced to a compact representation through *downsampling*, preserving mainly the relevant global anatomical information. This step significantly reduces the computational cost without compromising the overall structure of the image.

Subsequently, a representative vector (*fingerprint*) was calculated for each image. The fingerprints of the CT-MAR images were then compared with the fingerprints of the DeepLesion and UCLH Stroke EIT images using a normalized dot product-based vector similarity metric (cosine similarity). It was chosen for being invariant to global scale differences between vectors — a relevant property given that the images come from different formats with different intensity normalizations. For each CT-MAR image, the image with the highest similarity score was selected.

Since the majority of the CT-MAR images originated from DeepLesion, the screening process was initially prioritized on that dataset. Subsequently, the same methodology was applied to the DICOM exams from UCLH Stroke EIT to identify the remaining images. In this way, it was possible to implement a unified traceability strategy regardless of the storage format of the original datasets.

The matching was considered valid only when the score of similarity exceeded a previously defined empirically threshold. Cases with very high scores were automatically marked as safe matches, as it was observed that all pairs visually analyzed with a score above 0.993 effectively corresponded to the same image. To maintain an additional safety margin, the match was considered automatically valid only when the score exceeded 0.994.

Scores below this threshold indicated potentially ambiguous matches and were subjected to manual visual inspection. Scores below approximately 0.98 indicated clearly incorrect matches

and were automatically discarded without the need for manual review. Thus, all pairs with scores between 0.98 and 0.994 were visually analyzed to confirm the validity of the match. In order to avoid unnecessary exclusion of pairs, all pairs with a score below 0.98 for both datasets (DeepLesion and UCLH Stroke EIT) were subjected to visual inspection to ensure they were not erroneously discarded.

For the 530 pairs that remained without approved matches after direct search in both source datasets, a complementary indirect matching step was applied. In this step, each unmatched CT-MAR image was compared with already traced CT-MAR images; the most similar neighbors were used to select candidate exams from DeepLesion and then all *slices* of those exams were searched with 64×64 dimension *fingerprints*. The 530 proposals were visually inspected: 95 matches were approved and incorporated into the consolidated manifest, while 435 remained without a safe match and were excluded.

Moreover, the algorithm was designed to automatically ignore images that had already been verified as correct matches, reducing redundancies and accelerating future reprocessing. Given the high number of images present in the datasets, the processing was parallelized and optimized through the use of vectorized matrix operations. When available, the algorithm utilizes GPU acceleration to reduce execution time. The calculated fingerprints were also cached to avoid unnecessary recomputation in subsequent runs.

Although, the compact representation based exclusively on the mean of block intensities HU is efficient, it can theoretically generate false positives when different patients exams are acquired with identical acquisition protocols or present very similar anatomical structures. Since these conditions tend to produce similar intensity distributions and, consequently, fingerprints with high scores. To mitigate this risk, a conservative automatic acceptance threshold (0.994) was adopted and a systematic visual validation of all pairs with scores between 0.98 and 0.994 was performed. Additionally, as a complementary safeguard, all automatic matches were analyzed in terms of MS-SSIM, where the pairs with the most critical scores were subjected to an additional visual review to ensure the reliability of the matches.

Despite that, this limitation does not compromise the main objective of the methodology. Even though consecutive slices from the same exam often present high spatial redundancy and can therefore obtain very similar similarity scores. The purpose of the algorithm was not to perform rigorous voxel-to-voxel alignment, but rather to correctly identify the exam and patient of origin of each CT-MAR image. In this way, even potential ambiguities between adjacent slices remain restricted to the same anatomical volume and do not introduce a relevant risk of data leakage between training, validation, and test sets.

The literature demonstrates that the quality of the pairs used in supervised studies directly influences the performance and generalization capability of [MAR](#) models [9, 19]. In addition, structural metrics of image quality, such as [SSIM](#), have been widely used in medical imaging due to their higher sensitivity to structural preservation [23]. Although the implemented method does not directly use [SSIM](#), the adopted strategy seeks to precisely preserve the global structural information of the image during the matching process.

Moreover, the data utilization without any screening treatment would condition the entire project to an inherent data leakage from the dataset, compromising the evaluation of the models performance and the validity of the results. Thus, even if the screening method is not perfect, it represents a significant improvement over the total absence of screening, ensuring a more reliable basis.

Taking this into account, this step constituted an important quality control procedure for the experimental dataset and was fundamental to ensure an appropriate partitioning of the data used in this work.

4.1.3 Final Group and Exclusion of Unmatched Pairs

Not all slices could be traced, but most of the data was successfully associated with the original exams. At the end of the direct and indirect matching steps, and before the complementary analysis with [MS-SSIM](#), out of the 14,000 pairs provided by the [CT-MAR](#) Dataset, 11,942 were traced to DeepLesion (including the 95 recovered by indirect matching) and 1,623 to UCLH Stroke EIT, totaling 13,565 pairs with associated metadata [4.1](#).

Dataset	Traced	Matched	Unmatched
DeepLesion	12,374	11,942	432
UCLH Stroke	1,626	1,623	3
Total	14,000	13,565	435

Table 4.1 Resultados do rastreio antes da análise complementar com [MS-SSIM](#).

In the evaluation step through [MS-SSIM](#), among all pairs with a score below 0.90, only 3, previously associated with Deep Lesion, were rejected in the visual inspection [4.2](#).

Dataset	Traced	Matched	Unmatched
DeepLesion	12,374	11,939	435
UCLH Stroke	1,626	1,623	3
Total	14,000	13,562	438

Table 4.2 Pair matching results after the complementary MS-SSIM analysis.

Due to the fact that a significant amount of the data could be traced with high reliability, the remaining cases that could not be safely grouped were excluded from the training and validation process. In total, 438 pairs were removed for not having sufficient information to ensure safe grouping at the patient level. Among them, the 3 pairs rejected for being false positives in the evaluation step via [MS-SSIM](#). All pairs removed from this study were visually inspected to confirm the absence of correspondence, ensuring that the exclusion was justified.

	Excluded	
	Before	After
Total	435	438

Table 4.3 Unmatched pairs data that was excluded before and after the complementary MS-SSIM analysis.

Thus, out of the 14,000 pairs originally provided by the [CT-MAR](#) Dataset, 13,562 were included in this work, while 438 were excluded to preserve the integrity of the experimental partitioning and minimize the risk of data leakage.

4.1.4 Anatomical Distribution and Stratification

The anatomical proportion of 14,000 pairs in the [CT-MAR](#) Dataset changed little after the screening and exclusion process of pairs with low reliability. Of the 438 removed cases, 435 were of the *body* type and 3 of the *head* type. The anatomical distribution of the pairs included in the study is as follows:

- **Body:** approximately 88.03% ;
- **Head:** approximately 11.97% .

Stratification was implemented within each subset (train/val/test) to maintain the anatomical proportion at the patient level, ensuring that the relative proportion between categories is preserved across all subsets. In this way, the anatomical balance is preserved, ensuring that the

models are trained and evaluated on an appropriate representation of the anatomical diversity present in the original dataset.

At the patient level, the anatomical distribution is as follows:

- **Body:** approximately 95.9% (258/269 patients);
- **Head:** approximately 4.1% (11/269 patients).

After partitioning, the train, validation, and test subsets had the following distribution:

Split	Body	Head	% Body	% Head
Train	180	7	96.3%	3.7%
Val	39	2	95.1%	4.9%
Test	39	2	95.1%	4.9%
Total	258	11	–	–

Table 4.4 Patient-level anatomical distribution by split.

The patient-level partition was defined only once, with a fixed random seed, maintaining approximately 70% of the patients for training, 15% for validation, and 15% for testing. This fixed partition ensured independent subsets for validation and testing and was kept unchanged across all experimental repetitions.

Split	Patients	% Total
Train	187	69.52%
Val	41	15.24%
Test	41	15.24%
Total	269	100%

Table 4.5 Patient distribution by split.

Due to the large variation in the number of slices per patient, the anatomical distribution at the pair level is quite different from the distribution at the patient level (4.6 and 4.7). The same occurs for the total pairs distribution. Thus, the proportion of $\sim 70/15/15$ was maintained at the patient level, but at the pair level it does not follow exactly the same proportion.

Split	Body	Head	% Body	% Head
Train	8 563	953	89.98%	10.02%
Val	2 147	385	84.79%	15.21%
Test	1 229	285	81.17%	18.83%
Total	11 939	1 623	–	–

Table 4.6 Pair distribution by split and anatomical category.

Split	Pairs	% Total
Train	9 516	70.17%
Val	2 532	18.67%
Test	1 514	11.16%
Total	13 562	100%

Table 4.7 Pair distribution by split.

4.2 Pre-processing

4.2.1 Integrity Verification

All pairs were validated through an integrity report, which checks for the existence of both sides of the pair (artifact image and clean image). There were no pairs with missing data, ensuring that the dataset was complete and ready for subsequent pre-processing and model training steps.

4.2.2 Background Removal through Body Mask

The [CT](#) images often contain large background areas that do not contribute to the task of metal artifact removal. That creates a class imbalance in the training data, where irrelevant background voxels dominate over the anatomical regions of interest. This imbalance, in turn, hinders the learning and deviates the model focus from clinically relevant structures [38, 39]. To address this challenge, a pre-processing step also based on [DL](#) was implemented. A [U-Net](#) was trained to segment the body tissue from the background, generating binary masks that were applied to the original images to isolate the regions of interest. This model was trained using $\sim 1,500$ samples from the [CT-MAR](#), where the segmentation masks were automatically generated through a thresholding process. Additionally, a manual refinement was performed on

the obtained masks to ensure the quality of the training data and consequently the effectiveness of the segmentation model.

This step was essential to reduce the influence of background artifacts and improve the quality of features extraction during training.

4.2.3 Intensity Window and Normalization

The image intensities are expressed in HU from the CT-MAR. To reduce extreme variations, an intensity window appropriate to the anatomical context was applied. After windowing, the values were normalized to the range $[0, 1]$ and 2D tensors of dimension 256×256 were generated for each slice and its pair.

4.2.4 Artifact Mask for Auxiliary Supervision

Binary difference masks of the affected regions by artifacts were produced and used as an additional supervision signal in the dual architecture. These masks are exclusively used during the training and are not required during inference. The binary difference mask is formally defined as:

$$M_{diff}(i, j) = \begin{cases} 1; & \text{if } |I_{artifact}(i, j) - I_{clean}(i, j)| > \tau \\ 0; & \text{otherwise.} \end{cases} \quad (4.1)$$

where $I_{artifact}$ represents the image with artifacts, I_{clean} the clean reference image, and τ is the normalized difference threshold. The value of this threshold was empirically defined to capture regions with perceptible differences while maintaining robustness to low-amplitude noise.

4.3 Compared Architectures

4.3.1 U-Net Single Supervision

The first model consist of an architecture based on U-Net [17], composed of encoder-decoder with skip connections. Li et al [40], apply blocks of Batch Normalization (BN) and Rectified Linear Unit (ReLU) activation functions after each dynamic convolution. This work does something similar, but chooses to use standard convolutions with 3×3 kernel, followed by BN blocks and ReLU activation functions. The choice for standard convolutions is justified by simplicity and computational efficiency, in addition to maintaining the approach used by Ronneberger et al [17]. Like Li et al [40], the decoder also uses bilinear upsampling. This architecture has only one output.

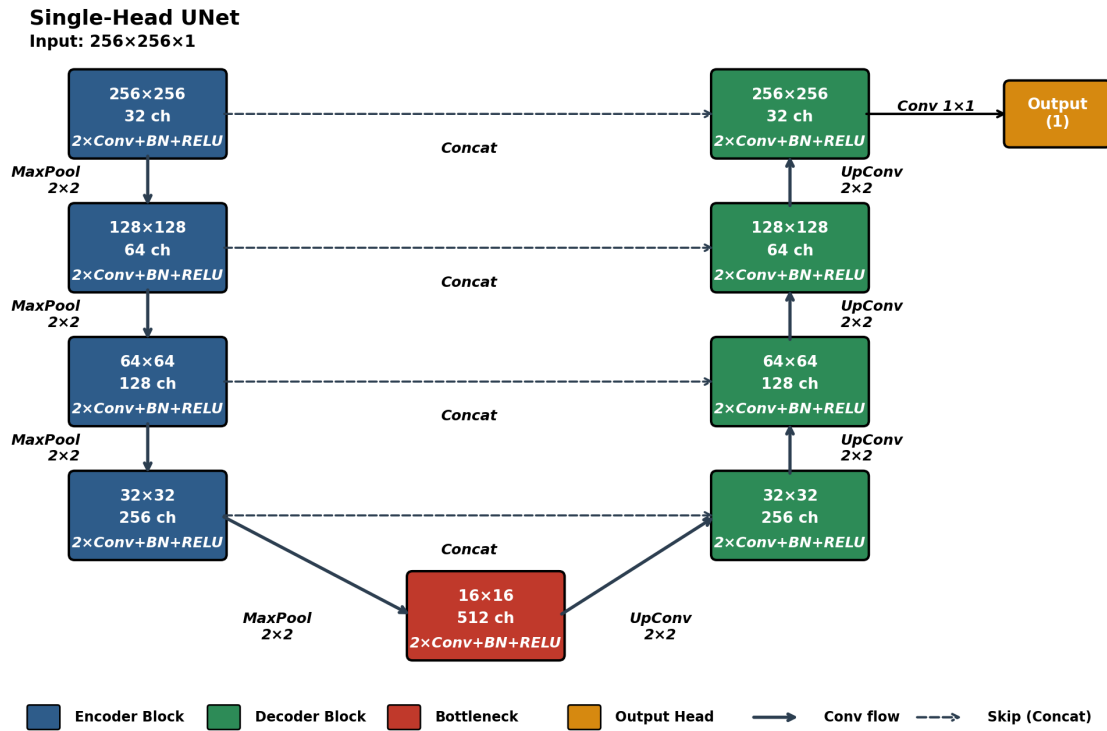


Fig.4.1 Single supervision architecture diagram.

4.3.2 U-Net Dual Supervision

A variation of the model described above was conceived, maintaining the shared *encoder* and *decoder* and adding an independent output in the last layer. Thus, there are two distinct outputs in the decoder, each with its own loss function. The first output is responsible for reconstructing the corrected image, while the second output is dedicated to predicting the artifact mask.

The rationale for adding the auxiliary supervision of the artifact mask is based on the premise that the presence of an additional loss focused on segmenting regions affected by artifacts can provide complementary information to the model, aiding in learning more robust and discriminative representations. This approach is inspired by previous works that have shown that multi-task supervision can improve performance in related tasks [41, 42].

The two architectures were evaluated against each other to determine the impact of the additional supervision with artifact mask on reconstruction quality and artifact reduction capability. All model bases are identical, with the presence of the artifact mask prediction output being the only difference in the second version. Thus, the comparison between the two architectures is fair and isolated, allowing for an exclusive evaluation of the impact of the additional supervision of the artifact mask.

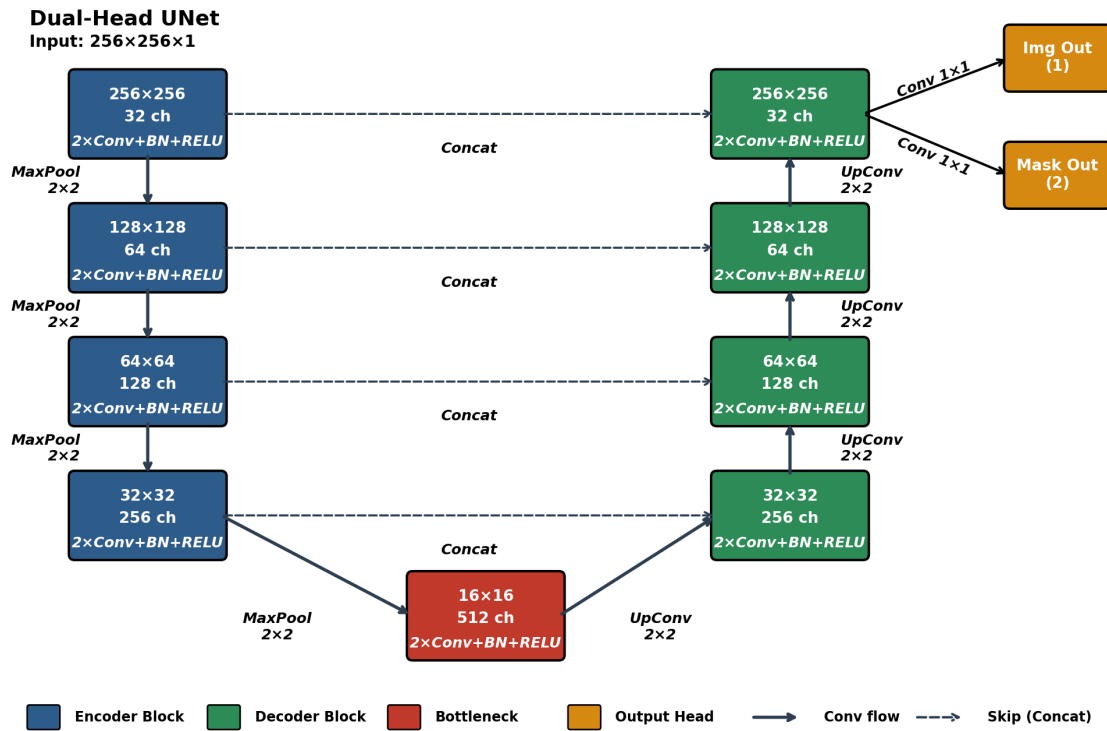


Fig.4.2 Dual supervision architecture diagram.

4.4 Reproducible Training

4.4.1 Settings, determinism and two-stage selection

Both models were trained under the same settings, with *float32* precision. To assess stability against different initializations and batch presentation orders, each architecture was trained with deterministic operations and with seeds 1337, 42, and 2024, always on the same patient partition. **Automatic Mixed Precision (AMP)** was disabled and batches were kept complete, including the last batch of each epoch, to avoid samples being excluded from training, validation, or evaluation.

The AdamW optimizer was used, with an initial learning rate of 2.26×10^{-4} , weight decay of 1.00×10^{-4} and gradient clipping with norm 1.00, identical for both architectures. The learning rate was adjusted by the ReduceLROnPlateau scheduler, with a reduction factor of 0.50, patience of two epochs, cooldown of one epoch and minimum value of 1.00×10^{-6} , monitoring the reconstructed image **SSIM** on the validation set in both models. The training had a maximum limit of 300 epochs and early stopping could only be triggered after at least five epochs and occurred when the monitored metric did not improve by at least 1.00×10^{-4} for 15 consecutive

epochs. In each run, the checkpoint corresponding to the best value of the monitored validation metric was retained.

Only the best checkpoint per run was preserved, saved in `.keras` format with a filename encoding the architecture, epoch, and monitored metric; the training CSV explicitly flags the corresponding row with `best_checkpoint=1`. Checkpoint selection relied exclusively on the validation metric chosen in advance: `val_ssim_hu` for the Single Model and `val_img_out_ssim_hu` for the reconstruction output of the Dual Model.

Each of the six validation-selected checkpoints was evaluated once on the fixed internal test partition. The candidate advanced to external evaluation was selected from this frozen set by the highest mean patient-level $\Delta SSIM$ in the Artifact Zone. As a preservation safeguard, the corresponding changes in the Distant Zone were required to be substantially smaller than the correction observed in the Artifact Zone. No specific numeric limit was set in advance for how small this Distant-Zone change had to be, so this comparison is only descriptive and is not, by itself, a formal safety check. Global PSNR, SSIM, and MAE, together with the remaining region-of-interest metrics, were complementary and did not override this selection rule. Thus, the test partition was used only to compare the six already frozen candidates. It was not used to choose a new checkpoint, tune hyperparameters, alter weights, or trigger further training. Once selected, the architecture, seed, weights, and hyperparameters of the external-evaluation candidate were frozen.

Run traceability was ensured through reproducibility snapshots in JSON format, preserved in `outputs/reproducibility/`. Six final internal evaluations were frozen, corresponding to the three random initializations (seeds 42, 1337, and 2024) of the Single Model and Dual Model. All snapshots were generated from commit `a950be7276a5585de45e8ae874c59c065a1985be`, on branch `projeto_final`, with a clean working tree (`is_dirty: false`).

Each snapshot records the operating system, Python version, and relevant dependency versions. The execution environment used Python 3.11.9 on Windows 10, including TensorFlow 2.15.0, NumPy 1.26.4, Pandas 2.2.2, scikit-image 0.21.0, and Matplotlib 3.7.5. In addition, SHA-256 hashes were computed for the configuration files, the canonical pair manifest, the patient-level partition definition, the tensor generation metadata, and the environment requirements file.

For each run, the snapshot preserves the path, size, and SHA-256 hash of the checkpoint, the full command used for evaluation on the test set, and the hashes of the output files, including aggregated and per-sample metrics. Thus, every value, table, or figure reported can

be unambiguously traced to the code, environment, input data, checkpoint, and results that produced it.

During the training, the following quantities were monitored:

- **SSIM, PSNR, and MAE in HU scale:** calculated in the body region to track the structural and radiometric fidelity of the reconstructions;
- **Reconstruction loss:** to track the optimization of the corrected image;
- **Segmentation loss:** only in the dual model, to track the auxiliary supervision of the artifact mask.

4.4.2 Loss Functions

- **Reconstruction:** Despite L2 being widely used for image restoration, Zhao et al. [43] highlight the best results studying different loss functions for this task and show that it consistently underperforms perceptually-motivated alternatives. The authors studied loss functions based on L1, SSIM and MS-SSIM, being that second achieves equal or tiny inferior results than L1, while the third achieves better performance than SSIM but still does not clearly outperform L1. The paper also tests a mix of L1 and MS-SSIM, and obtained superior results in all metrics evaluated by the authors. This combination is justified by the complementarity between the two functions, where L1 favors local fidelity and MS-SSIM promotes structural preservation, resulting in a more balanced correction of artifacts. Although the best result reported by the authors involved L1 and MS-SSIM, in this work the weighted sum of L1 and SSIM was adopted. This choice maintains the general principle of combining a pixel fidelity component with a structural component, without reproducing the specific formulation or weights used by Zhao et al.

The weights used for the experimental protocol were 0.7 for L1 and 0.3 for SSIM. The relation adopted for the combination of the loss functions is as follows:

$$\mathcal{L}_{Recon} = 0.7 \cdot \mathcal{L}_{L1} + 0.3 \cdot (1 - \text{SSIM})$$

Zhao et al. [43] applied a Gaussian spatial weighting to the L1 component in the hybrid loss, in order to make it compatible with the spatial distribution of the gradients of the MS-SSIM component. In this work, the L1 component was calculated directly as the mean absolute error in the body region, without additional Gaussian weighting. Since the normalization used is linear with respect to HU values, this component maintains

a penalty directly proportional to the radiometric error, while **SSIM** contributes to local structural preservation.

This combination of loss function is adopted for the Single Model and also for the reconstruction output of the Dual Model. In this way, it is ensured that the comparison between the two versions is fair, isolating the effect of the additional supervision of the mask.

- **Mask auxiliary supervision:** The mask artifact prediction output, present in Dual Model, was trained with the sum of **Dice Similarity Coefficient (DICE)** loss and **Binary Cross-Entropy (BCE)**. This combination provides simultaneous supervision for global overlap between masks and pixel-wise classification, being suitable for segmentation problems with class imbalance [38]. The relation adopted for the combination of the loss functions is as follows:

$$\mathcal{L}_{Seg} = \mathcal{L}_{Dice} + \mathcal{L}_{BCE}$$

The total loss function for the dual-head version is composed of the weighted sum of the reconstruction and segmentation losses, and is defined as:

$$\mathcal{L} = 1.0 \cdot \mathcal{L}_{Recon} + 0.045 \cdot \mathcal{L}_{Seg}$$

4.5 Internal Evaluation

In the inference phase, the models were evaluated using quantitative metrics widely adopted in the literature, in comparison with the Ground Truth. The metrics used to evaluate reconstructed image quality were:

- **PSNR** ;
- **SSIM** ;
- **MAE**.

Recent studies have shown that **MS-SSIM** and **SSIM** capture image fidelity better than other image quality metrics [44]. Especially for medical images, where the fidelity of predicted images is crucial, the choice of metrics that reflect human perception of quality and structural preservation is fundamental. Therefore, **SSIM** was selected as the primary metric for quantitative evaluation of the quality of reconstructed images, ensuring a robust and relevant analysis for the clinical context. And **PSNR** was used as a complementary metric to assess the radiometric fidelity of reconstructions, especially important in computed tomography where the accuracy of **HU** values is essential for diagnosis and clinical planning.

The internal validation was conducted on the validation and test subsets, evaluating consistency, stability, and intra-dataset generalization capability.

4.5.1 Global Metrics

For global evaluation, the predicted and reference images are denormalized to the window $[-1000, 400]$ HU, corresponding to a range $MAX = 1400$ HU. Additionally, the evaluation region is cropped to the smallest bounding box of the body mask before SSIM evaluation, eliminating scan windows composed exclusively of background — which would artificially score $SSIM = 1,0$ and inflate the global metric.

As an additional, purely descriptive check for any overall shift in average HU values, a Bland–Altman plot was made comparing the reconstruction with the clean reference. For each patient, the mean HU of each image was calculated over all valid pixels of the body mask of the respective slices, weighted by the number of pixels in each slice. The graph plots the mean difference per patient ($Model - GT$) against the average of the two measurements, together with the average difference (bias) and the range within which most differences fall ($\bar{d} \pm 1,96 SD_d$, the limits of agreement). This plot is descriptive only, one point per patient, and does not replace the accuracy metrics above or the region-based analysis below.

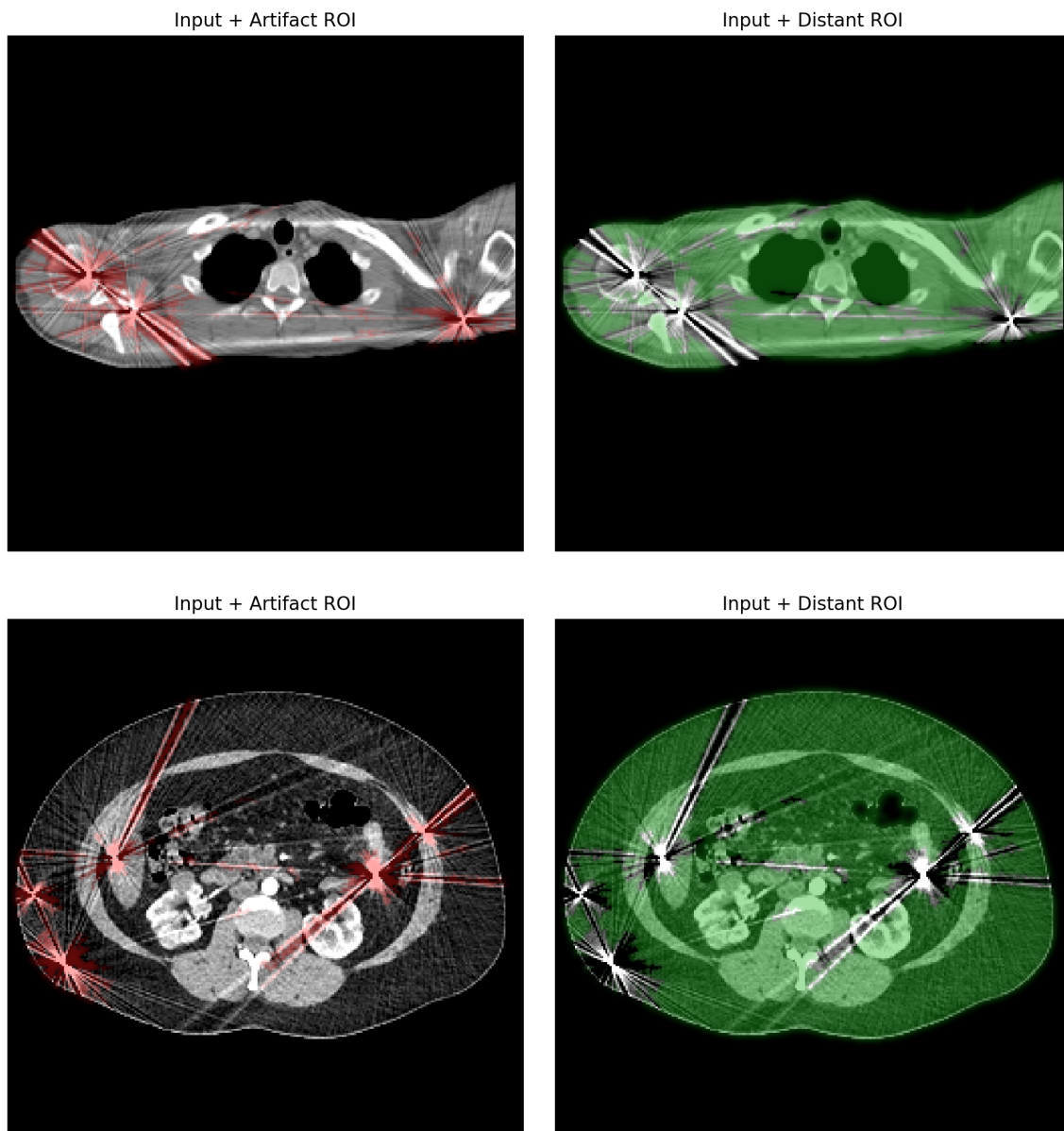
4.5.2 ROI Metrics

The global evaluation does not distinguish the model's behavior in regions directly affected by artifacts from that in body regions distant from the implant. Therefore, each slice was divided into two complementary zones based on how much the artifact-corrupted input differs from the clean reference. Metal artifacts do not only cause strong local streaks near the implant; they are typically accompanied by a faint, almost uniform intensity shift across the whole slice, a side effect of beam hardening. To avoid mistaking this global shift for a local artifact, the pixel-wise difference between the clean reference and the input was first computed, and its typical (median) value over the body region was estimated and subtracted out, so that only deviations beyond this baseline shift were considered.

The Artifact Zone was then defined as the set of body pixels whose bias-corrected difference exceeds a fixed threshold, equivalent to about 42 HU on the normalized intensity scale used ($\tau = 0.03$). Connected-component filtering (using SciPy's image-labeling utilities) was applied to discard components smaller than 20 pixels, which correspond to isolated CT quantum-noise pixels rather than genuine artifact streaks. No additional morphological dilation was applied, preserving the observed geometry of the *streaks*. The Distant Zone was defined simply as

the remaining body region outside the Artifact Zone. This zone quantifies changes outside the directly affected region. The clinical interpretation of anatomical preservation was reserved for external evaluation by readers.

The [Figure 4.5.2](#) illustrates the application of the two zones in representative slices of the dataset, covering different artifact patterns.



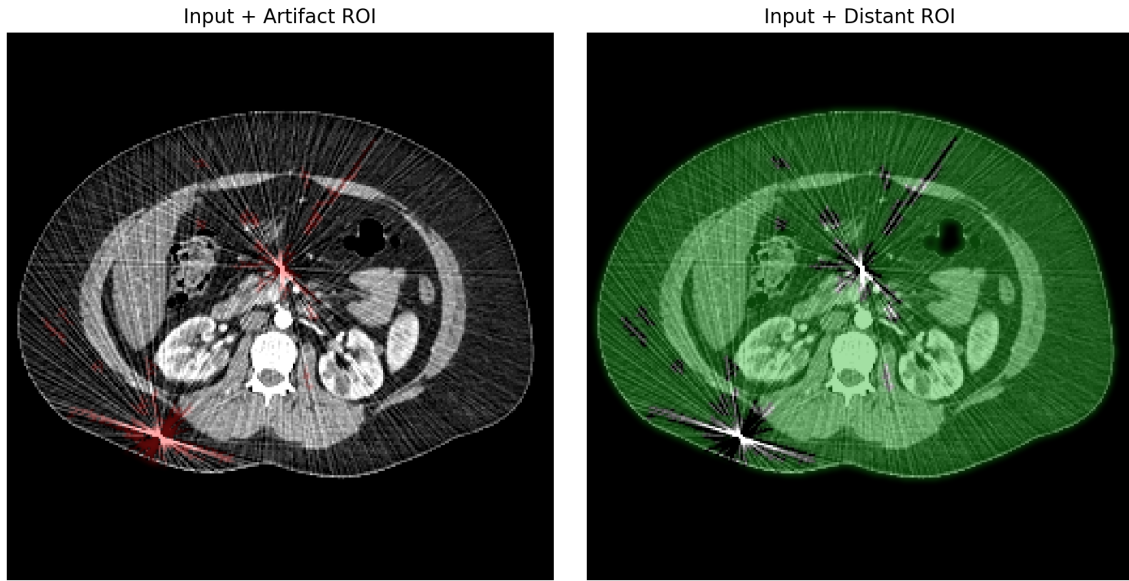


Fig.4.3 Artifact Zone ROI (left) and Clean Zone ROI (right) examples.

In every evaluation context, the metrics were expressed as variations relative to the performance of the corrupted input. Let Ω be the evaluation domain, which can correspond to the global region defined above (Ω_{global}), the Artifact Zone ($\mathcal{R}_{\text{art}}$), or the Distant Zone ($\mathcal{R}_{\text{dist}}$), and let $\hat{I}$ be the reconstruction produced by the model. The variations were defined by:

$$\Delta\text{PSNR}_{\Omega} = \text{PSNR}(\hat{I} \mid_{\Omega}, I_{\text{ref}} \mid_{\Omega}) - \text{PSNR}(I_{\text{in}} \mid_{\Omega}, I_{\text{ref}} \mid_{\Omega})$$

$$\Delta\text{SSIM}_{\Omega} = \text{SSIM}(\hat{I} \mid_{\Omega}, I_{\text{ref}} \mid_{\Omega}) - \text{SSIM}(I_{\text{in}} \mid_{\Omega}, I_{\text{ref}} \mid_{\Omega})$$

$$\Delta\text{MAE}_{\Omega} = \text{MAE}(I_{\text{in}} \mid_{\Omega}, I_{\text{ref}} \mid_{\Omega}) - \text{MAE}(\hat{I} \mid_{\Omega}, I_{\text{ref}} \mid_{\Omega}).$$

Thus, $\Delta\text{PSNR} > 0$ and $\Delta\text{SSIM} > 0$ indicate improvement over the corrupted input. Since lower MAE represents better fidelity, the order of subtraction was reversed for this metric: $\Delta\text{MAE} > 0$ indicates error reduction and, therefore, improvement; $\Delta\text{MAE} < 0$ indicates error increase. Regular SSIM was used, calculated with a local window of 11×11 pixels and cropping with context around the evaluated region.

Metric	Train	Global	Artifact Zone	Clean Zone
PSNR	✓	✓	✓	✓
SSIM	✓	✓	✓	✓
MAE	✓	✓	✓	✓

Table 4.8 Evaluation metrics by context.

4.5.3 Statistical Analysis

Every statistical comparison in this work used the patient, not the slice, as the unit of analysis. A patient with more slices should not count more than a patient with fewer slices, so before any comparison, the per-slice metrics were averaged within each patient, giving one paired value per patient.

To evaluate the improvement over the corrupted input, for each patient the output's value was compared with the corrupted input's value, for both **PSNR** and **SSIM**, so that a positive difference always represents improvement. For **MAE**, whose reduction represents better performance, the subtraction was reversed so that a positive difference also represents improvement. In every test, the question asked was whether there was typically no difference at all between the paired values. The direct comparison between the two architectures followed the same logic, subtracting the Single Model's value from the Dual Model's value for each patient, so that negative differences favor the Single Model.

The p -values presented in the comparison tables were adjusted using the Holm sequential procedure, which controls the probability of at least one false positive across a family of related tests without assuming that the tests are independent: the smallest raw p -values are penalized the most, while the relative ordering between tests is preserved. This correction was computed in Python with the `multipletests` function from the `statsmodels` library, using the 'holm' method.

In the reconstruction improvement table, the correction was applied jointly to the four corresponding tests for **PSNR** and **SSIM** for each of the two architectures. In the paired comparison Single Model vs Dual Model, the correction was applied jointly to the two tests, one for **PSNR** and another for **SSIM**.

The confidence interval of 95% reported for the mean differences was obtained by patient-level percentile *bootstrap*. For each metric, the vector of $n = 41$ paired differences per patient was resampled with replacement, maintaining n observations in each resample. The mean difference was then calculated in each of the 10,000 resamples. The lower and upper limits of the interval correspond, respectively, to the 2.5 and 97.5 percentiles of the empirical distribution of these means. This procedure does not assume normality of the differences and preserves the patient, rather than the slice, as the unit of inference. Seed 1337 was used to make the resampling deterministic.

4.6 External Evaluation

The external evaluation was realized in independent data provided by the [The Cancer Imaging Archive \(TCIA\)](#) [45], allowing to assess the impact of domain shift and the robustness of the model in scenarios not seen during training.

4.6.1 Design and Cases

The study consists in putting doctors to analyse the images produced by the model compared to the original images with artifacts. Every doctor looks at the same set of cases, one at a time, and for each one says which of the two images they prefer on a few simple questions. The evaluation was designed to be blind, with the doctors not knowing which image is the original and which is the model output. It was used a set of 15 samples from diverse [TCIA](#) datasets, with 2 additional hidden samples repeated for consistency check and 2 training samples excluded from the analysis.

The cases were chosen from a larger pool of 48 [TCIA](#) exams, mostly from the chest (31 cases), with a smaller number from the abdomen (6), the pelvis (1), and other or mixed regions (10). Each case corresponds to a single axial CT slice. The 17 cases used (15 main and 2 training) were drawn from five [TCIA](#) collections: TCGA-LUAD [46], LIDC-IDRI [47], VAREPOP-APOLLO [48], CPTAC-SAR [49], and CPTAC-PDA [50]. The 2 hidden repeats were shown again later in the same session, at least five cases after they first appeared, without telling the reader that it was a repeat.

In accordance with the data-usage policy of each collection, this work acknowledges: the results shown here are in whole or in part based upon data generated by the TCGA Research Network (<http://cancergenome.nih.gov/>), used in the TCGA-LUAD cases; the National Cancer Institute and the Foundation for the National Institutes of Health, for their critical role in the creation of the free publicly available LIDC/IDRI Database used in the LIDC-IDRI cases; the Veterans Health Administration's Research for Precision Oncology Program and the Applied Proteogenomics Organizational Learning and Outcomes (APOLLO) Research Network, whose data were used in the VAREPOP-APOLLO cases; and the National Cancer Institute Clinical Proteomic Tumor Analysis Consortium (CPTAC), whose data were used in the CPTAC-SAR and CPTAC-PDA cases.

A web application was developed to present the images in a randomized and counterbalanced manner, allowing the doctors to zoom jointly the pair for better visualization, and enabling

two anatomical windowing options (soft tissue and lung). Whether the original or the model output appeared on the left or the right was shuffled between doctors, but kept the same when a case was repeated, so that this choice could not by itself explain any preference.

The link to the application was sent to the physicians invited by the researcher, rather than through an open call, with a target of seven participants and at least five required for the results to be useful. To take part, a doctor only had to confirm that they use CT in their clinical practice at least once a month. No name, contact details, professional license number, or other proof of qualification was collected. The only information kept about each doctor is their country, medical specialty, how often they use CT, years of experience, previous experience with artifact reduction, and monitor type — none of which can identify a specific person.

4.6.2 Quantitative Characterization of Input–Output

As a complementary characterization of the alteration introduced by the model in external cases, SSIM, PSNR, and MAE were calculated directly between the input image and the respective output of the model selected for that. The metrics were evaluated only in the 15 main cases of the frozen external set. The two training cases were excluded. The calculation domain was the effective body mask, manually corrected during case preparation. For PSNR, the intensity window $[-1000, 400]$ HU with amplitude 1400 HU was used.

There is no clean reference for these real cases. Thus, higher SSIM and PSNR, as well as lower MAE, describe less global alteration between input and output, and not greater accuracy of artifact removal or improvement compared to a reference. The statistics were summarized per case, such as mean and standard deviation, and the 95% confidence intervals for the mean were obtained by percentile *bootstrap* with 10,000 resamplings and seed 1337. The interpretation of the clinical acceptability of these alterations remained reserved for blind evaluation by readers.

4.6.3 Presentation and Evaluation Instrument

For each case, the two images, the artifact input and the model output, are shown side by side at the same time, labelled only “Image A” and “Image B”, with nothing on screen indicating which is which. After looking at a case, the doctor answers four short questions, always with the same three possible answers: “Image A”, “Image B”, or “No difference”. In turn, these questions ask which image has less visible metal artifact, which image better shows the anatomy hidden behind the artifact (this is the main question of the study), which image has fewer artifacts or distortions in the rest of the picture, away from the artifact, and which image the doctor

would prefer to use for a clinical read. The third question exists specifically to catch a possible unwanted effect of the model: if doctors tend to prefer the artifact input in regions away from the implant, that would suggest the model is introducing changes where none were needed. At the very end of the session, each doctor answers one extra question, on a five-point scale from “Definitely not” to “Definitely yes”, about whether a processed version like this could be a useful addition to a normal CT exam, and may optionally leave a short written comment.

4.6.4 Outcomes and Analysis

The main question of the study, the primary outcome, is which image better shows the anatomy under the metal artifact, taken from the corresponding question described above. Since each doctor answers this question for every case, and every case is judged by every doctor, a statistical model was used that takes both of these repetitions into account at the same time, instead of simply counting how many times each answer was given. The other three per-case questions, together with the final usefulness question, are treated as secondary outcomes and are summarized more simply, as the frequency of each answer with a 95% confidence interval.

Because two cases are shown a second time to each doctor later in the same session, without saying so, it is possible to check how consistent each doctor is with themselves. For this consistency check, the intraclass correlation coefficient ICC(2,1) was used for answers on a scale, and Fleiss’s κ was used for the simple output-versus-input preference. Both are standard ways of answering the question “if the same case is shown twice to the same doctor, how often do they answer the same way”, adapted here to work with only two repeats per doctor. Given the small number of doctors expected to take part, all of these analyses were planned to be simple and robust to a small sample, favoring frequencies and confidence intervals over more elaborate models wherever possible.

4.6.5 Operational Status

Before any real doctor took part, the application was first tested with a technical pilot using placeholder data, to check that the questions, the images, and the randomization all worked as intended. Moving from this rehearsal stage to inviting real doctors was only allowed once a documented checklist was fully satisfied, covering an approved and tested hosting setup, a defined data retention and deletion plan, an approved participant information text, and a working, secure deployment of the application.

Chapter 5

Results

5.1 Final Group and Experimental Integrity

5.1.1 Automatic Approval and Manual Review Outcomes

The final curation outcome of every pair in the [CT-MAR](#) dataset was recovered from the canonical manifest. Retained pairs were classified as manually reviewed when either source-level decision was `manual_approved`; all remaining retained pairs were classified as automatically retained. Pairs not included in the final group were classified as excluded after verification. This final category comprises 435 pairs without a secure correspondence and three pairs rejected after the complementary [MS-SSIM](#) review.

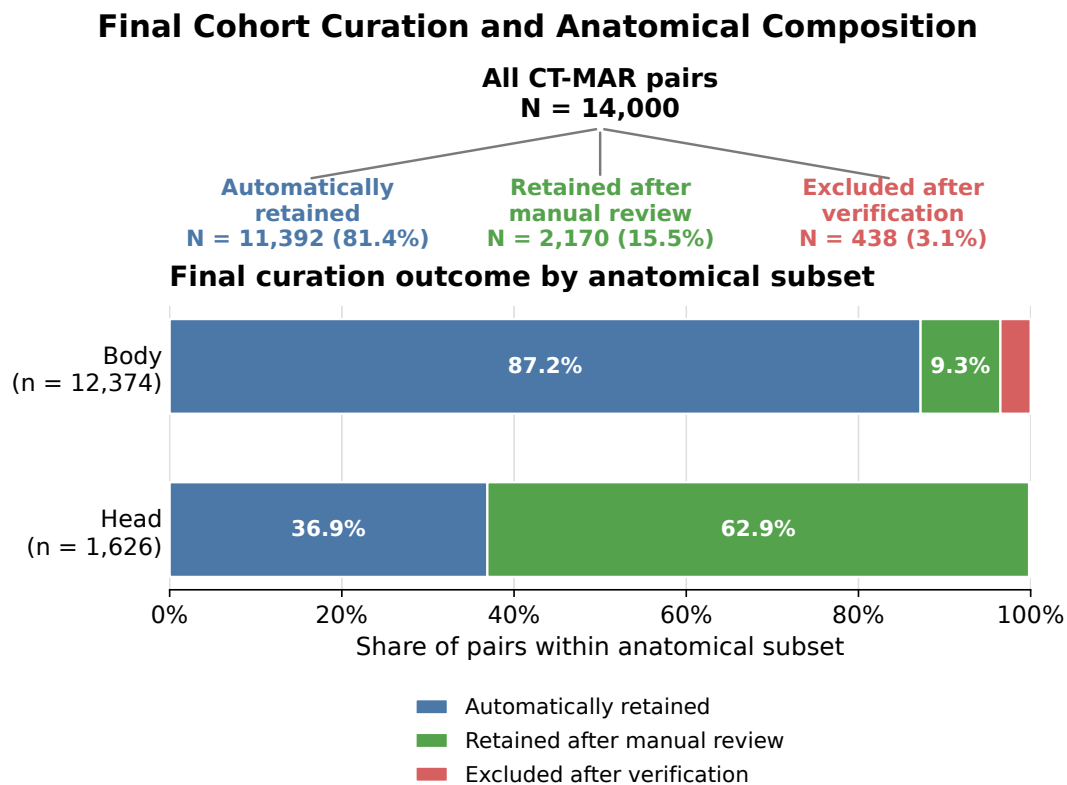


Fig.5.1 Final curation outcomes for all 14,000 CT-MAR pairs. The upper panel reports the global distribution of outcomes; the lower panel reports the percentage within each anatomical subset. The manifest labels distinguish *Body* and *Head* subsets, rather than the precise anatomical location of a metallic implant.

5.1.2 Complementary Analysis with MS-SSIM

Before advancing to the analysis of the results of the artifact reduction models, another evaluation of the CT-MAR x DeepLesion/Stroke EIT results was conducted to verify the quality of the automatic matches. This step is crucial to ensure an extra layer of confidence in the results obtained by the structural similarity comparison algorithm. Every pair that had its match automatically approved due to its score being above 0.994 was reviewed by a new algorithm that calculates the MS-SSIM between the corresponding images. A histogram of the MS-SSIM distribution was generated, and pairs with scores below 0.9 were manually reviewed and either approved or rejected based on human evaluation.

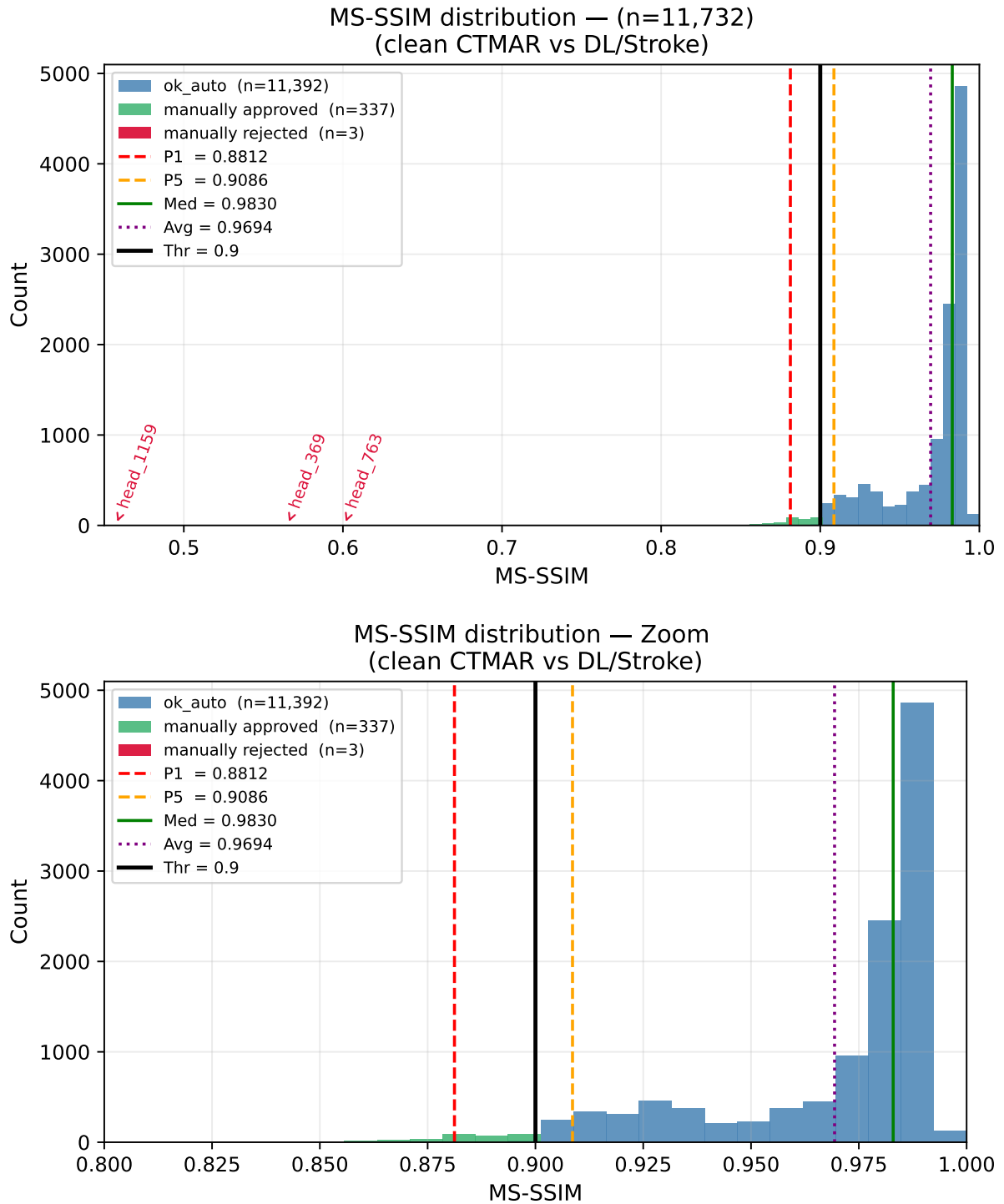


Fig.5.2 MS-SSIM distribution between corresponding pairs (CT-MAR vs DL/Stroke, automatically approved). Above: histogram with percentiles P1 and P5, median, mean, and threshold of 0.9. Below: zoom of the histogram for the region of greatest interest, highlighting pairs with MS-SSIM below 0.9, which were subject to manual review.

The distribution analysis between the corresponding pairs reveals that most pairs exhibit high structural similarity, with a mean value of 0.9694 and a median of 0.9830, indicating that the automatic matching process is generally reliable and effective in identifying corresponding pairs between the CT-MAR and DeepLesion/Stroke EIT datasets.

The pairs with **MS-SSIM** scores below 0.9 totaled 340, of which only 3 were rejected after human visual inspection 5.3. In this inspection, it was observed that even cases with quite low **MS-SSIM** scores were not false positives but rather due to differences in post-processing or general image treatment that generate local alterations to which **MS-SSIM** is more sensitive. This can be visualized in figure 5.4. This fact highlights the effectiveness of the global structural similarity matching strategy adopted, which managed to identify pairs accurately, with only 3 false positives identified even after rigorous analysis 5.5. This shows that the methodology is robust and performed better in detecting slices in this work than using **MS-SSIM** itself. While local differences affected **MS-SSIM** more significantly, the cosine similarity-based methodology was able to identify correct matches with high global scores, even in cases of pronounced local differences.

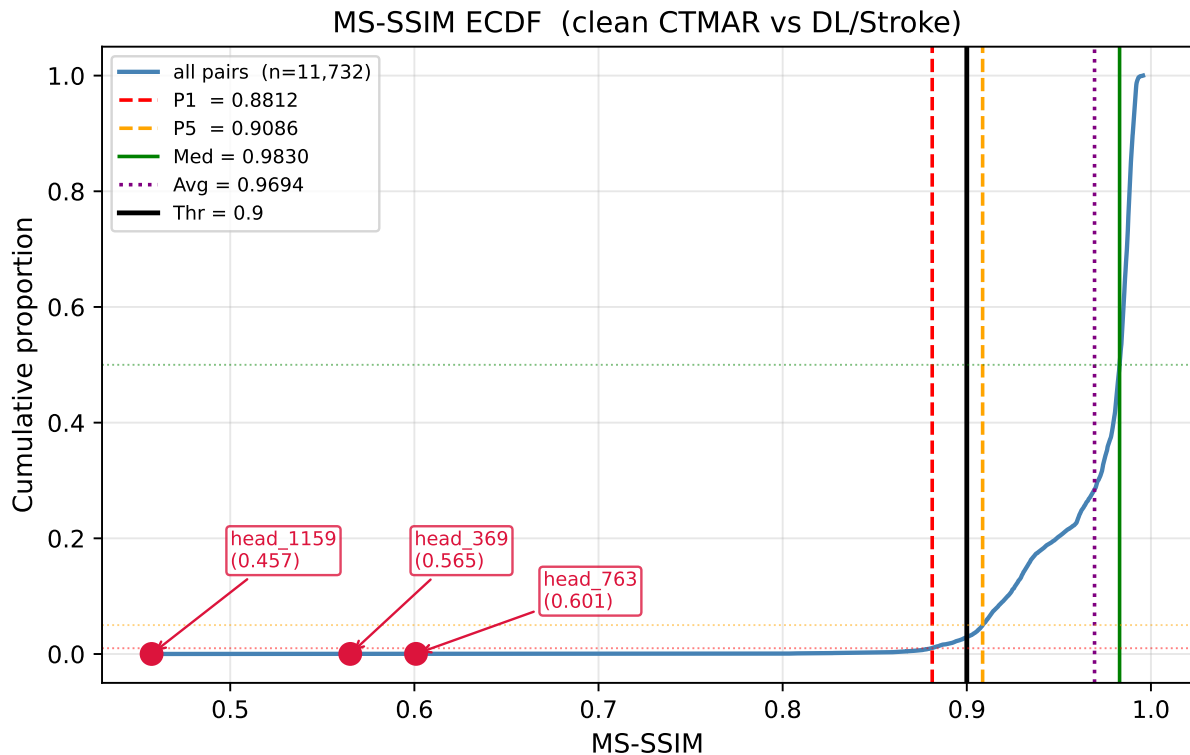


Fig.5.3 **MS-SSIM** distribution between corresponding pairs (**CT-MAR** vs DL/Stroke, automatically approved). Left: histogram with percentiles P1 and P5, median, mean, and threshold of 0.9. Right: corresponding ECDF. Pairs with **MS-SSIM** below 0.9 were subject to manual review.



Fig.5.4 Examples of corresponding pairs with low **MS-SSIM** values (0.7597, 0.7694, and 0.7717, respectively) approved after human visual inspection despite low scores.

Clearly, the reprovved pairs presented **MS-SSIM** values significantly below the standard (< 0.61), with extremely low scores, as shown in figure 5.5. Additionally, they represented pairs with scores closer to the automatic acceptance threshold of 0.994 5.6. This demonstrates that the cases where the method fails are clear outliers in **MS-SSIM** and remain precisely in the gray area prone to errors in the empirical determination of the automatic acceptance threshold. Furthermore, the reprovved pairs have very particular visual characteristics, with

backgrounds dominating the image and few visible anatomical structures, making structural similarity assessment more challenging due to the high homogeneity of visual features.

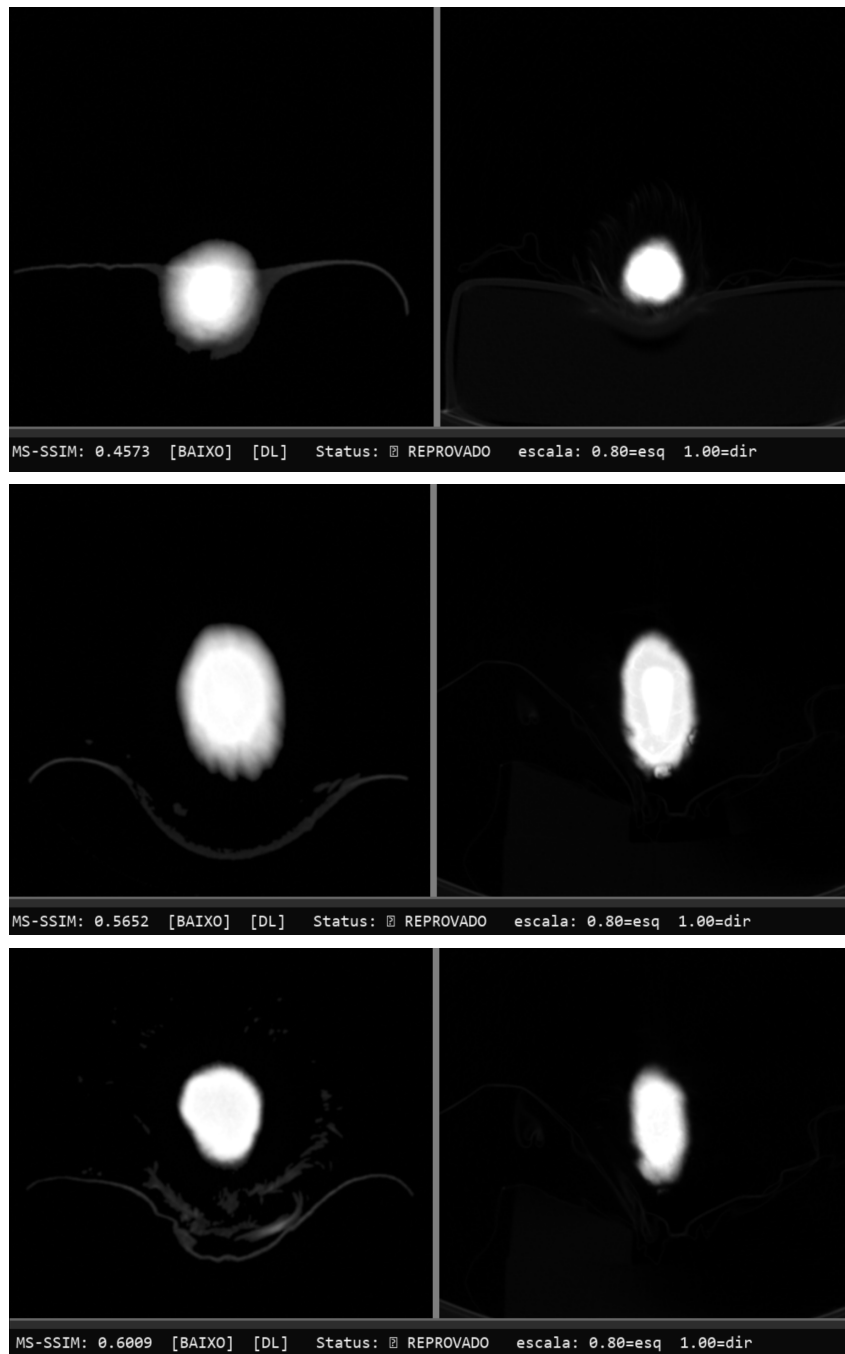


Fig.5.5 Three reprovod pairs after human visual inspection with [MS-SSIM](#) values (head_1159: 0.4573, head_369: 0.5652, and head_763: 0.6009, respectively). The cosine similarity scores for these pairs were head_1159: 0.9960, head_369: 0.9952, and head_763: 0.9951.

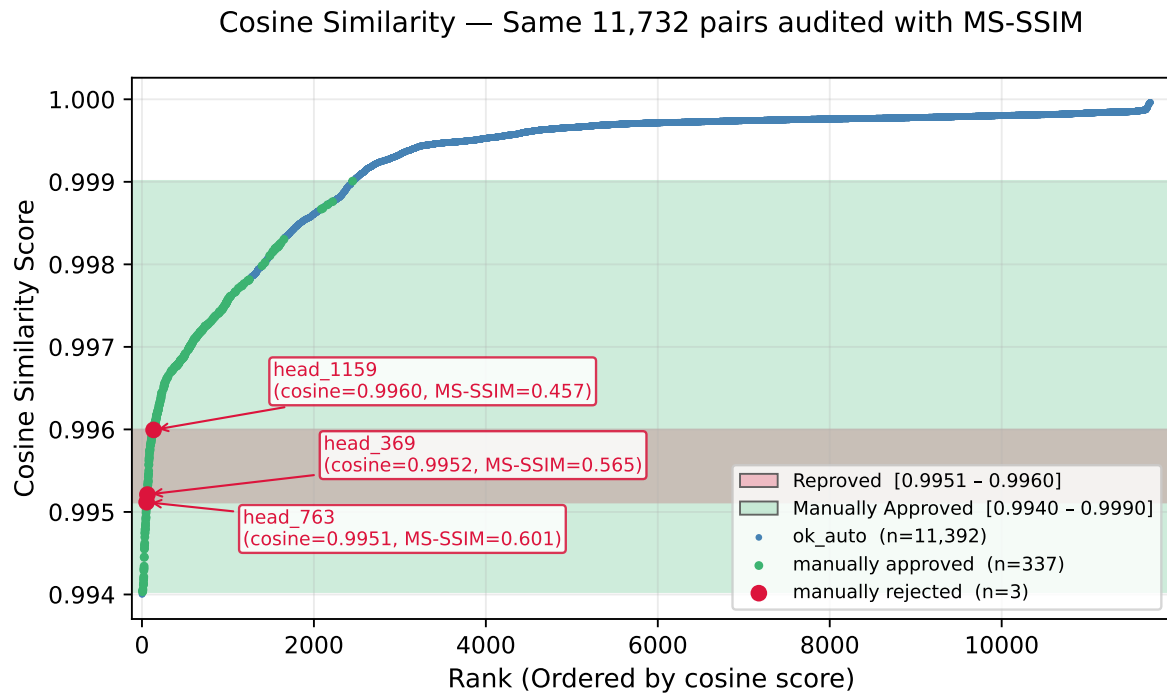


Fig.5.6 Cosine similarity distribution of visually inspected pairs.

The pairs with the most critical **MS-SSIM** scores just above the reproved ones had values greater than 0.75 [5.4](#), demonstrating a clear distance between false positives and visually approved pairs. All inspected pairs with scores above this **MS-SSIM** value passed the visual inspection test [5.7](#). Since the pairs that did not undergo this scrutiny had scores above 0.90 in **MS-SSIM**, it is safe to assume that automatically approved pairs with scores above this threshold are reliable.

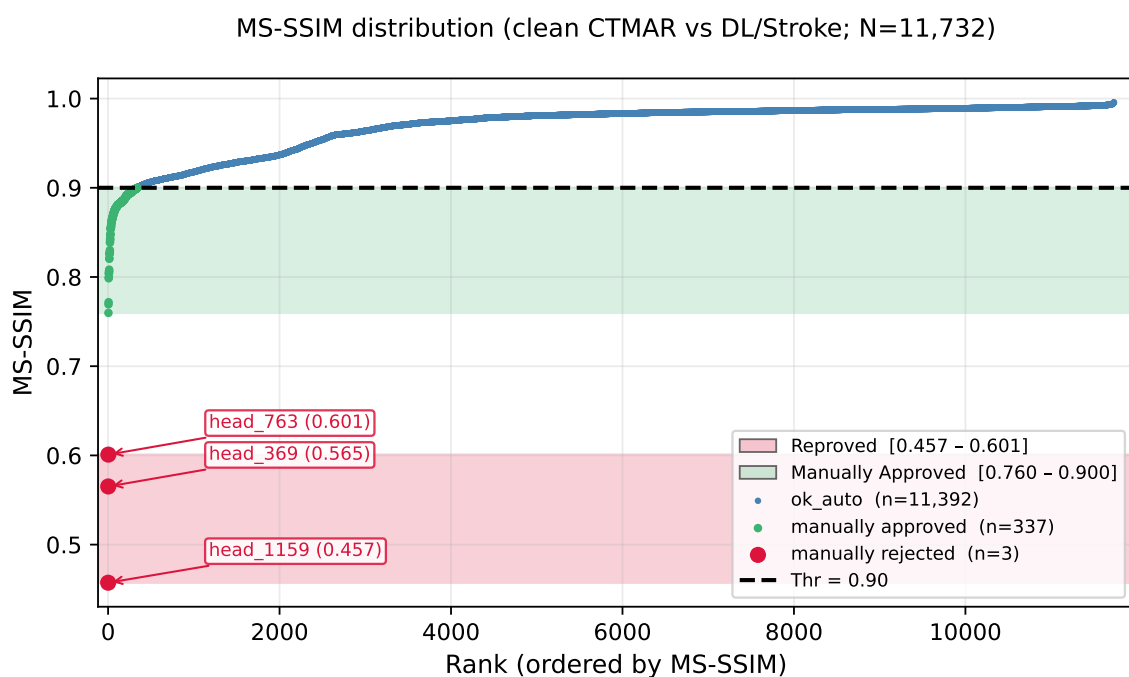


Fig.5.7 **MS-SSIM** values distribution for visually inspected pairs.

In the direct search, 530 pairs did not obtain approved matches in either of the two source datasets. The complementary indirect matching step recovered 95 of these cases after visual inspection, leaving 435 pairs without secure matches.

With relation to possible false negatives that could have resulted in unfair exclusion of pairs in this study, the adopted methodology was even more robust. Since there were few pairs that did not receive approval in either of the two source datasets, it was easy to visually verify each of these cases to confirm whether there was really no match or if the method failed to identify it. Of the 435 excluded pairs, 245 were excluded due to the automated criterion based on the threshold of 0.98. None of these cases were really false negatives, confirming once again the effectiveness of the adopted automatic matching method. The others 190 pairs fell into the low-reliability score range (0.98 – 0.994) and were marked as false matches at this stage, highlighting the importance of having determined a gray area of scores for which the automatic matching method is not efficient, and where human visual review becomes indispensable [5.8](#).

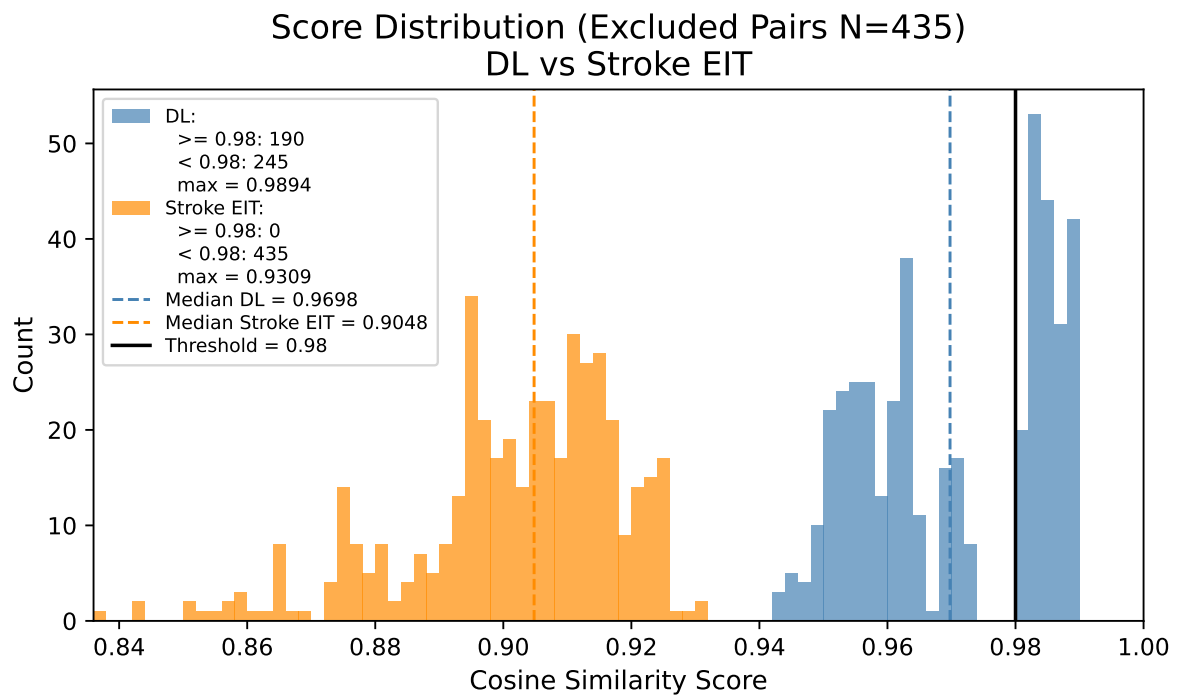


Fig.5.8 Score distribution of excluded pairs (DL x Stroke EIT) for pairs without approved matches in either of the two source datasets (N=435). The orange histogram shows the distribution of scores for the 435 pairs rejected in the Stroke EIT Dataset. The blue histogram shows the distribution of the same 435 pairs rejected in the Deep Lesion Dataset.

5.2 Model Training

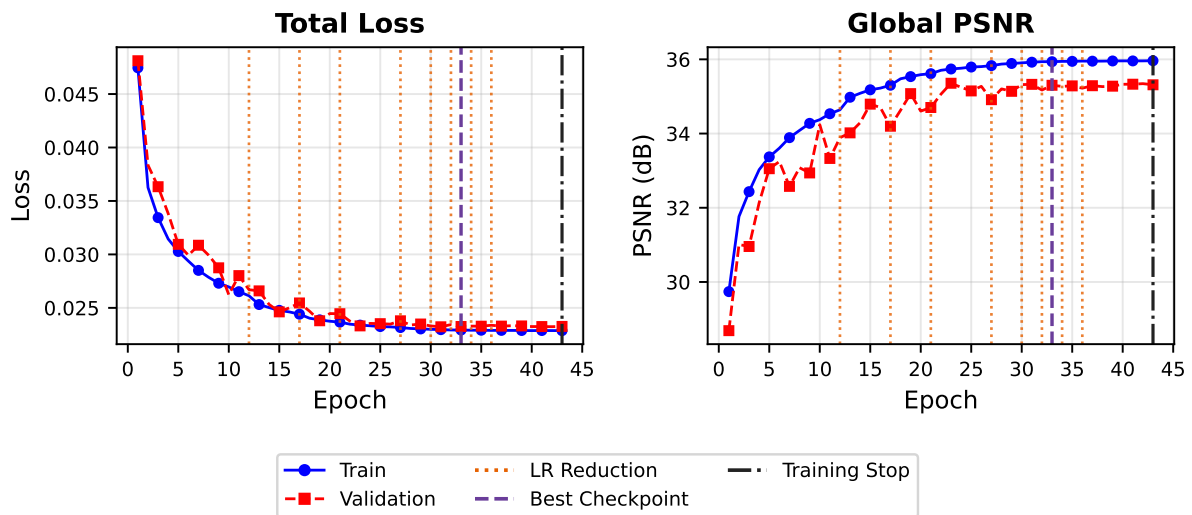
5.2.1 Training Convergence and Per-Run Checkpoint Selection

All six deterministic runs converged and terminated between epochs 38 and 47, after the learning rate reached its lower bound. The best validation SSIM values occurred between epochs 23 and 44, before training termination, supporting restoration of the best checkpoint weights rather than use of the final epoch.

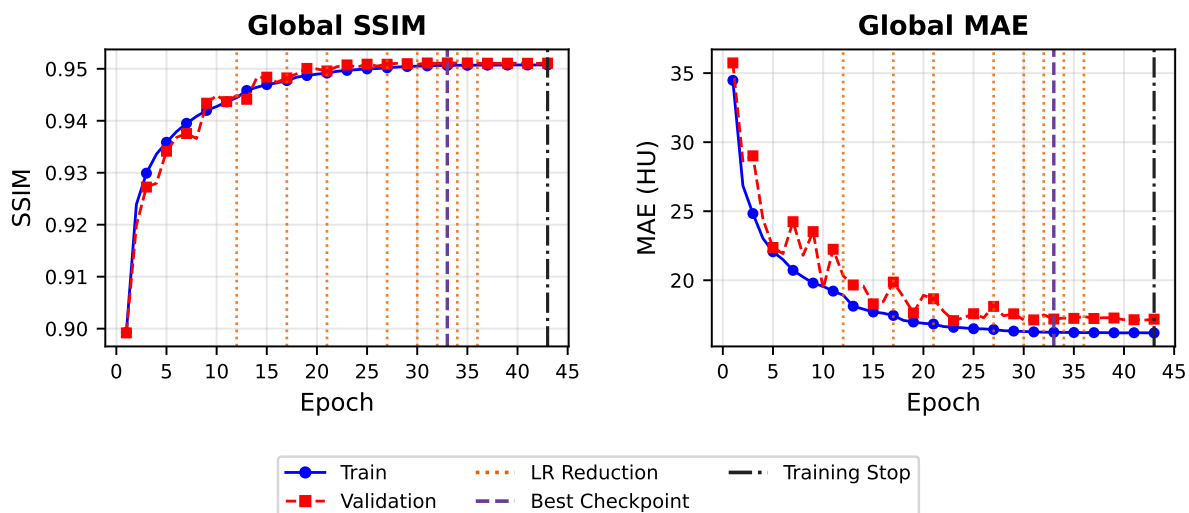
The highest validation reconstruction SSIM was 0.95113, obtained by the Single Model with seed 2024 at epoch 33. At that same checkpoint, validation PSNR was 35.30099 dB and validation MAE was 17.19180 HU. These values show which checkpoint was chosen for this individual run. On their own, they did not decide which run would go forward to external evaluation: that choice was made afterwards, by comparing all six checkpoints on the internal test partition using the Artifact-ROI rule agreed on in advance.

The convergence curves for the two seed-2024 architectures are shown below as page-breakable chart pairs. The complete set of six runs and the validation tables for all model–seed combinations are provided in [section A.1](#).

Single Model — Training Convergence



Single Model — Training Convergence



Single Model — Training Convergence

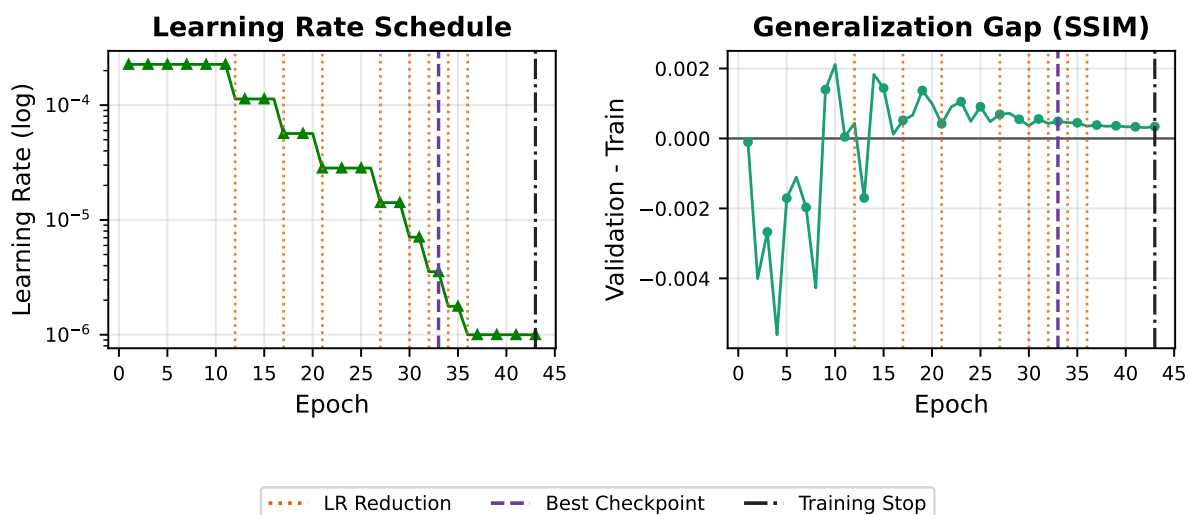
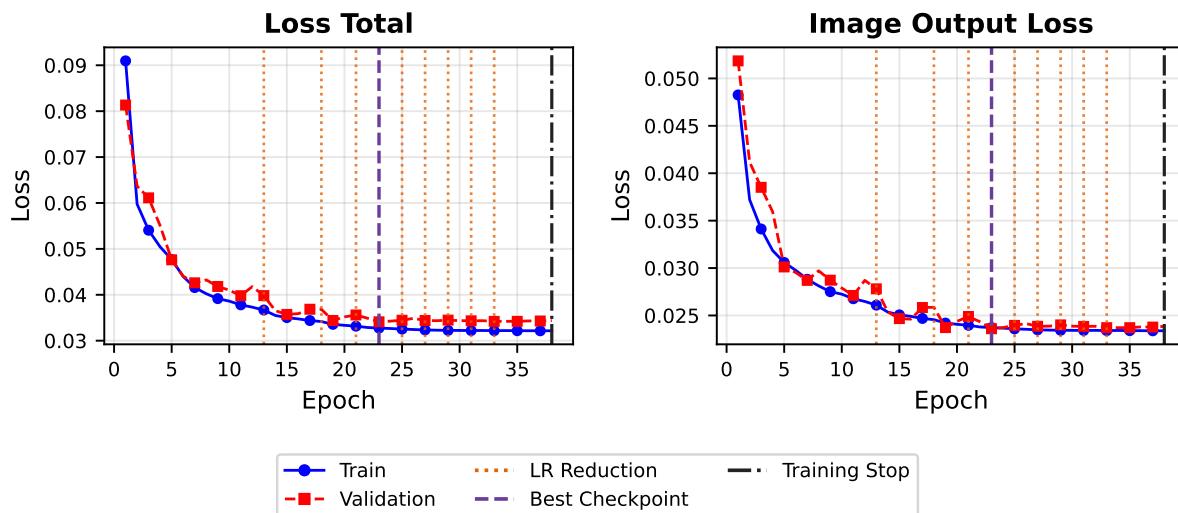
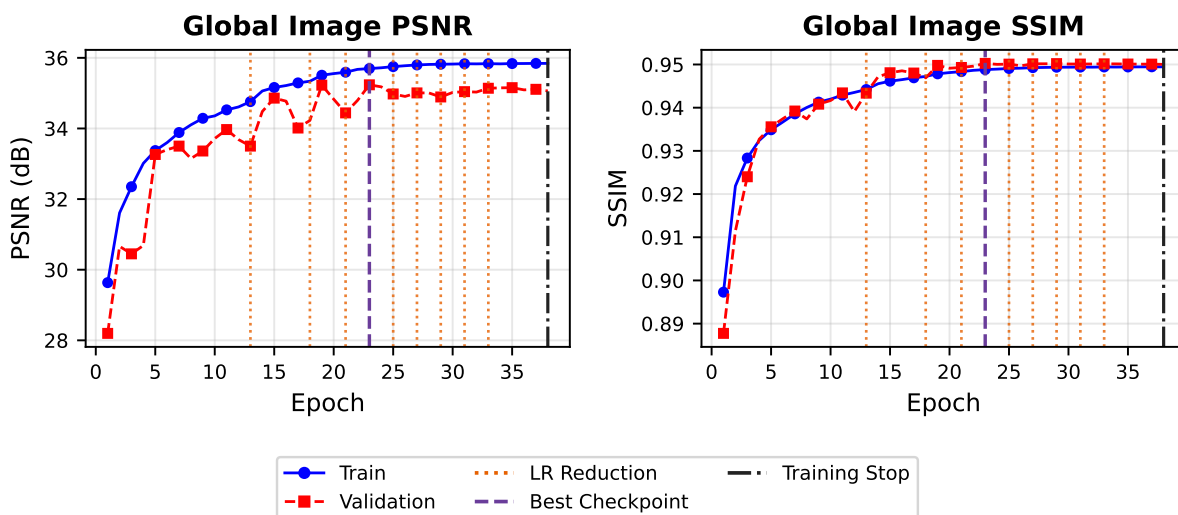


Fig.5.9 Training convergence for the Single Model, seed 2024. Each chart pair repeats the shared legend so it remains interpretable if the group spans multiple pages. Orange dotted vertical lines indicate learning-rate reductions; the purple dashed line marks the epoch of the best validation checkpoint; and the black dash-dotted line marks training termination. The *generalization-gap* panel reports validation minus training SSIM and supports inspection for persistent divergence compatible with overfitting.

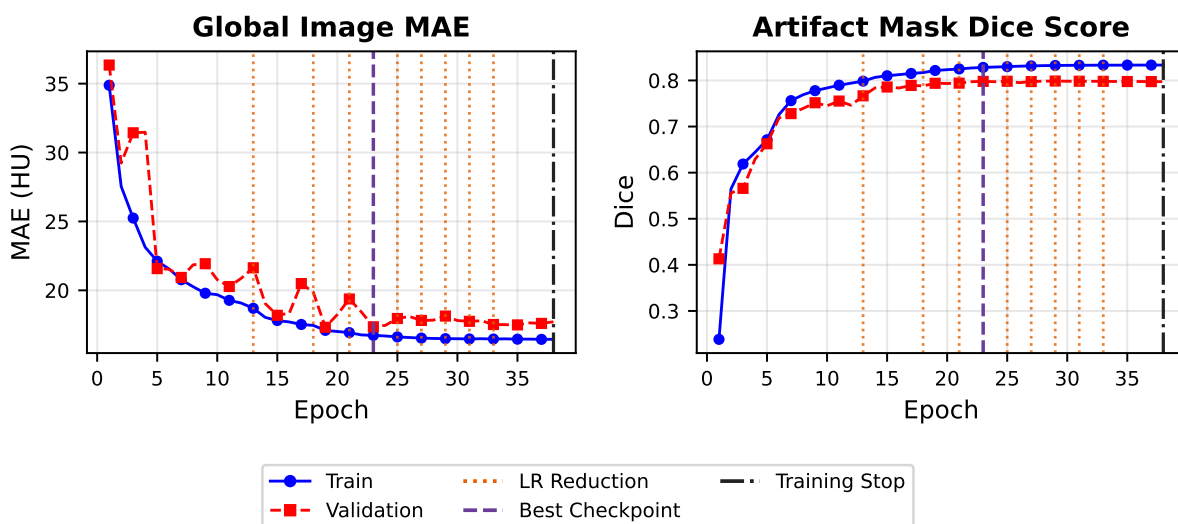
Dual Model — Training Convergence



Dual Model — Training Convergence



Dual Model — Training Convergence



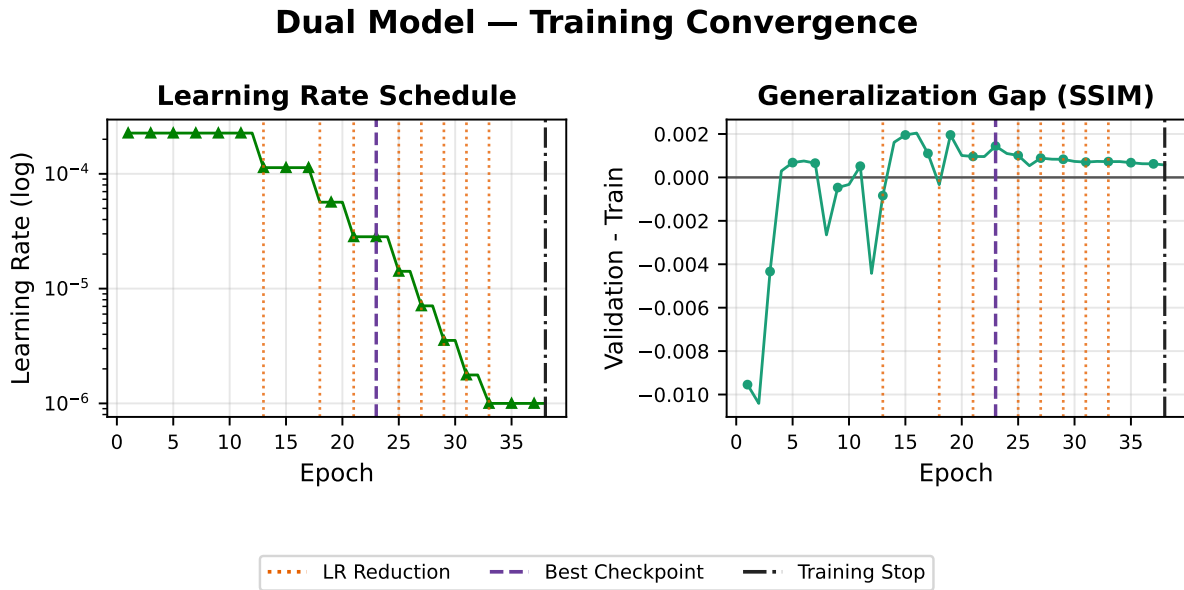


Fig.5.10 Training convergence for the Dual Model, seed 2024. Each chart pair repeats the shared legend so it remains interpretable if the group spans multiple pages. Orange dotted vertical lines indicate learning-rate reductions; the purple dashed line marks the epoch of the best validation checkpoint; and the black dash-dotted line marks training termination. The *generalization-gap* panel reports validation minus training SSIM and supports inspection for persistent divergence compatible with overfitting.

5.3 Internal Evaluation in Test Subset

5.3.1 Global Outcomes

The internal evaluation was conducted on the fixed test partition, composed of 1478 slices from 41 independent patients. For each checkpoint previously selected in validation, PSNR, SSIM, and MAE were calculated between the reconstruction and the clean reference. All statistical comparisons used the patient, not the slice, as the unit of analysis: metrics were first averaged within each patient, and 95% confidence intervals were obtained by resampling patients 10,000 times (a bootstrap procedure). The test set was not used to change any run's checkpoint; it was used once to compare the six validation-frozen candidates for the external-evaluation decision described in subsection 5.3.3.

The Table 5.2, Table 5.1, and Table 5.3 present the slice-wise means. Since all runs were evaluated on the same test partition, the input value is common to both models for each seed. In all runs, reconstruction increased the global PSNR from 27.39 dB to 34.95–35.20 dB, increased SSIM from 0.8361 to 0.9421–0.9442, and reduced MAE from 36.6 HU to 17.3–17.8 HU.

The seed-wise global tables show that the Single Model, seed 2024, was not an isolated run: the three Single Model runs had very similar global performance. Seed 2024 was at the

SSIM global

Seed	Input	Single Model	Dual Model
42	0.8361 $\pm$ 0.0880	0.9439 $\pm$ 0.0289	0.9424 $\pm$ 0.0292
1337	0.8361 $\pm$ 0.0880	0.9441 $\pm$ 0.0286	0.9421 $\pm$ 0.0289
2024	0.8361 $\pm$ 0.0880	0.9442 $\pm$ 0.0288	0.9432 $\pm$ 0.0292

All input and model values are computed against the clean reference (*ground truth*).

Table 5.1 Global **SSIM** per slice in test subset ($n = 1478$), expressed as mean $\pm$ standard deviation.

PSNR global (dB)

Seed	Input	Single Model	Dual Model
42	27.39 $\pm$ 4.14	35.09 $\pm$ 2.82	35.06 $\pm$ 2.84
1337	27.39 $\pm$ 4.14	35.13 $\pm$ 2.81	34.95 $\pm$ 2.82
2024	27.39 $\pm$ 4.14	35.20 $\pm$ 2.84	35.16 $\pm$ 2.82

All input and model values are computed against the clean reference (*ground truth*).

Table 5.2 Global **PSNR** per slice in test subset ($n = 1478$), expressed as mean $\pm$ standard deviation.

MAE global (HU)

Seed	Input	Single Model	Dual Model
42	36.569 $\pm$ 21.906	17.649 $\pm$ 5.689	17.654 $\pm$ 5.905
1337	36.569 $\pm$ 21.906	17.485 $\pm$ 5.624	17.842 $\pm$ 5.789
2024	36.569 $\pm$ 21.906	17.341 $\pm$ 5.573	17.475 $\pm$ 5.563

All input and model values are computed against the clean reference (*ground truth*).

Table 5.3 Global **MAE** per slice in test subset ($n = 1478$), expressed as mean $\pm$ standard deviation.

favorable extreme for global **SSIM**, **PSNR**, and **MAE**. This comparison between seeds is only descriptive and did not decide which run went forward for external evaluation; that decision was based on the Artifact-ROI Δ **SSIM** rule agreed on in advance, reported below.

The patient-level deltas and their confidence intervals are shown in Table 5.5, Table 5.4, and Table 5.6. All six runs showed positive mean improvement for the three global metrics, and the corresponding confidence intervals excluded the null value. These global results provide complementary evidence of improvement over the corrupted input; they were not used to rank the candidates for external evaluation.

 Δ SSIM Global (IC 95%)

Seed	Single Model	Dual Model
42	0.0887 [0.0728; 0.1063]	0.0870 [0.0713; 0.1043]
1337	0.0891 [0.0734; 0.1065]	0.0867 [0.0710; 0.1041]
2024	0.0891 [0.0732; 0.1066]	0.0879 [0.0719; 0.1053]

Note: Δ SSIM = SSIM(Model, GT) – SSIM(Input, GT).

Table 5.4 Δ **SSIM** global aggregated per patient in the test subset ($n = 41$). IC: 95% confidence interval.

For the selected run, Single Model with seed 2024, the histograms in Figure 5.11 show the per-slice distribution of the global deltas. Although these plots are descriptive and slices from the same patient are not independent, they make the few unfavorable cases visible: 12 of 1478

Δ PSNR Global (dB, IC 95%)

Seed	Single Model	Dual Model
42	6.96 [6.22; 7.68]	6.96 [6.24; 7.65]
1337	7.05 [6.33; 7.75]	6.85 [6.12; 7.55]
2024	7.11 [6.36; 7.83]	7.06 [6.31; 7.78]

Note: Δ PSNR = PSNR(Model, GT) – PSNR(Input, GT).

Table 5.5 Δ PSNR global aggregated per patient in the test subset ($n = 41$). IC: 95% confidence interval.

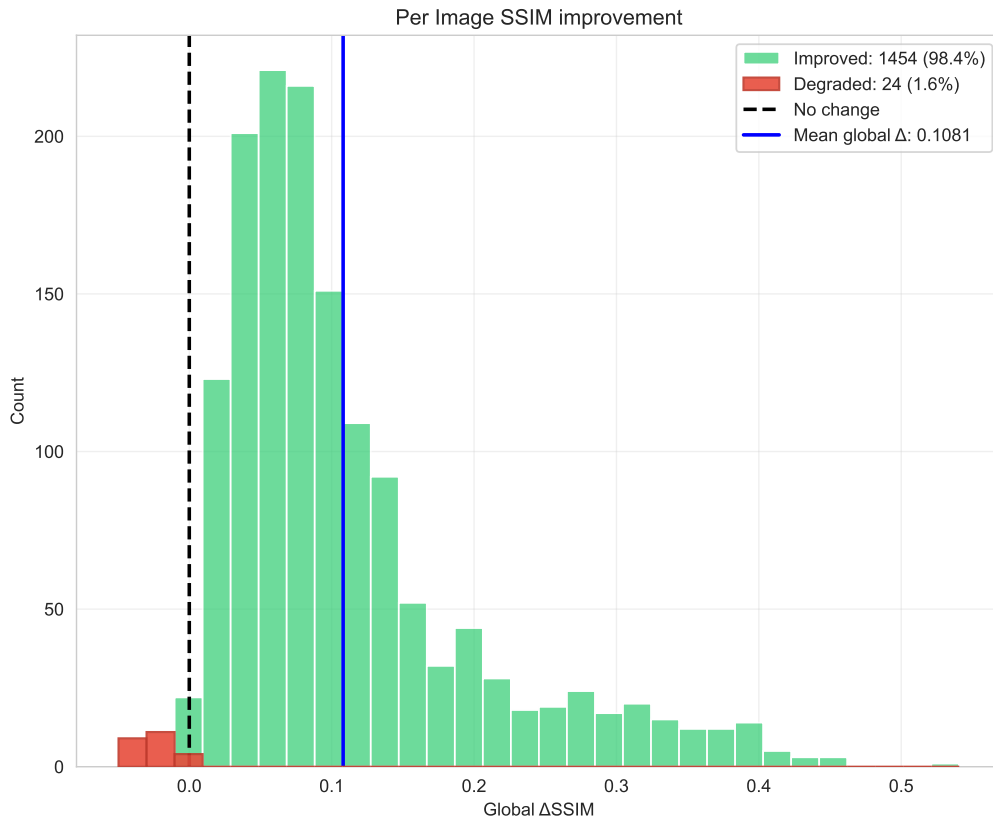
 Δ MAE Global (HU, IC 95%)

Seed	Single Model	Dual Model
42	15.3 [12.2; 19.0]	15.3 [12.3; 18.9]
1337	15.6 [12.4; 19.3]	15.2 [12.1; 18.8]
2024	15.7 [12.5; 19.5]	15.5 [12.4; 19.3]

Note: Δ MAE = MAE(Input, GT) – MAE(Model, GT); positive values indicate error reduction.

Table 5.6 Δ MAE global aggregated per patient in the test subset ($n = 41$). Values > 0 indicate improvement; IC: 95% confidence interval.

slices (0.8%) showed a negative Δ PSNR and 24 (1.6%) showed a negative Δ SSIM. Statistical inference remains based on the patient-level means presented in the preceding tables.

Single Model — Global per Image Improvement

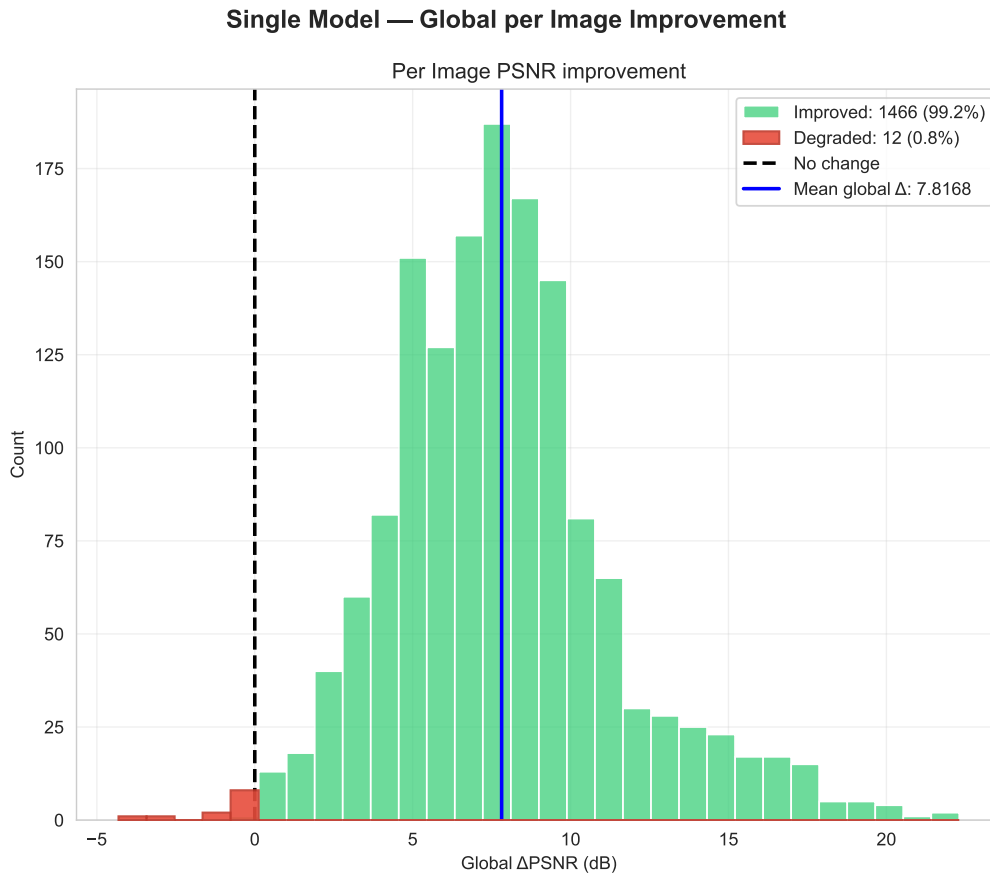


Fig.5.11 Per-image global improvement over the corrupted input for the Single Model, seed 2024: Δ SSIM (top) and Δ PSNR (bottom). Green bars indicate improvement and red bars indicate degradation. These plots are only descriptive; the formal statistical comparisons instead use per-patient averages.

The Single Model runs with seeds 1337 and 2024 had virtually identical global [SSIM](#). The seed-2024 run was slightly more favorable for global [PSNR](#) and [MAE](#); however, these global metrics were complementary and did not decide which run went forward for external evaluation. That decision was instead based on the region-of-interest analysis, which directly measures where the structural correction happens.

As an additional, purely descriptive check of the overall HU calibration, [Figure 5.12](#) presents the agreement between the mean HU of the reconstruction and that of the clean reference, aggregated per patient. Each point corresponds to one of the 41 patients; each patient's mean was calculated over all valid body-mask pixels, weighted by the number of pixels in each slice. The mean bias was +4.19 HU, with 95% limits of agreement between +1.45 and +6.93 HU. This analysis describes a small positive shift in global mean intensity, but does not quantify local artifact removal, nor does it replace the global and [ROI](#) metrics.

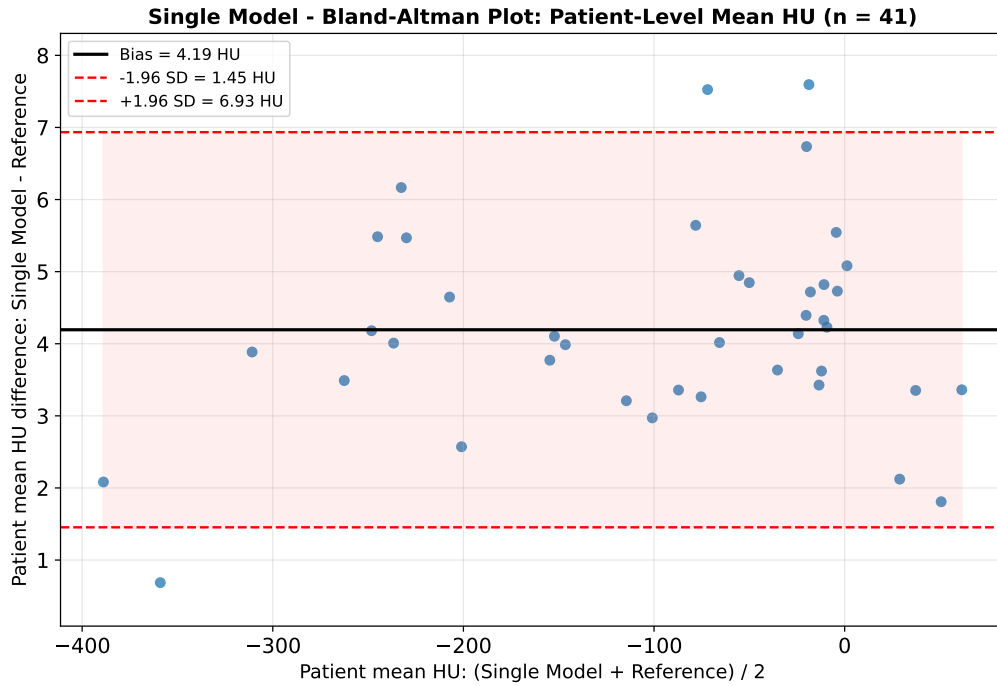


Fig.5.12 Bland-Altman agreement between the Single Model reconstruction and the clean reference for patient-level mean HU ($n = 41$). Each patient mean was calculated over all valid body-mask pixels; positive differences indicate higher mean HU in the reconstruction. The limits of agreement are descriptive.

5.3.2 Region of Interest Evaluation

The regions of interest and the per-zone metric calculation followed the procedure described in subsection 4.5.2. This stratification makes it possible to separately quantify the correction in the region directly affected by artifacts and the changes introduced in body regions distant from the implant.

The final results, first aggregated per patient and then summarized over 41 independent patients, are presented in Table 5.8, Table 5.7, and Table 5.9. All six runs showed substantial improvement in the Artifact Zone: Δ PSNR between 11.47 and 11.86 dB, Δ SSIM between 0.322 and 0.328, and MAE reduction between 96.9 and 98.5 HU. In the Distant Zone, the mean changes were smaller (Δ PSNR between 1.36 and 1.61 dB; Δ SSIM between 0.057 and 0.058; MAE reduction between 3.2 and 3.6 HU), and should be interpreted as a quantitative description rather than as clinical validation of anatomical preservation.

5.3.3 Selection of the External-Evaluation Candidate

Among the six checkpoints previously frozen by validation SSIM, the Single Model with seed 2024 had the highest mean patient-level Artifact-ROI Δ SSIM (0.3279 ± 0.0955). Its mean Distant-ROI Δ SSIM was 0.0581 ± 0.0422 , substantially smaller than the improvement observed in the Artifact Zone. This meets the preservation check that was planned in advance; but since

Δ SSIM

Seed	Model	Artifact ROI	Clean ROI
42	Single Model	0.3276 ± 0.0954	0.0579 ± 0.0426
	Dual Model	0.3233 ± 0.0937	0.0568 ± 0.0421
1337	Single Model	0.3278 ± 0.0933	0.0582 ± 0.0421
	Dual Model	0.3224 ± 0.0953	0.0566 ± 0.0422
2024	Single Model	0.3279 ± 0.0955	0.0581 ± 0.0422
	Dual Model	0.3266 ± 0.0954	0.0576 ± 0.0422

Note: Δ SSIM = SSIM(Model, GT) – SSIM(Input, GT) within the indicated ROI.

Table 5.7 Δ SSIM by ROI on the test set. Values are mean $\pm$ standard deviation across patients ($n = 41$).

 Δ PSNR (dB)

Seed	Model	Artifact ROI	Clean ROI
42	Single Model	11.75 ± 2.59	1.46 ± 1.43
	Dual Model	11.65 ± 2.53	1.48 ± 1.43
1337	Single Model	11.84 ± 2.54	1.53 ± 1.35
	Dual Model	11.47 ± 2.57	1.36 ± 1.52
2024	Single Model	11.86 ± 2.69	1.61 ± 1.41
	Dual Model	11.84 ± 2.62	1.53 ± 1.48

Note: Δ PSNR = PSNR(Model, GT) – PSNR(Input, GT) within the indicated ROI.

Table 5.8 Δ PSNR by ROI on the test set. Values are mean $\pm$ standard deviation across patients ($n = 41$).

 Δ MAE (HU)

Seed	Model	Artifact ROI	Clean ROI
42	Single Model	97.8 ± 19.6	3.3 ± 3.2
	Dual Model	97.5 ± 19.1	3.4 ± 3.1
1337	Single Model	98.5 ± 19.6	3.5 ± 3.1
	Dual Model	96.9 ± 19.9	3.2 ± 3.3
2024	Single Model	98.4 ± 20.2	3.6 ± 3.1
	Dual Model	98.3 ± 19.8	3.5 ± 3.3

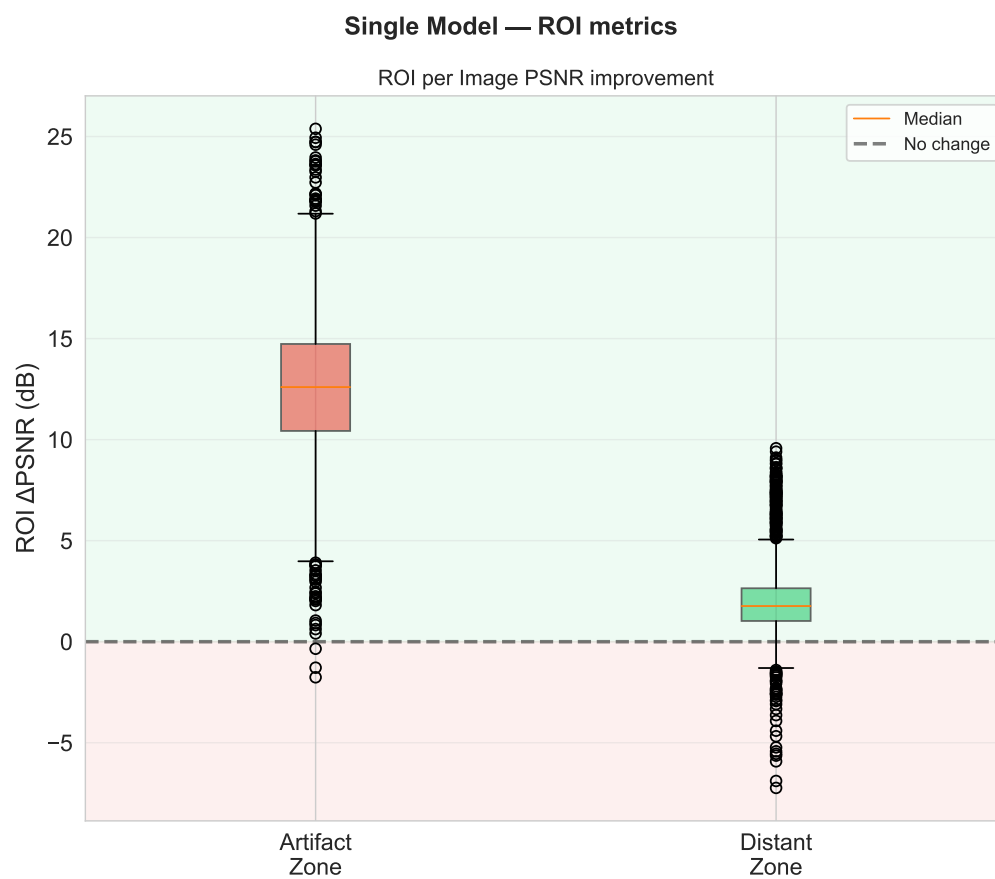
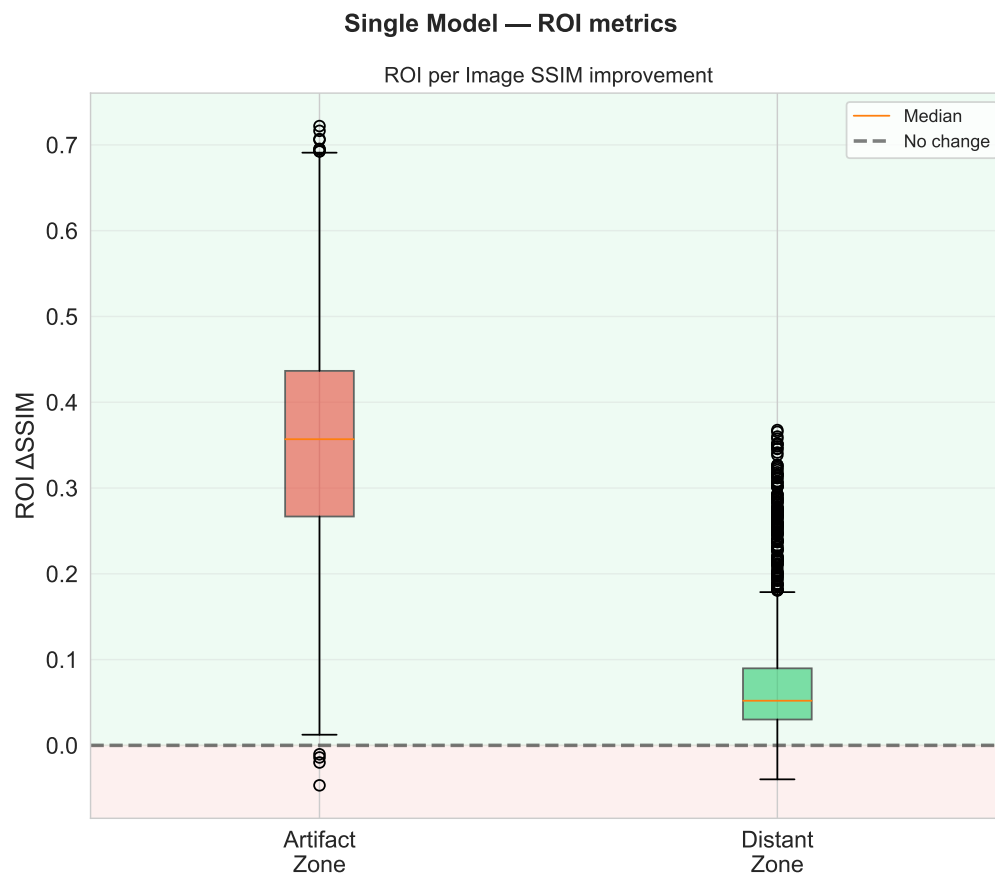
Note: Δ MAE = MAE(Input, GT) – MAE(Model, GT) within the indicated ROI; positive values indicate error reduction.

Table 5.9 MAE reduction by ROI on the test set. Values are mean $\pm$ standard deviation across patients ($n = 41$); positive values correspond to error reduction.

no specific numeric limit was set for how small the Distant-Zone change should be, it should not be read as formal proof of anatomical safety. The global results and the other ROI metrics reported above were complementary and did not change this decision. The Single Model, seed 2024, was therefore selected for external evaluation, after which its architecture, seed, weights, and hyperparameters remained unchanged.

The Figure 5.13 and Figure 5.14 show the image-level distributions for the selected run. For clarity, the remaining plots present only this model; data for all six runs are available in Appendix A.

The box plots highlight the separation between the correction in the Artifact Zone and the smaller change in the Clean Zone. The scatter plot allows identification of heterogeneity between slices, including individual cases with unfavorable changes in the Clean Zone.



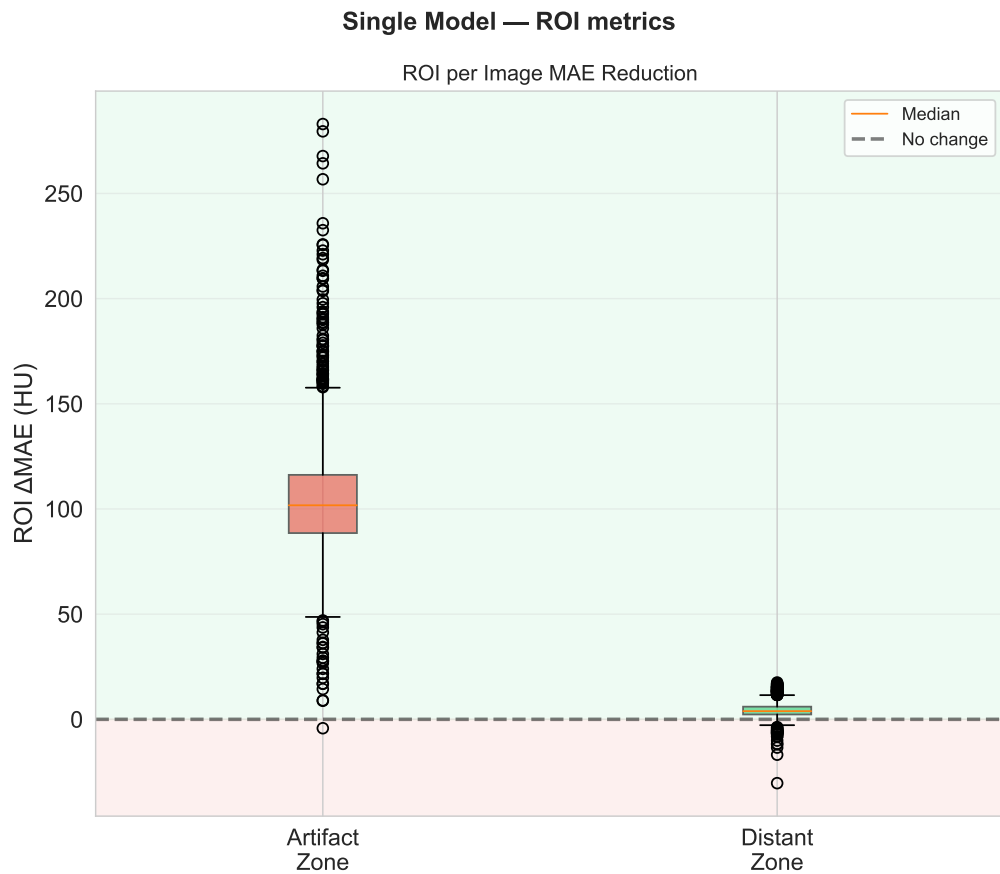
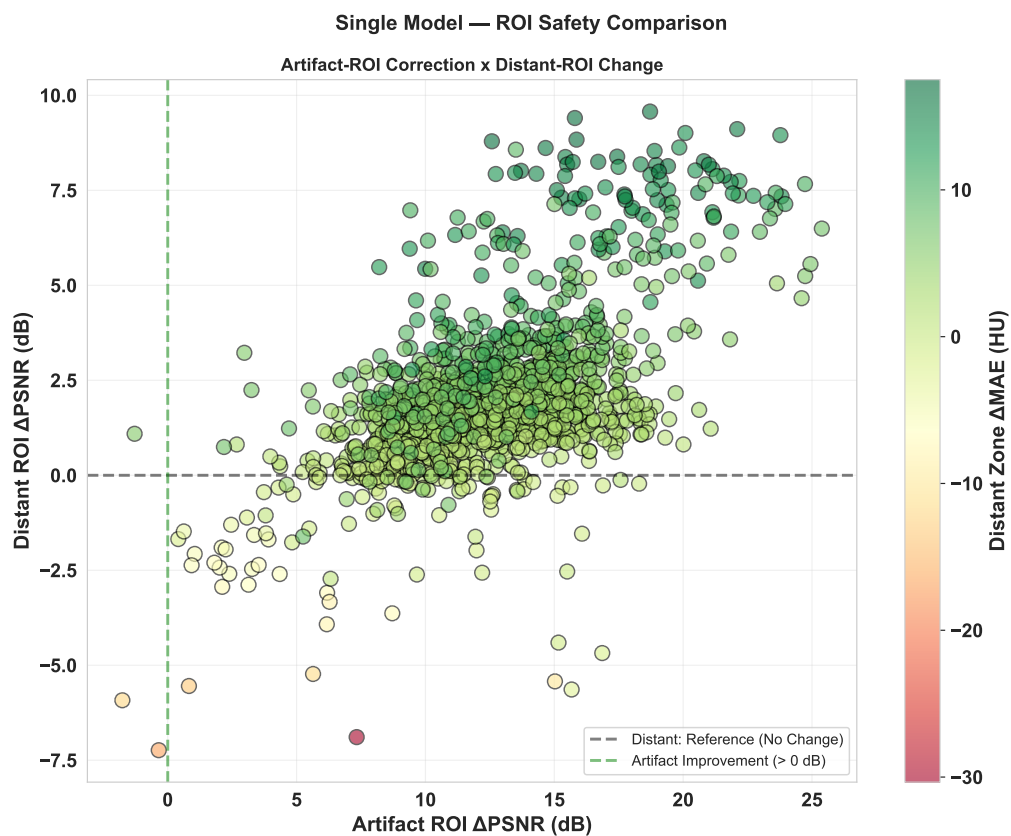


Fig.5.13 Per-image distributions of SSIM, PSNR, and MAE changes in the artifact and clean zones for the Single Model, seed 2024.



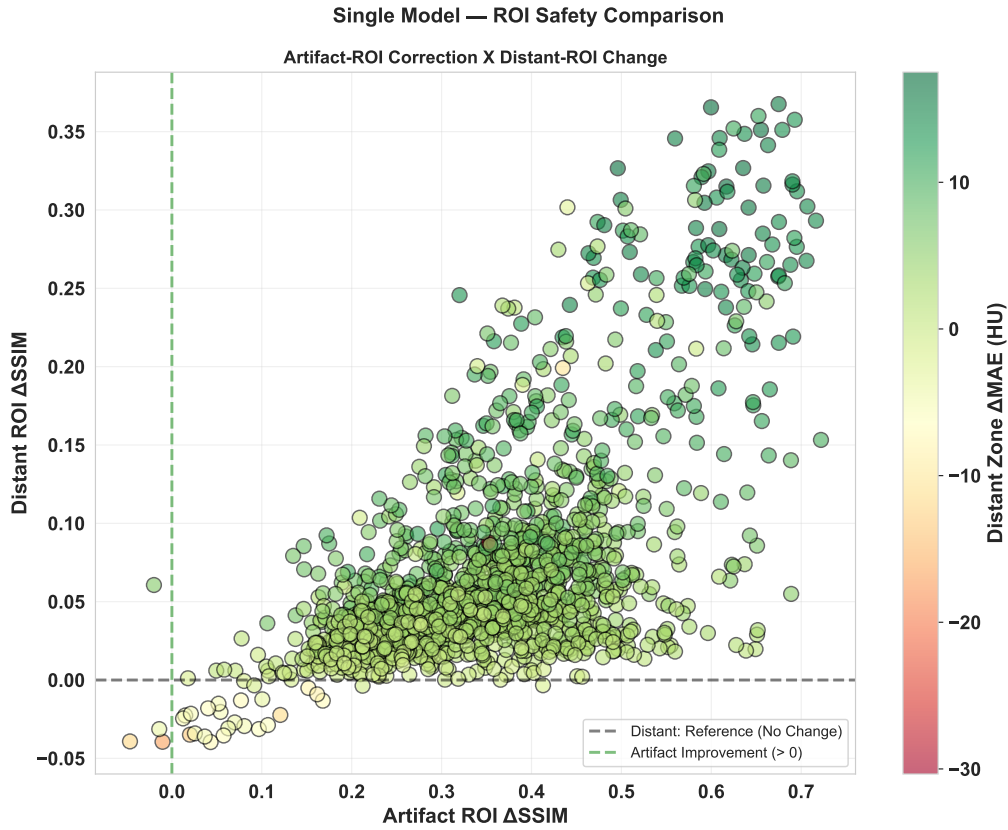


Fig.5.14 Per-image relationship between the $\Delta SSIM \setminus \Delta PSNR$ improvement in the Artifact Zone and the $\Delta SSIM \setminus \Delta PSNR$ change in the Clean Zone for the Single Model, seed 2024. Color encodes the ΔMAE reduction in the Clean Zone.

5.3.4 Residual Maps in Extreme Cases

To complement the results, six individual cases from the Single Model, seed 2024, were inspected in the test partition. The selection was made exclusively by the per-slice $\Delta SSIM$ in the Artifact Zone: the three highest values correspond to the greatest structural improvement in that region relative to the input, and the three lowest values to the smallest structural improvement. Thus, the terms “greatest” and “smallest” improvement in this subsection refer only to this quantitative criterion and do not constitute a clinical classification of the cases.

In each map, the first row shows the change required relative to the clean reference ($r_{\text{target}} = GT - \text{Input}$), the change produced by the model ($r_{\text{model}} = \text{Output} - \text{Input}$), and the remaining residual error ($r_{\text{diff}} = r_{\text{model}} - r_{\text{target}} = \text{Output} - GT$). The second row shows the absolute error of the input, the absolute error of the output, and their difference, where green indicates error reduction and red indicates worsening. The yellow outline delimits the Artifact Zone. Color scales are adjusted individually for each slice to preserve detail in cases with small residuals. However, comparison between cases should prioritize the spatial patterns and the statistics presented in each panel, rather than the apparent color intensity alone. Below each map, the

corresponding visualization of the artifact input, the model output, and the clean reference is shown.

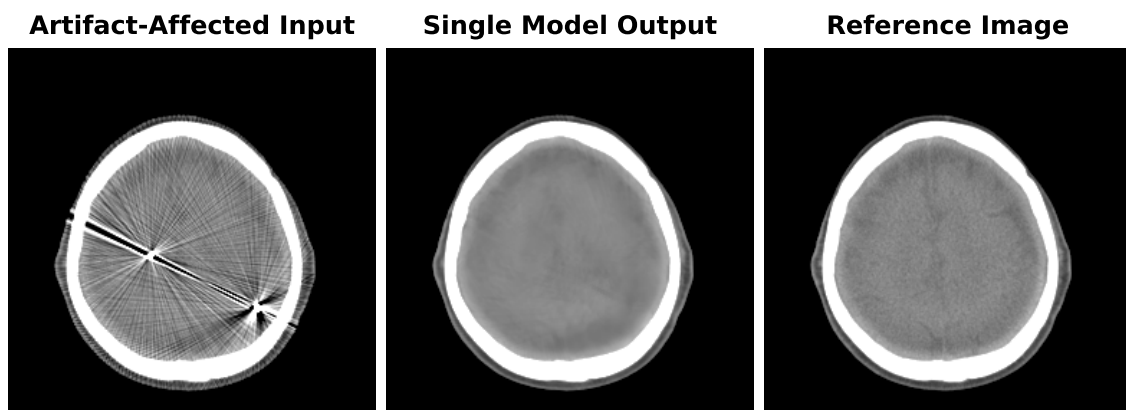
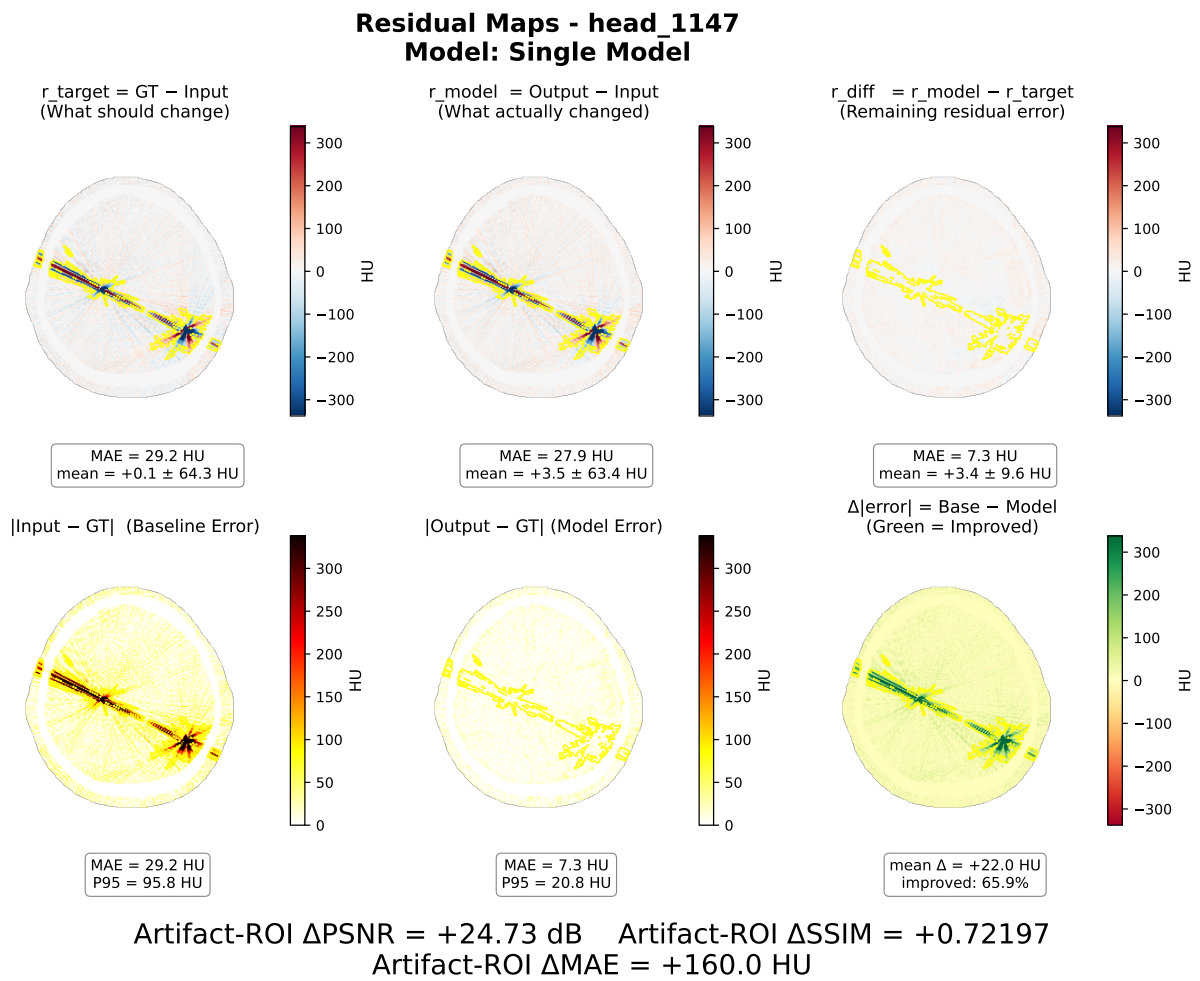


Fig.5.15 Residual map (top) and reconstruction visualization (bottom) for case head_1147 (test-set index 1250), selected among the three highest artifact-ROI ΔSSIM values ($\Delta\text{SSIM}_{\text{ROI}} = +0.7220$; global $\Delta\text{SSIM} = +0.1649$; $\Delta\text{PSNR} = +15.99$ dB). The yellow contour delimits the artifact ROI; all error maps are restricted to the body mask.

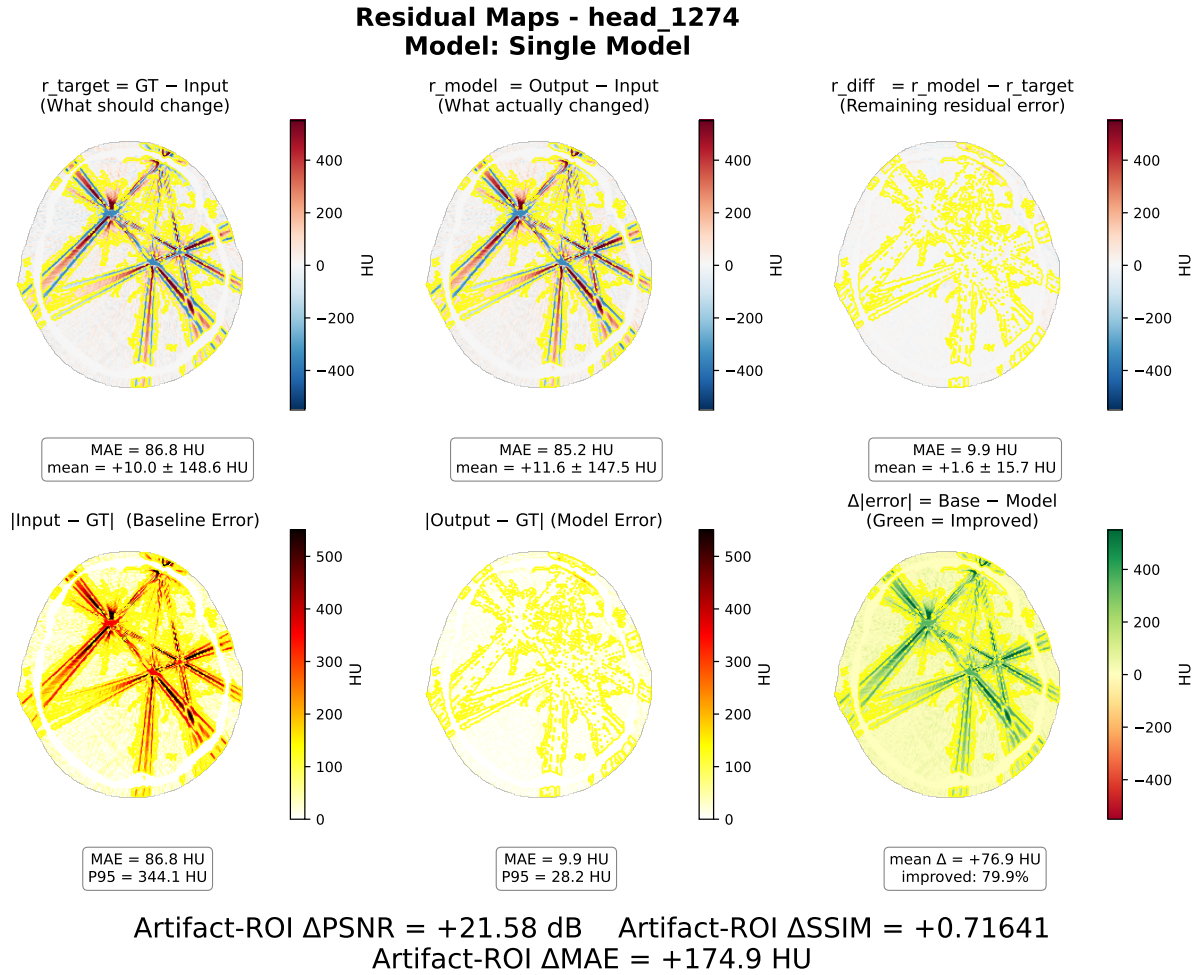


Fig.5.16 Residual map (top) and reconstruction visualization (bottom) for case head_1274 (test-set index 1364), selected among the three highest artifact-ROI ΔSSIM values ($\Delta\text{SSIM}_{\text{ROI}} = +0.7164$; global $\Delta\text{SSIM} = +0.3752$; $\Delta\text{PSNR} = +19.48$ dB). The yellow contour delimits the artifact ROI; all error maps are restricted to the body mask.

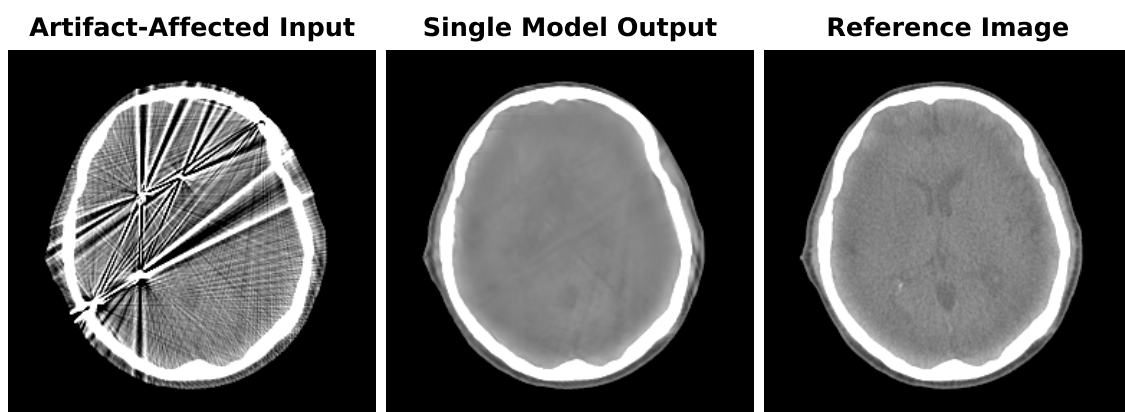
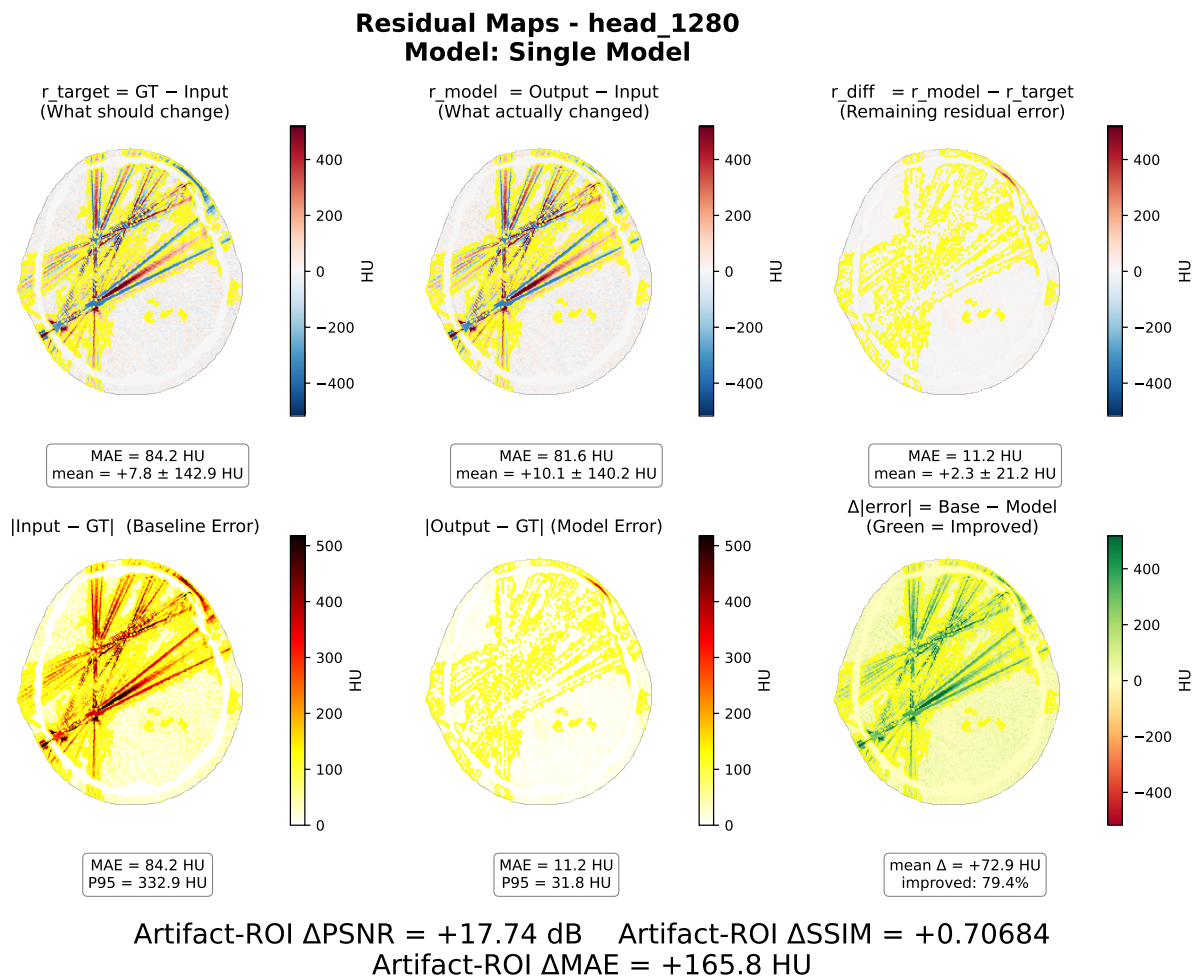


Fig.5.17 Residual map (top) and reconstruction visualization (bottom) for case head_1280 (test-set index 1369), selected among the three highest artifact-ROI ΔSSIM values ($\Delta\text{SSIM}_{\text{ROI}} = +0.7068$; global $\Delta\text{SSIM} = +0.3837$; $\Delta\text{PSNR} = +16.53$ dB). The yellow contour delimits the artifact ROI; all error maps are restricted to the body mask.

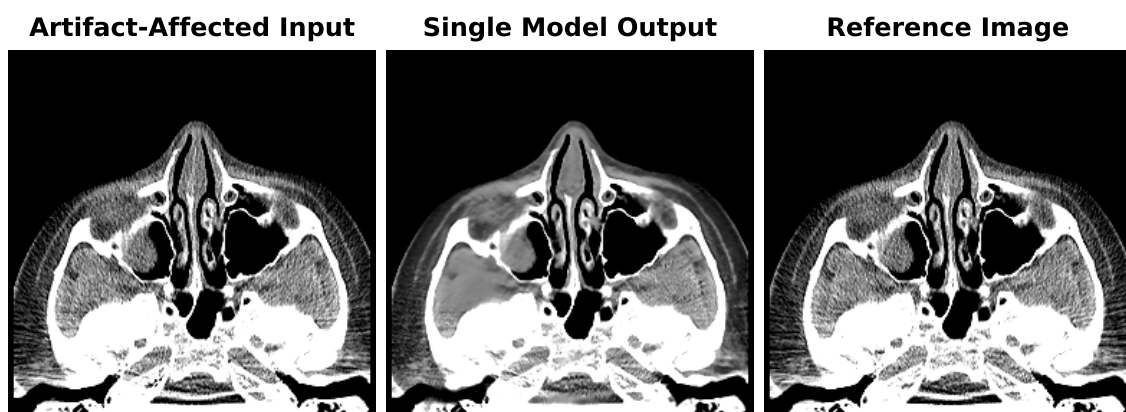
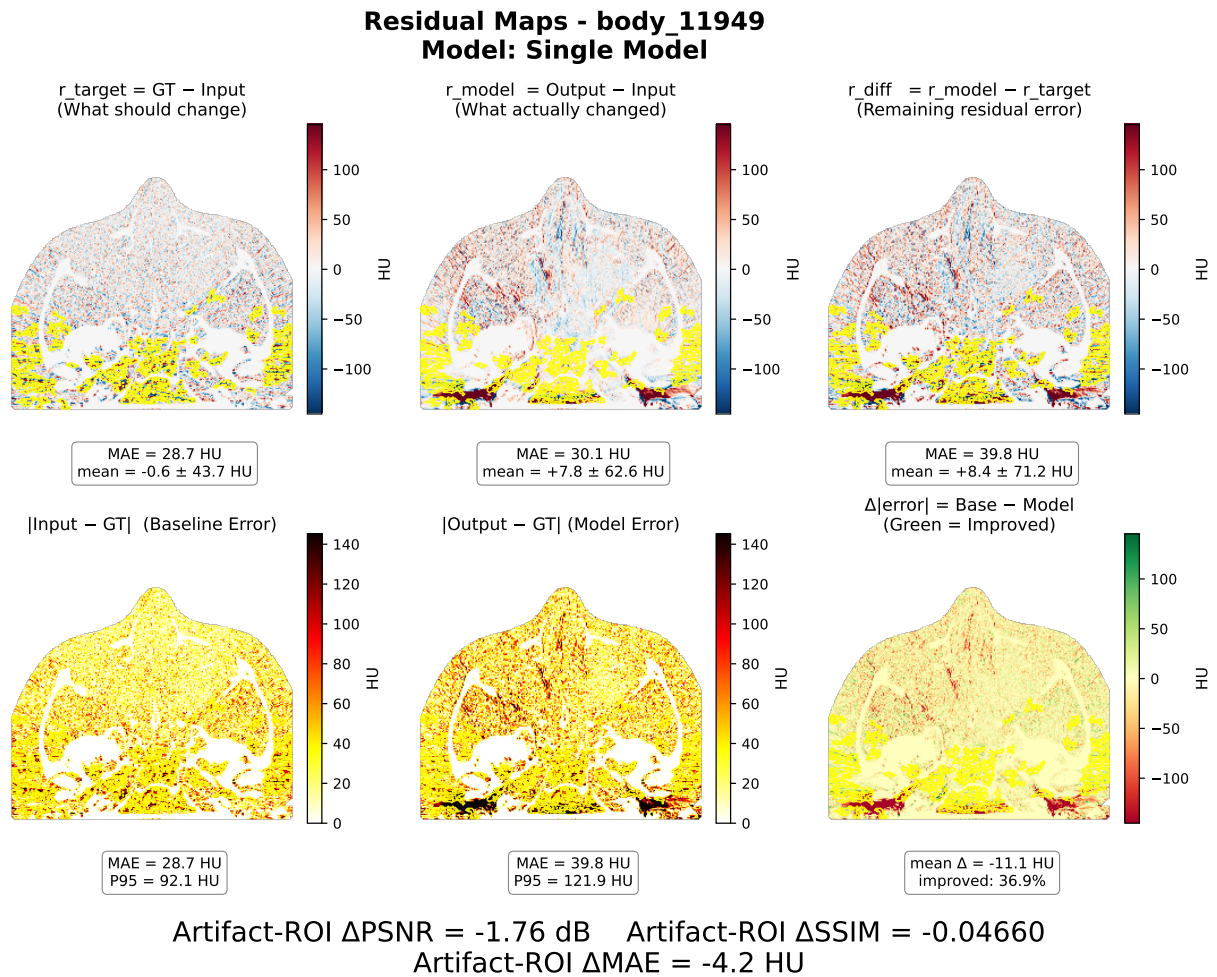


Fig.5.18 Residual map (top) and reconstruction visualization (bottom) for case body_11949 (test-set index 0716), selected among the three lowest artifact-ROI ΔSSIM values ($\Delta\text{SSIM}_{\text{ROI}} = -0.0466$; global $\Delta\text{SSIM} = -0.0497$; $\Delta\text{PSNR} = -4.31$ dB). The yellow contour delimits the artifact ROI; all error maps are restricted to the body mask.

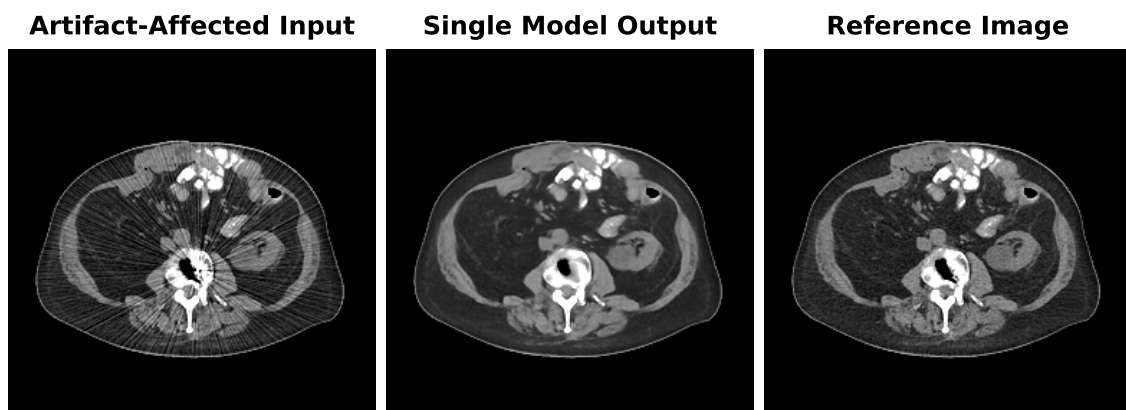
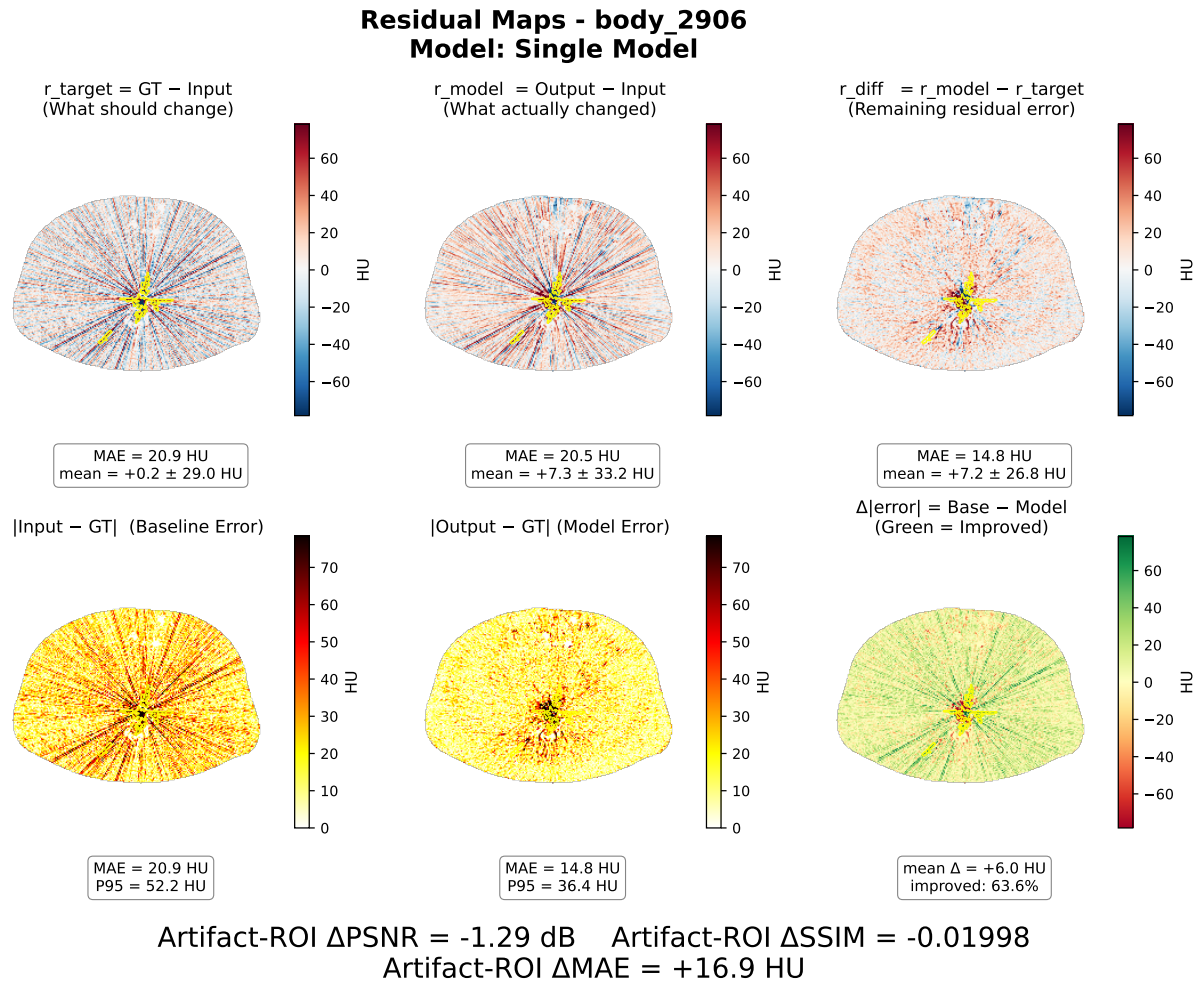


Fig.5.19 Residual map (top) and reconstruction visualization (bottom) for case body_2906 (test-set index 1122), selected among the three lowest artifact-ROI ΔSSIM values ($\Delta\text{SSIM}_{\text{ROI}} = -0.0200$; global $\Delta\text{SSIM} = +0.0740$; $\Delta\text{PSNR} = +0.38$ dB). The yellow contour delimits the artifact ROI; all error maps are restricted to the body mask.

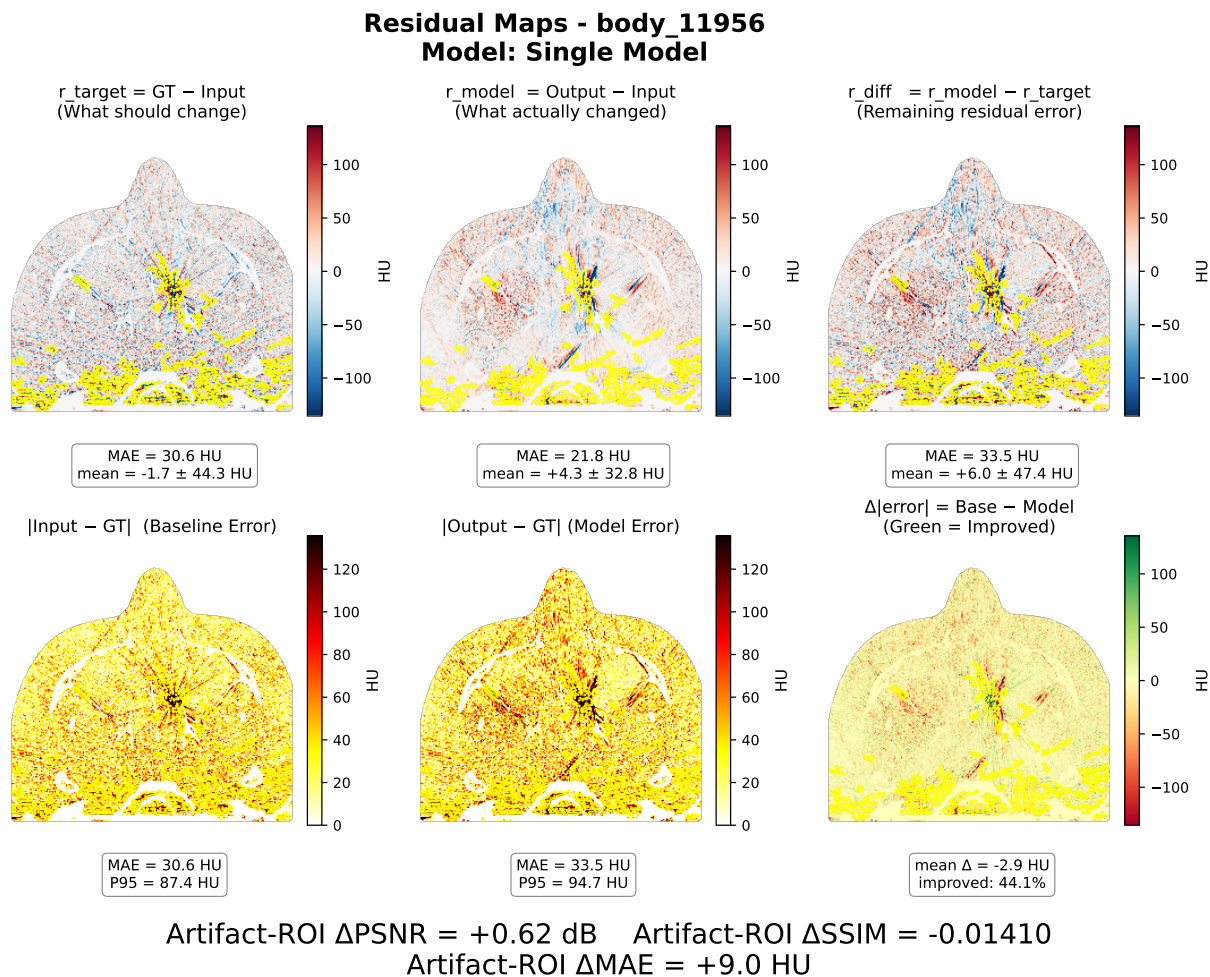


Fig.5.20 Residual map (top) and reconstruction visualization (bottom) for case body_11956 (test-set index 0723), selected among the three lowest artifact-ROI $\Delta SSIM$ values ($\Delta SSIM_{ROI} = -0.0141$; global $\Delta SSIM = -0.0345$; $\Delta PSNR = -0.64$ dB). The yellow contour delimits the artifact ROI; all error maps are restricted to the body mask.

5.4 Paired Comparison Single x Dual Model

5.4.1 Global Statistical Analysis

This analysis compares only the two final checkpoints trained with seed 2024, one per architecture, each chosen within its own run using validation [SSIM](#). The earlier region-based comparison of all six checkpoints had already selected the Single Model, seed 2024, for external evaluation. The architecture comparison here is therefore complementary: it did not influence that choice and did not change any checkpoint, hyperparameter, weight, or the architecture used for external evaluation.

Per-slice metrics were averaged per patient before any summary or statistical test, giving $n = 41$ independent patients. So the means, 95% confidence intervals, and p -values in this section's tables all refer to per-patient averages, with each patient counted equally.

The tables in this section analyze the global metrics, calculated over all pixels of the body mask of each slice (background excluded), as a complementary confirmation of the main hypothesis of structural improvement, which is formally tested in the Artifact Zone in the following section.

Checking the distribution of the per-patient values for ΔSSIM and ΔMAE showed that they were skewed, not evenly spread around the middle. Because of this, the main evidence for these two metrics came from the sign test, which simply asks how often the model did better than the input, without assuming the improvements are symmetrically distributed. For ΔPSNR , whose per-patient values showed no such skew, the two-sided one-sample t -test was used instead, to test the average improvement directly.

This distribution check was done separately for each model before applying the same rule to both. For ΔSSIM , the Shapiro–Wilk test showed the Single Model's values were not evenly distributed ($W = 0.9158$; $p = 0.0050$; skewness = 1.087), and the same was true for the Dual Model ($W = 0.9162$; $p = 0.0052$; skewness = 1.056), supporting the use of the sign test for both architectures. For ΔMAE , the same pattern was found in the Single Model ($W = 0.7607$; $p = 8.95 \times 10^{-7}$; skewness = 2.281) and in the Dual Model ($W = 0.7686$; $p = 1.27 \times 10^{-6}$; skewness = 2.200). For ΔPSNR , neither model's values showed this kind of unevenness (Single Model: $W = 0.9742$; $p = 0.4675$; Dual Model: $W = 0.9734$; $p = 0.4427$), which supported using the t -test for the average improvement in both models.

[Table 5.10](#), [Table 5.11](#), and [Table 5.12](#) show that both architectures substantially improved the global reconstruction quality relative to the corrupted input. The Single Model showed a mean ΔSSIM of 0.0891 (95% BCa CI: [0.0744;0.1075]), a mean ΔPSNR of 7.11 dB (95% t CI:

Reconstruction Quality: SSIM

Model	Mean improvement	95% BCa CI	p (Holm)
Single Model	0.0891	[0.0744; 0.1075]	7.64×10^{-11}
Dual Model	0.0879	[0.0731; 0.1062]	7.64×10^{-11}

Global body-mask-aware metric, aggregated at the patient level ($n = 41$); no ROI restriction. Two-sided exact binomial sign-test p -values, Holm-adjusted separately within each model across SSIM, PSNR, and MAE.

Table 5.10 Global SSIM improvement over the corrupted input on the test set.

Reconstruction Quality: PSNR

Model	Mean improvement (dB)	95% t CI	p (Holm)
Single Model	7.11	[6.35; 7.87]	4.53×10^{-21}
Dual Model	7.06	[6.30; 7.81]	4.42×10^{-21}

Global body-mask-aware metric, aggregated at the patient level ($n = 41$); no ROI restriction. Two-sided one-sample t -test p -values, Holm-adjusted separately within each model across SSIM, PSNR, and MAE.

Table 5.11 Global PSNR improvement over the corrupted input on the test set.

Reconstruction Quality: MAE

Model	Mean reduction (HU)	95% BCa CI	p (Holm)
Single Model	15.72	[12.94; 20.14]	7.64×10^{-11}
Dual Model	15.54	[12.79; 19.88]	7.64×10^{-11}

Global body-mask-aware metric, aggregated at the patient level ($n = 41$); no ROI restriction. Positive Δ MAE denotes error reduction. Two-sided exact binomial sign-test p -values, Holm-adjusted separately within each model across SSIM, PSNR, and MAE.

Table 5.12 Global MAE reduction over the corrupted input on the test set.

[6.35; 7.87]), and a mean MAE reduction of 15.72 HU (95% BCa CI: [12.94; 20.14]). The Dual Model showed a Δ SSIM of 0.0879 (95% BCa CI: [0.0731; 0.1062]), a Δ PSNR of 7.06 dB (95% t CI: [6.30; 7.81]), and a mean MAE reduction of 15.54 HU (95% BCa CI: [12.79; 19.88]). All six tests of global improvement over the input remained significant after the Holm adjustment applied separately within each architecture. The p -values for Δ SSIM and Δ MAE measure how consistently the model did better than the input from patient to patient, while those for Δ PSNR measure the average improvement; so their sizes should not be compared directly.

These tests confirm that each architecture improves upon the corrupted input, but they do not constitute a direct comparison between architectures. This distinction is essential: small p -values in Table 5.10, Table 5.11, Table 5.12 do not, by themselves, allow one to claim that one architecture is superior to the other.

The improvement deltas underlying the paired comparison are likewise aggregated per patient, giving equal weight to each of the 41 patients.

Dual Model – Single Model: Improvement Deltas

Metric	Mean Difference	95% BCa CI	p (sign test; Holm)
Δ SSIM	−0.0012	[−0.0015; −0.0010]	2.35×10^{-9}
Δ PSNR (dB)	−0.054	[−0.107; −0.017]	2.75×10^{-2}
Δ MAE (HU)	−0.186	[−0.338; −0.091]	8.62×10^{-4}

Differences of patient-level input-to-output improvement deltas (Dual Model – Single Model), with 95% BCa bootstrap CIs (10 000 resamples). Two-sided exact binomial sign-test p -values are Holm-adjusted across PSNR, SSIM, and MAE. Negative differences favor the Single Model for all three metrics.

Table 5.13 Patient-level paired comparison of improvement deltas between architectures ($n = 41$).

The direct comparison between architectures is presented in Table 5.13. For this analysis, the improvement difference (Dual Model minus Single Model) was calculated for each patient, for Δ SSIM, Δ PSNR, and Δ MAE, and the sign test was applied to each, with a joint Holm correction across the three tests. This asks whether one architecture wins in more patients than would be expected by chance, without assuming the differences are symmetrically distributed. The 95% confidence intervals, obtained by resampling patients 10,000 times, describe the size of the average difference, while the p -values describe how consistently one architecture won. For all metrics, negative values favor the Single Model.

The Single Model showed a mean advantage of 0.054 dB in Δ PSNR, 0.0012 in Δ SSIM, and 0.186 HU in Δ MAE. The confidence intervals did not include zero, and the adjusted p -values were 2.75×10^{-2} for PSNR, 2.35×10^{-9} for SSIM, and 8.62×10^{-4} for MAE. The differences favored the Single Model in 28 of the 41 patients for PSNR, in 39 for SSIM, and in 32 for MAE. Although these differences are statistically significant, they are small in absolute terms. So the results point to a modest overall numerical advantage for the Single Model, but do not, on their own, show that it is clinically better. The bar chart summarizes the similar global improvement obtained by both architectures.

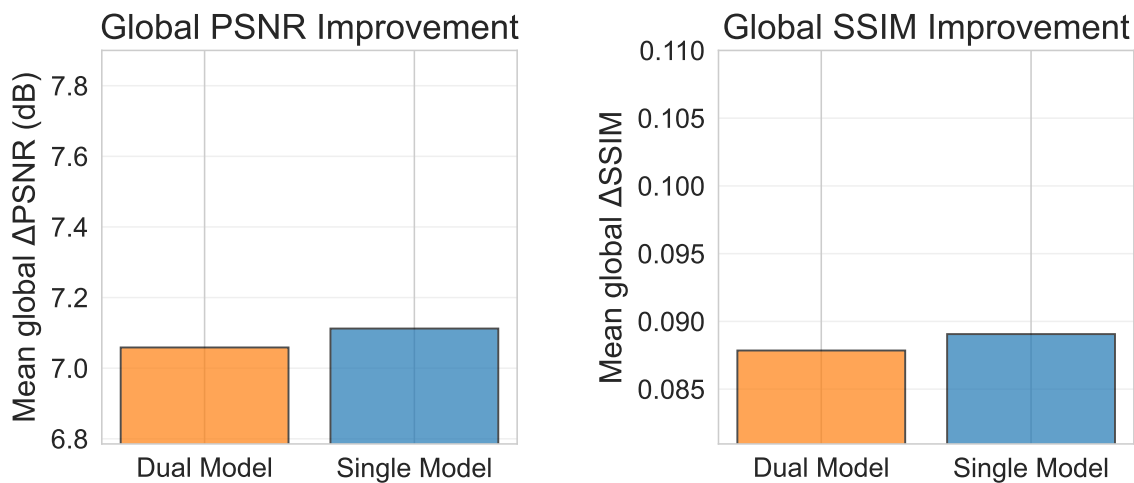


Fig.5.21 Global PSNR and SSIM improvement over the corrupted input for the Single and Dual Models on the test set.

5.4.2 Statistical Analysis per ROI

This section formally tests this study’s main hypothesis – that structure improves in the Artifact Zone – and describes preservation in the Distant Zone as a complement. As before, per-slice metrics were always averaged per patient before any summary or test ($n = 41$), and 95% confidence intervals were obtained by resampling patients 10,000 times. In the Artifact Zone, the question is simply whether the model improved on the input, so the signed changes in PSNR, SSIM, and MAE were analyzed. For each model, the sign test was applied to each of the three metrics, with a joint Holm correction across them.

Artifact ROI: SSIM

Model	Mean improvement	95% BCa CI	p (Holm)
Single Model	0.3279	[0.2963; 0.3548]	2.73×10^{-12}
Dual Model	0.3266	[0.2947; 0.3534]	2.73×10^{-12}

Patient-level Artifact-ROI improvement ($n = 41$). Two-sided exact binomial sign-test p -values, Holm-adjusted across PSNR, SSIM, and MAE separately for each model.

Table 5.14 SSIM improvement over the corrupted input in the Artifact ROI.

Artifact ROI: PSNR

Model	Mean improvement (dB)	95% BCa CI	p (Holm)
Single Model	11.86	[10.90; 12.59]	2.73×10^{-12}
Dual Model	11.84	[10.91; 12.55]	2.73×10^{-12}

Patient-level Artifact-ROI improvement ($n = 41$). Two-sided exact binomial sign-test p -values, Holm-adjusted across PSNR, SSIM, and MAE separately for each model.

Table 5.15 PSNR improvement over the corrupted input in the Artifact ROI.

Artifact ROI: MAE

Model	Mean reduction (HU)	95% BCa CI	p (Holm)
Single Model	98.38	[91.75; 104.24]	2.73×10^{-12}
Dual Model	98.33	[91.84; 104.06]	2.73×10^{-12}

Patient-level Artifact-ROI improvement ($n = 41$). Positive Δ MAE denotes error reduction. Two-sided exact binomial sign-test p -values, Holm-adjusted across PSNR, SSIM, and MAE separately for each model.

Table 5.16 MAE reduction over the corrupted input in the Artifact ROI.

Table 5.14, Table 5.15, and Table 5.16 show improvement in the three metrics in the Artifact Zone for both models. The mean gains were approximately 0.32 for SSIM, 11.8 dB for PSNR, and 98 HU for MAE. These results confirm, for both architectures, this study’s primary hypothesis of statistically significant structural improvement in the Artifact Zone.

In the direct comparison of improvement deltas in the Artifact Zone (Table 5.17), the difference in SSIM was small but remained significant after the Holm correction ($p_{\text{Holm}} = 0.0345$), favoring the Single Model. The differences in the PSNR and MAE deltas were not significant after adjustment.

Metric	$\Delta_{\text{Dual}} - \Delta_{\text{Single}}$	95% BCa CI	p (Holm)
ΔSSIM	-0.0013	$[-0.0025; 0.0002]$	0.0345
$\Delta\text{PSNR (dB)}$	-0.019	$[-0.085; 0.060]$	0.533
$\Delta\text{MAE (HU)}$	-0.049	$[-0.318; 0.345]$	0.422

Patient-level paired comparison of input-to-output improvement deltas ($n = 41$). Differences are $\Delta_{\text{Dual}} - \Delta_{\text{Single}}$; negative differences favor the Single Model for all metrics. The 95% BCa bootstrap CIs use 10 000 reamostragens; two-sided exact binomial sign-test p -values are Holm-adjusted across the three Artifact-ROI metrics.

Table 5.17 Patient-level paired comparison of improvement deltas in the Artifact ROI.

In the distant region, the question is different: the goal is to see how well the tissue away from the artifact was left unchanged, since ideally there should be no change at all relative to the input. So the average per-slice $|\Delta|$ was calculated for each patient, where lower values mean the reconstruction stayed closer to the input. Because no specific “close enough to zero” limit was set in advance, it is not possible to formally test or conclude that the change is equivalent to zero. Thus, Table 5.18, Table 5.19, Table 5.20 report the size of the change and its confidence intervals, without a p -value against zero.

Distant ROI: SSIM

Model	Mean $ \Delta $	95% BCa CI
Single Model	0.0599	$[0.0503; 0.0754]$
Dual Model	0.0595	$[0.0499; 0.0748]$

Patient-level mean absolute per-slice input-to-output change ($n = 41$). Lower values are closer to zero. No equivalence margin or hypothesis test against zero was prespecified.

Table 5.18 Magnitude of SSIM input-to-output change in the Distant ROI.

Distant ROI: PSNR

Model	Mean $ \Delta $ (dB)	95% BCa CI
Single Model	1.99	$[1.66; 2.34]$
Dual Model	1.97	$[1.64; 2.33]$

Patient-level mean absolute per-slice input-to-output change ($n = 41$). Lower values are closer to zero. No equivalence margin or hypothesis test against zero was prespecified.

Table 5.19 Magnitude of PSNR input-to-output change in the Distant ROI.

Distant ROI: MAE

Model	Mean $ \Delta $ (HU)	95% BCa CI
Single Model	4.30	$[3.60; 5.16]$
Dual Model	4.24	$[3.54; 5.08]$

Patient-level mean absolute per-slice input-to-output change ($n = 41$). Lower values are closer to zero. No equivalence margin or hypothesis test against zero was prespecified.

Table 5.20 Magnitude of MAE input-to-output change in the Distant ROI.

To compare the architectures on preservation in the distant region, the Dual Model’s advantage was defined for each patient as $|\Delta|_{\text{Single}} - |\Delta|_{\text{Dual}}$, and the sign test was applied, with a joint Holm correction across the three metrics. This is a separate comparison from the Artifact-Zone

improvement above: here, what matters is the size of the change, and the better value is the one closest to zero. Positive values indicate a smaller change introduced by the Dual Model. This test answers only which model introduced a smaller change, not whether that change is clinically negligible.

Metric	Dual preservation advantage	95% BCa CI	p (Holm)
SSIM	0.000390	[0.000056; 0.000642]	3.37×10^{-4}
PSNR (dB)	0.020	[-0.104; 0.069]	0.0275
MAE (HU)	0.066	[-0.050; 0.155]	0.00865

Patient-level paired preservation advantage ($n = 41$), defined as $|\Delta|_{\text{Single}} - |\Delta|_{\text{Dual}}$. Positive values favor the Dual Model because its mean absolute input-to-output change is smaller. The 95% BCa bootstrap CIs use 10 000 reamstragens, two-sided exact binomial sign-test p -values are Holm-adjusted across the three Distant-ROI metrics.

Table 5.21 Relative preservation advantage of the Dual Model in the Distant ROI.

In the relative comparison (Table 5.21), the Dual Model showed a smaller absolute change in PSNR, SSIM, and MAE after the Holm correction. This suggests the Dual Model preserved the distant tissue somewhat better than the Single Model, but it does not prove the change was equivalent to none at all.

The computational performance (inference latency and throughput) of the model selected for external evaluation is characterized in section A.3.

5.5 External Evaluation for Doctors

5.5.1 Capture Status and Participants

The external evaluation consists of a paired comparison blinded to the processing label: in each case, readers simultaneously view the artifact input and the model output, identified only as “Image A” and “Image B”. The left/right position is counterbalanced across readers and kept constant between a case and its hidden repeat. Each session presents 19 comparisons: 2 training cases (excluded from the analysis), 15 main cases, and 2 hidden repeats of previously presented cases, used exclusively to assess intra-reader consistency. Physicians were recruited directly by the researcher, with self-declaration of clinical CT use with a minimum monthly frequency. No name, contact information, professional registration number, or proof of qualification was collected.

Table 5.22 summarizes the status of recruitment and response completion.

The self-reported characteristics of the readers who completed the evaluation are presented in Table 5.23. None of these variables allow individual identification of a participant.

Indicator	Value
Total Readers	[to fill]
Total valid responses (readers $\times$ 15 main cases)	[to fill]
Data collection start date	2026-08-27
Data collection closing date	[to fill]

Table 5.22 Status of recruitment and response completion for the external evaluation.

Variable	Category	N	%
Jurisdiction	Brazil	[to fill]	[to fill]
	Portugal	[to fill]	[to fill]
CT usage frequency	Monthly	[to fill]	[to fill]
	Weekly	[to fill]	[to fill]
	Daily	[to fill]	[to fill]
CT experience (years)	< 2	[to fill]	[to fill]
	2–5	[to fill]	[to fill]
	6–10	[to fill]	[to fill]
	> 10	[to fill]	[to fill]
Experience with artifact reduction	None	[to fill]	[to fill]
	Occasional	[to fill]	[to fill]
	Frequent	[to fill]	[to fill]
Monitor type	Diagnostic	[to fill]	[to fill]
	Conventional	[to fill]	[to fill]
	Unknown	[to fill]	[to fill]

Table 5.23 Self-reported characteristics of the readers who completed the external evaluation.

The distribution by medical specialty is presented separately in [Table 5.24](#), given the larger number of categories.

Specialty	N	%
Radiology and Diagnostic Imaging	[to fill]	[to fill]
Pulmonology	[to fill]	[to fill]
Thoracic or Cardiothoracic Surgery	[to fill]	[to fill]
Oncology	[to fill]	[to fill]
Internal Medicine or General Medicine	[to fill]	[to fill]
Intensive Care Medicine	[to fill]	[to fill]
Emergency Medicine	[to fill]	[to fill]
Cardiology	[to fill]	[to fill]
Gastroenterology or Hepatology	[to fill]	[to fill]
General Surgery or other surgical specialty	[to fill]	[to fill]
Urology	[to fill]	[to fill]
Orthopedics or Traumatology	[to fill]	[to fill]
Infectious Diseases	[to fill]	[to fill]
Other medical specialty	[to fill]	[to fill]

Table 5.24 Distribution by medical specialty of the readers who completed the external evaluation.

5.5.2 Quantitative Characterization of Input–Output

Before the professional evaluation, the global change produced by the *Single Model* was characterized in the 15 main cases of the external set. The two training cases were excluded from this summary. Since these real exams do not have a clean reference, the metrics in Table 5.25 compare exclusively the input image with the model output, within the body mask.

Input–Output Similarity and Change		
Metric	Mean $\pm$ SD	95% CI
SSIM (Input, Output)	0.8953 ± 0.0729	[0.8572; 0.9290]
PSNR (Input, Output) (dB)	33.14 ± 6.49	[29.95; 36.21]
MAE (Input, Output) (HU)	21.12 ± 15.26	[14.54; 29.26]

No artifact-free reference is available. Metrics were calculated within the manually corrected effective body mask and quantify input–output similarity or change; they do not measure artifact-removal accuracy or improvement.

Table 5.25 Global input–output metrics in the frozen external main set ($n = 15$) for `singlehead-033-0.9511-ssim_hu.keras`. Values are mean $\pm$ standard deviation; 95% CIs are percentile bootstrap intervals for the mean.

In the external set, the mean input–output similarity was 0.8953 for SSIM and 33.14 dB for PSNR, while the mean magnitude of change was 21.12 HU in MAE. The dispersion between cases, expressed by the standard deviations and confidence intervals in the table, evidences heterogeneity in the degree of change produced by the reconstruction. These results describe the preservation and overall modification of the external images, but do not allow inference, without a clean reference, of artifact-removal accuracy or clinical benefit. This interpretation is addressed by the blinded evaluation by physicians in the following subsections.

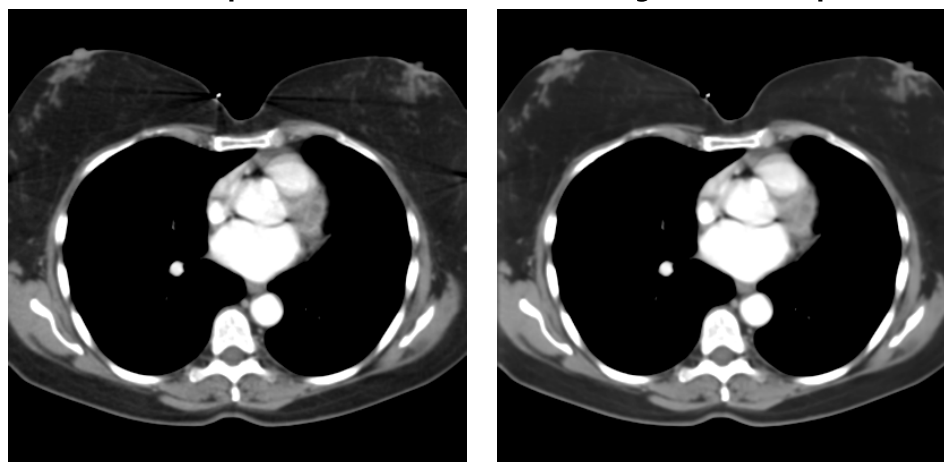
To make this heterogeneity visible, the three cases with the highest and the three with the lowest input–output SSIM were selected descriptively. The former represent the pairs with the smallest overall structural change, while the latter represent those with the greatest change.

Illustrative External Cases Selected by Input–Output SSIM				
Group	Case	SSIM	PSNR (dB)	MAE (HU)
Least changed	Case 003	0.9809	41.19	8.76
	Case 002	0.9756	42.35	8.10
	Case 012	0.9586	37.91	9.64
Most changed	Case 015	0.7278	23.32	43.10
	Case 005	0.7770	20.77	63.90
	Case 008	0.8334	33.53	20.39

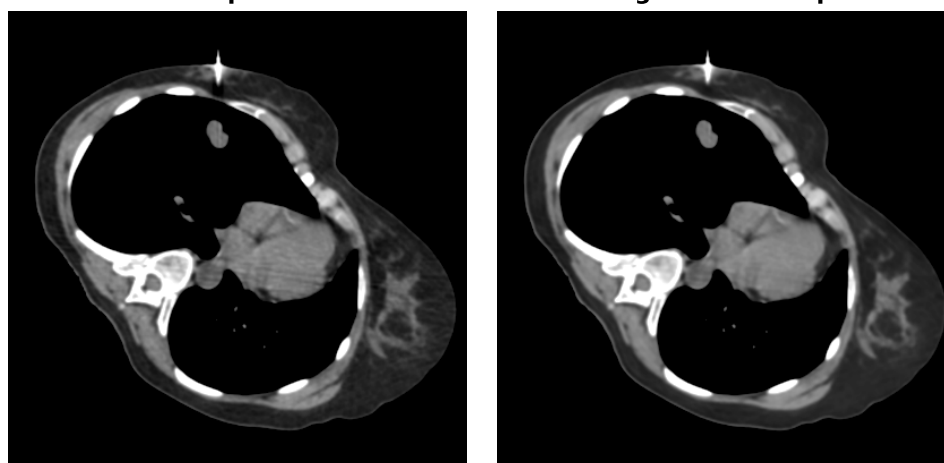
Cases were selected only by the extremes of SSIM(Input, Output). High SSIM denotes less structural change between input and output; this ranking does not define artifact-removal quality or clinical acceptability.

Table 5.26 Descriptive external cases with the highest and lowest input–output SSIM values.

Case 003 — Input-Output Pair
SSIM=0.9809; PSNR=41.19 dB; MAE=8.76 HU



Case 002 — Input-Output Pair
SSIM=0.9756; PSNR=42.35 dB; MAE=8.10 HU



Case 012 — Input-Output Pair
SSIM=0.9586; PSNR=37.91 dB; MAE=9.64 HU

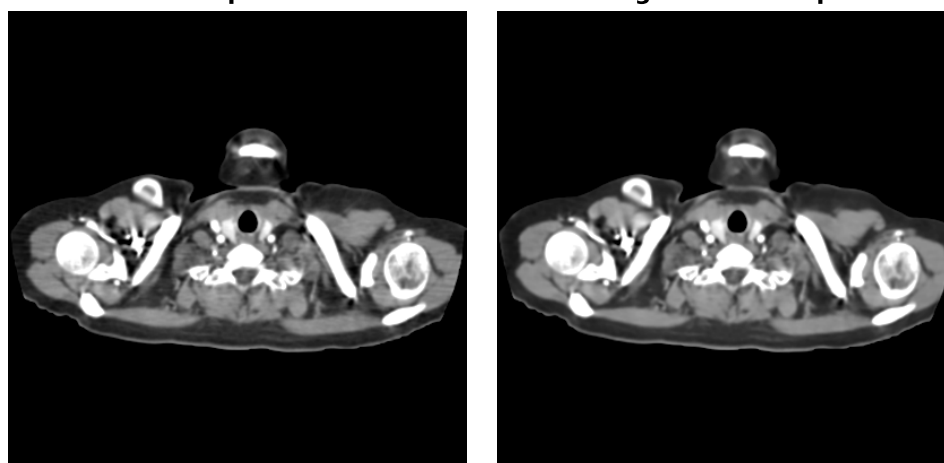
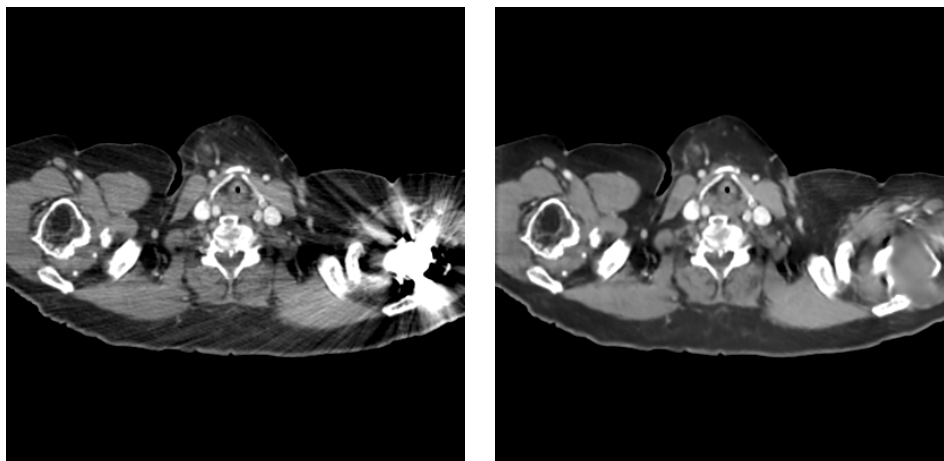
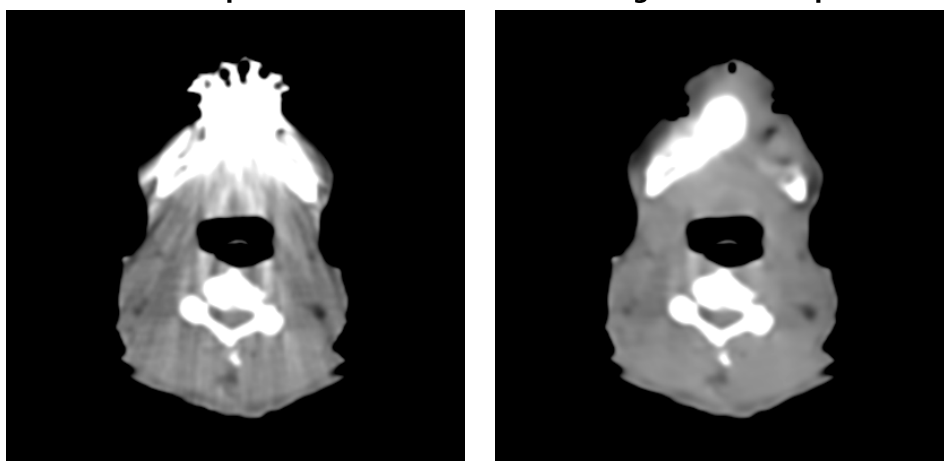


Fig.5.22 External input-output pairs with the three highest SSIM values, representing the least structural change. The soft-tissue window is identical for input and output within each case.

Case 015 — Input-Output Pair
SSIM=0.7278; PSNR=23.32 dB; MAE=43.10 HU



Case 005 — Input-Output Pair
SSIM=0.7770; PSNR=20.77 dB; MAE=63.90 HU



Case 008 — Input-Output Pair
SSIM=0.8334; PSNR=33.53 dB; MAE=20.39 HU
Input **Single Model output**

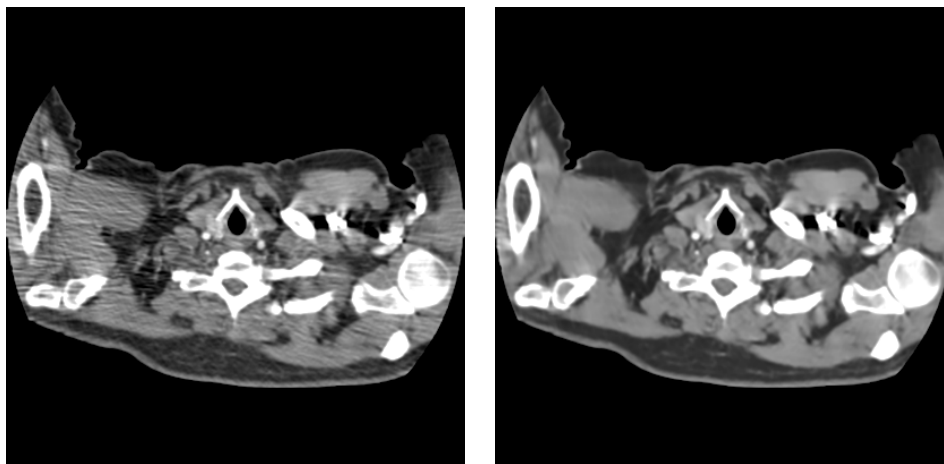


Fig.5.23 External input-output pairs with the three lowest SSIM values, representing the greatest structural change. The soft-tissue window is identical for input and output within each case.

5.5.3 Primary Outcome: Anatomical Visualization

The primary outcome is the paired preference, per case, regarding which image better preserves the anatomical region underlying the metal artifact (“Image A” / “Image B” / “No difference”). Answers are converted back to input/output using the left/right position recorded for each case, without the reader ever knowing this mapping while answering. Training cases and hidden repeats are not part of this analysis ($n = 15$ main cases per reader).

Because each doctor judges many cases, and each case is judged by many doctors, a statistical model was used that accounts for both of these repetitions at the same time (a model with a random effect for the reader and for the case), rather than treating every answer as fully independent. In interpreting the result, the 95% confidence interval is given more weight than the p -value on its own.

Response	N	%	95% CI
Preferred output (Single Model)	[to fill]	[to fill]	[to fill]
Preferred input (with artifact)	[to fill]	[to fill]	[to fill]
No difference	[to fill]	[to fill]	[to fill]

Pooled frequencies over all valid responses (readers $\times$ 15 main cases); 95% CI by *bootstrap* or exact method to be specified.

Table 5.27 Paired preference regarding preservation of the anatomical region underlying the metal artifact.

Effect	Estimate	95% CI	p
Condition (output vs. input)	[to fill]	[to fill]	[to fill]

Mixed-effects ordinal model with random effects for reader and case; scale and parametrization of the coefficient to be specified (e.g., *log-odds* or predicted marginal proportion).

Table 5.28 Mixed-effects ordinal model estimate for the primary outcome.

5.5.4 Secondary Outcomes and Safety Perception

The secondary outcomes use the same per-case format (“Image A” / “Image B” / “No difference”), converted back to input/output, and are reported simply as the frequency of each answer with a 95% confidence interval per question; because this summary is descriptive, no correction for multiple comparisons was applied.

Response	N	%	95% CI
Preferred output (lower artifact intensity)	[to fill]	[to fill]	[to fill]
Preferred input	[to fill]	[to fill]	[to fill]
No difference	[to fill]	[to fill]	[to fill]

Table 5.29 Paired preference regarding the perceived intensity of metal artifacts.

Response	N	%	95% CI
Preferred output (less distortion)	[to fill]	[to fill]	[to fill]
Preferred input	[to fill]	[to fill]	[to fill]
No difference	[to fill]	[to fill]	[to fill]

This item functions as an exploratory signal of possible harm: preference for the input suggests perceived distortions or artifacts introduced by the model outside the artifact zone.

Table 5.30 Paired preference regarding artifacts or anatomical distortions introduced in regions distant from the artifact.

Response	N	%	95% CI
Preferred output for clinical interpretation	[to fill]	[to fill]	[to fill]
Preferred input	[to fill]	[to fill]	[to fill]
No difference	[to fill]	[to fill]	[to fill]

Table 5.31 Paired preference regarding the image chosen for clinical interpretation.

Additionally, at the end of the session, each reader answered once a 5-point ordinal question about the complementary usefulness of a version processed with this pattern relative to the original CT exam (Table 5.32).

Category	N	%
Definitely not	[to fill]	[to fill]
Probably not	[to fill]	[to fill]
Uncertain	[to fill]	[to fill]
Probably yes	[to fill]	[to fill]
Definitely yes	[to fill]	[to fill]

One response per reader (not per case); n = number of readers who completed the evaluation.

Table 5.32 Perceived complementary usefulness of a processed version relative to the original CT exam.

Intra-reader consistency was assessed through two hidden repeats per reader: the same case is re-presented later in the session, without signaling, in the same left/right position as the first presentation. Agreement between the original response and the repeat was quantified

using the ICC(2,1) statistic, a standard way of measuring how consistent repeated ratings are, and, for the simple output-versus-input preference, using Fleiss's κ (Table 5.33).

Statistic	Value	95% CI	Interpretation
ICC(2,1)	[to fill]	[to fill]	[to fill]
Fleiss's κ (output vs. input)	[to fill]	—	[to fill]

$n = 2$ hidden repeats per reader. 95% CI of ICC(2,1) by the F -distribution-based method (Shrout & Fleiss, 1979).

Table 5.33 Intra-reader consistency in the hidden repeats.

5.5.5 Limited Interpretation to the Observational Design

The design of this evaluation will make it possible to conclude, once the data are available, which image readers prefer along each assessed dimension, but it does not allow inference of causality regarding clinical outcomes (for example, diagnostic or therapeutic impact), nor generalization beyond the set of 15 cases and the panel of readers actually recruited. The absence of a clean reference in these external cases means that the assessment of anatomical accuracy depends entirely on readers' judgment, rather than on an objective error metric. Finally, the number of recruited readers limits the precision of the estimates and the possibility of subgroup analyses by specialty or experience.

Chapter 6

Discussion

6.1 Summary of the Main Findings

The overarching objective of this dissertation was to develop, evaluate, and rigorously compare a **CNNs**-based approach for metal artifact reduction in **CT**, prioritizing anatomical preservation, robustness to domain variation, and interpretability of the correction. The internal evaluation results support the primary hypothesis of structural improvement in the Artifact Zone relative to the corrupted input: across the six runs (two architectures, three seeds), **SSIM** in this zone increased by 0.322 to 0.328 (Table 5.7). For the run selected for external evaluation (Single Model, seed 2024), this improvement was statistically significant at the patient level ($\Delta \text{SSIM} = 0.3279$, 95% BCa CI [0.2963; 0.3548], $p_{\text{Holm}} = 2.73 \times 10^{-12}$; Table 5.14), with an equivalent pattern for the Dual Model ($\Delta \text{SSIM} = 0.3266$, $p_{\text{Holm}} = 2.73 \times 10^{-12}$). This localized structural improvement was accompanied by a consistent and equally significant global improvement: across all six runs, reconstruction increased global **PSNR** from 27.39 dB to 34.95–35.20 dB, global **SSIM** from 0.8361 to 0.9421–0.9442, and reduced global **MAE** from 36.6 HU to 17.3–17.8 HU (Table 5.1, Table 5.2, Table 5.3). At the patient level, the secondary global-improvement tests were significant for all three metrics in every run ($p < 0.001$; Table 5.4, Table 5.5, Table 5.6), with 95% confidence intervals excluding the null value.

Descriptive slice-level inspection (Figure 5.11) showed that unfavorable cases were rare: only 0.8% of slices had negative ΔPSNR and 1.6% had negative ΔSSIM , out of 1,478 test slices. This consistency is reinforced by the **ROI** analysis (subsection 5.3.2), which showed that correction was concentrated in the Artifact Zone ($\Delta \text{PSNR} \approx 11.8$ dB, $\Delta \text{SSIM} \approx 0.33$, **MAE** reduction ≈ 98 HU), with substantially smaller changes in the Distant Zone ($|\Delta \text{PSNR}| \approx 2$ dB, $|\Delta \text{SSIM}| \approx 0.06$, $|\Delta \text{MAE}| \approx 4$ HU). This contrast indicates that the model did not apply indiscriminate image smoothing, but rather concentrated changes where artifacts were present.

Within each run, the checkpoint was selected exclusively by validation reconstruction [SSIM](#). The six resulting checkpoints were then frozen and compared once on the internal test partition to select the external-evaluation candidate by mean patient-level Artifact-ROI Δ [SSIM](#), subject to the descriptive Distant-Zone preservation check. So the internal test analysis included a step of picking a candidate, rather than being a strict confirmation test. It did not, however, change a checkpoint, tune hyperparameters, alter weights, or trigger further training after the six validation-selected checkpoints had been fixed.

6.2 Architectural Comparison Interpretation

The paired comparison between architectures ([section 5.4](#)) addressed the second specific objective—whether auxiliary artifact-mask supervision improves reconstruction relative to the version without mask supervision—with a nuanced result. Globally, the Single Model showed a small but statistically significant advantage for all three metrics (Δ PSNR +0.054 dB, Δ SSIM +0.0012, Δ MAE +0.186 HU; $p_{\text{Holm}} = 2.75 \times 10^{-2}$, 2.35×10^{-9} , and 8.62×10^{-4} , respectively; [Table 5.13](#)), favoring the Single Model in 28, 39, and 32 of the 41 patients. In the Artifact Zone, this advantage was limited to [SSIM](#) ($p_{\text{Holm}} = 0.0345$), with no significant difference in [PSNR](#) or [MAE](#) ([Table 5.17](#)).

Conversely, in the Distant Zone, the Dual Model produced smaller absolute changes from the input for all three metrics, with a statistically significant preservation advantage after Holm correction ($p_{\text{Holm}} = 3.37 \times 10^{-4}$ for [SSIM](#), 0.0275 for [PSNR](#), and 0.00865 for [MAE](#); [Table 5.21](#)). This pattern can be interpreted as a trade-off: the additional segmentation supervision appears to spatially constrain correction, making the Dual Model marginally more conservative outside the Artifact Zone, at the cost of a slightly smaller correction magnitude both globally and within the Artifact Zone itself. However, in this work the decoder was kept almost entirely shared, with the segmentation output added only at the final layer. A deeper separation between reconstruction and segmentation decoders might enable a better exploration of this trade-off, but was not tested here.

In neither direction do the observed differences appear clinically distinguishable on their own—they are tenths of a dB in [PSNR](#), thousandths in [SSIM](#), and fractions of a HU in [MAE](#). Its statistical significance comes mainly from one architecture winning consistently across patients (sign tests with $n = 41$), not from a large difference in size. Thus, the answer to the architectural-comparison objective is that auxiliary mask supervision did not produce decisively superior reconstruction, but redistributed the correction–preservation trade-off in a measurable way. The external-evaluation candidate was selected from the six validation-frozen checkpoints

by Artifact-ROI Δ SSIM, with the Distant Zone used as a descriptive preservation check. The global paired comparison describes this architectural trade-off, but it did not change which checkpoint was selected or alter any decision already made during training.

6.3 Anatomical Preservation and Reconstruction Safety

Taken together, these additional descriptive analyses point to a consistent picture, though not one that is clinically conclusive on its own. The ROI partition already showed that changes in the Distant Zone were a small fraction of the correction observed in the Artifact Zone (Figure 5.13, Figure 5.14). Patient-level Bland–Altman agreement for mean body HU (Figure 5.12) showed a small but non-zero positive bias of +4.19 HU, with 95% limits of agreement from +1.45 to +6.93 HU—a slight systematic shift in mean intensity that should be monitored in future work, although it is small relative to the HU window range used.

As an additional characterization of the lower-performance tail, the 74 of 1,478 test slices (5%) with the lowest Δ PSNR (threshold 2.59 dB) retained positive mean improvement for all three metrics (mean Δ PSNR = 1.23 dB; mean Δ SSIM = 0.0071; mean Artifact-Zone Δ MAE = 62.0 HU; Figure A.12), despite substantial dispersion and individual cases of degradation relative to the input. This suggests that unfavorable cases tend to be graded rather than catastrophic, although they exist and warrant qualitative inspection—provided for the extreme Artifact-Zone Δ SSIM cases in subsection 5.3.4.

A complementary exploratory analysis (Figure A.9) related artifact burden (artifact-area/body-area fraction, averaged per patient) to global improvement and found a positive association: $\rho = 0.479$ for Δ PSNR (95% CI [0.156; 0.717], $p = 0.00154$) and $\rho = 0.569$ for Δ SSIM (95% CI [0.225; 0.794], $p = 0.000104$). In other words, the model tended to correct proportionally more in patients with a greater artifact burden—a reassuring finding from the perspective of targeted correction. However, because this analysis was not one of the tests planned in advance and was not corrected for multiple comparisons, it should be read as suggesting a pattern worth testing further, not as independent statistical proof.

Together, ROI stratification, Bland–Altman calibration, characterization of the lower-performance tail, and the association with artifact burden provide a convergent but still descriptive picture: correction is concentrated where needed, distant tissue is largely preserved with a small calibration shift, and the lower-performing slices do not show systematic anatomical distortion. Definitive assessment of clinical safety, however, remains reserved for the external reader evaluation (section 5.5).

6.4 Generalization and External Evaluation

The fourth specific objective—to investigate the generalization capacity of the selected version on real data external to training, obtained from [TCIA](#)—can only be answered partially with the technical evidence currently available. The 15 main cases in the external set have no clean reference; therefore, only the magnitude of change between input and output could be characterized ([Table 5.25](#)): mean [SSIM](#) of 0.8953, mean [PSNR](#) of 33.14 dB, and mean [MAE](#) of 21.12 HU, with meaningful variability between cases.

These values are not directly comparable with the internal improvement deltas, which require a clean reference to quantify correction accuracy. They only indicate that the model produced changes of plausible magnitude and neither collapsed to the identity nor produced extreme distortions in the sampled external cases. Picking out, purely for illustration, the three cases with the highest and the three with the lowest input–output [SSIM](#) shows this variability without being a formal statistical comparison.

Thus, the available external technical evidence supports only that model behavior on real data outside the training domain is qualitatively plausible and consistent in order of magnitude with that observed internally. It does not establish artifact-removal accuracy or validated generalization without a clean reference. The decisive answer to this question—whether human readers judge the external correction to be accurate and safe—depends on the blinded physician evaluation discussed in the next section.

6.5 Reader Evaluation and Clinical Implications

The blinded multi-reader evaluation ([section 5.5](#)) was designed with anatomical visualization as its primary outcome and artifact interference, paired preference, potential-harm flagging, complementary usefulness, and intra-reader consistency as secondary outcomes. This design suits a small panel of readers built specifically to compare input and output side by side.

At the time of writing, data collection and participant characterization for this evaluation had not yet been completed ([section 5.5](#)). For this reason, this section does not interpret perceived improvement, perceived safety, or clinical effectiveness: doing so before the primary and secondary outcomes are available would risk conclusions that exceed the technical evidence assembled to this point. Once completed, the evaluation should be read alongside the preceding section on anatomical preservation and interpreted strictly within the limits of its design—that

is, as a paired input–output comparison by a reader panel, without claiming, beyond what this design can show, that one version is causally better in clinical terms.

6.6 Limitations

Nature of the clean reference. The [CT-MAR](#) Dataset provides real slices with metal artifacts, but no paired clean reference from the same acquisition. The reference used in this work was obtained by structural-similarity matching (*fingerprint* and cosine similarity, automatic-approval threshold 0.994) to independent artifact-free slices from DeepLesion and UCLH Stroke EIT, audited using [MS-SSIM](#) and human visual review (only 3 confirmed false positives among 340 inspected borderline pairs). This strategy enabled training and evaluation against real rather than simulated artifacts, but the matching is similarity-based retrieval rather than guaranteed pixel-level registration from the same acquisition; small residual anatomical differences between matched pairs cannot be fully excluded despite the audit.

Anatomical imbalance. The anatomical distribution was approximately 88% *body* and 12% *head* at the pair level, and 95.9% *body* and only 4.1% *head* (11 of 269 patients) at the patient level. Conclusions specific to the head region therefore rely on a small patient subset and should be interpreted cautiously.

Patient-level sample size. The internal test set’s statistical comparisons treated the patient as the unit of analysis ($n = 41$), which was enough for the main paired comparisons performed, but limited for more detailed subgroup analyses (by anatomical region, artifact severity, or implant type) beyond the exploratory association with artifact burden already identified as non-primary.

Single-environment computational performance. The latency *benchmark* characterizes only the experimental configuration used (one GPU, batch size one, one runtime environment) and does not automatically generalize to other hardware, batch configurations, or the latency of a complete clinical workflow.

External evaluation. The external set is small and purposefully curated (15 main cases), has no clean reference, and is restricted to one model and seed selected after the internal test comparison of the validation-frozen candidates; the results describe change magnitude rather than correction accuracy.

Reader evaluation still in progress. The multi-reader evaluation compares input and output side by side, using a small panel of readers; data collection for it had not been completed at the time of writing, as discussed in the preceding section.

6.7 Implications and Future Work

The most immediate practical implications follow directly from the identified limitations. First, completion and full reporting of the blinded physician evaluation are prerequisites for any claim about perceived clinical benefit or perceived reconstruction safety; no conclusion of this nature should precede those results. Second, prospective validation in larger and more diverse external patient groups—particularly with greater representation of the head region—would test generalization more robustly than the 15 external cases currently available. Third, complementing the structural-similarity matching reference strategy with sources of paired reference from the same acquisition (for example, synthetic artifact insertion in the projection domain or dual-energy data) could reduce reliance on matching between distinct *datasets*, albeit at the cost of potentially less realistic artifacts.

From an architectural perspective, the identified trade-off between the Single Model (a slight advantage in correction magnitude) and the Dual Model (a slight preservation advantage in the Distant Zone) suggests that future work could explore ways to combine both effects—for example, adaptive weighting of reconstruction loss by the predicted artifact mask rather than independent auxiliary supervision—to seek greater correction and greater preservation simultaneously. Finally, expansion of the observer protocol (more readers, calibration or prior-training rounds, and a larger external-case panel) should accompany any future extension of the clinical evaluation, while keeping the same care taken throughout this dissertation to separate exploratory findings from confirmed ones.

Chapter 7

Conclusions

7.1 Study Conclusions

[A preencher: conclusão proporcional às evidências finais e aos resultados confirmatórios.]

7.2 Dissertation Contributions

- *[A preencher: curadoria rastreável e partição por paciente.]*
- *[A preencher: comparação controlada entre duas arquiteturas U-Net.]*
- *[A preencher: avaliação por paciente e produção reprodutível de resultados.]*
- *[A preencher: protocolo de avaliação externa por médicos e respetivo estado de execução.]*

Bibliography

- [1] November 8, 1895: Roentgen's Discovery of X-Rays. <https://www.aps.org/apsnews/2001/11/1895-roentgens-discovery-xrays>. [Cited on page 1.]
- [2] Emico Okuno and Elisabeth Yoshimura. *Física das Radiações*. Oficina de Textos, 2010. [Cited on page 1.]
- [3] JOEL D. HOWELL. EARLY CLINICAL USE OF THE X-RAY. *Transactions of the American Clinical and Climatological Association*, 127:341–349, 2016. [Cited on page 1.]
- [4] Allen Taylor. Illuminating Health: The Evolution and Significance of X-ray Imaging. *Imaging in Medicine*, 15(6):133–134, December 2023. [Cited on page 1.]
- [5] Reza Vali. Unveiling the Marvels of Computed Tomography (CT) Imaging. *Imaging in Medicine*, 16(1):150–151, February 2024. [Cited on page 1.]
- [6] Mohd Noor. An Overview of Computed Tomography: What it is and How it Works? Applications of Computed Tomography. *Imaging in Medicine*, 15(5):93–94, October 2023. [Cited on page 1.]
- [7] Raymond A. Schulz, Jay A. Stein, and Norbert J. Pelc. How CT happened: The early development of medical computed tomography. *Journal of Medical Imaging*, 8(5):052110, September 2021. [Cited on page 2.]
- [8] F. Edward Boas and Dominik Fleischmann. CT artifacts: Causes and reduction techniques. *Imaging in Medicine*, 4(2):229–240, April 2012. [Cited on page 2.]
- [9] Cecile E.J. Kleber, Ramez Karius, Lucas E. Naessens, Coen O. Van Toledo, Jochen A. C. Van Osch, Martijn F. Boomsma, Jan W.T. Heemskerk, and Aart J. Van Der Molen. Advancements in supervised deep learning for metal artifact reduction in computed tomography: A systematic review. *European Journal of Radiology*, 181:111732, December 2024. [Cited on pages 2, 6, 7, and 13.]

- [10] Sathyathas Puvanasunthararajah, Davide Fontanarosa, Marie-Luise Wille, and Saskia M. Camps. The application of metal artifact reduction methods on computed tomography scans for radiotherapy applications: A literature review. *Journal of Applied Clinical Medical Physics*, 22(6):198–223, May 2021. [Cited on page 2.]
- [11] Asutosh Sahu, Shobhit Mathur, Hiroyuki Takaoka, Joji Ota, Felix G. Meinel, Benjamin Böttcher, Benoît Magnin, Ankush Jajodia, and Corey T. Jensen. Transforming CT imaging with deep learning: Noise reduction, artifact management, and clinical applications – A comprehensive review. *European Journal of Radiology Artificial Intelligence*, 4:100042, December 2025. [Cited on pages 2, 6, and 7.]
- [12] Mohanad Alkhodari, Eman Alefishat, Herbert F. Jelinek, Ahmed Kaabneh, and Panos Liatsis. Enhancing CCTA image quality: A review of deep learning approaches for advanced artifact correction and denoising. *Artificial Intelligence Review*, 58(10):331, August 2025. [Cited on page 3.]
- [13] Julia F. Barrett and Nicholas Keat. Artifacts in CT: Recognition and Avoidance. *RadioGraphics*, 24(6):1679–1691, November 2004. [Cited on page 6.]
- [14] Mark Selles, Jochen A.C. Van Osch, Mario Maas, Martijn F. Boomsma, and Ruud H.H. Wellenberg. Advances in metal artifact reduction in CT images: A review of traditional and novel metal artifact reduction techniques. *European Journal of Radiology*, 170:111276, January 2024. [Cited on page 6.]
- [15] Lars Gjestebj, Qingsong Yang, Yan Xi, Bernhard E. H. Claus, Yannan Jin, Bruno De Man, Ge Wang, and Hongming Shan. Deep learning methods for CT image-domain metal artifact reduction. In Bert Müller and Ge Wang, editors, *Developments in X-Ray Tomography XI*, page 31, San Diego, United States, September 2017. SPIE. [Cited on pages 6 and 7.]
- [16] Yanbo Zhang and Hengyong Yu. Convolutional Neural Network Based Metal Artifact Reduction in X-Ray Computed Tomography. *IEEE Transactions on Medical Imaging*, 37(6):1370–1381, June 2018. [Cited on pages 6 and 7.]
- [17] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional Networks for Biomedical Image Segmentation, May 2015. [Cited on pages 7 and 17.]
- [18] Esther Meyer, Frank Bergner, Rainer Raupach, Thomas Flohr, and Marc Kachelriess. Normalized metal artifact reduction (NMAR) in computed tomography. In *2009 IEEE Nuclear Science Symposium Conference Record (NSS/MIC)*, pages 3251–3255, Orlando, FL, October 2009. IEEE. [Cited on page 7.]

- [19] Wei-An Lin, Haofu Liao, Cheng Peng, Xiaohang Sun, Jingdan Zhang, Jiebo Luo, Rama Chellappa, and Shaohua Kevin Zhou. DuDoNet: Dual Domain Network for CT Metal Artifact Reduction. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10504–10513, Long Beach, CA, USA, June 2019. IEEE. [Cited on pages 7 and 13.]
- [20] Lequan Yu, Zhicheng Zhang, Xiaomeng Li, Hongyi Ren, Wei Zhao, and Lei Xing. Metal Artifact Reduction in 2D CT Images with Self-supervised Cross-domain Learning. *Physics in Medicine & Biology*, 66(17):175003, September 2021. [Cited on page 7.]
- [21] Zhiwei Huang, Guo Zhang, Jinzhao Lin, Yu Pang, Huiqian Wang, Tong Bai, and Lisha Zhong. Multi-modal feature-fusion for CT metal artifact reduction using edge-enhanced generative adversarial networks. *Computer Methods and Programs in Biomedicine*, 217:106700, April 2022. [Cited on page 7.]
- [22] Xin Yi, Ekta Walia, and Paul Babyn. Generative Adversarial Network in Medical Imaging: A Review. *Medical Image Analysis*, 58:101552, December 2019. [Cited on page 7.]
- [23] Zhou Wang, A.C. Bovik, H.R. Sheikh, and E.P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, April 2004. [Cited on pages 8, 11, and 13.]
- [24] Z. Wang, E.P. Simoncelli, and A.C. Bovik. Multiscale structural similarity for image quality assessment. In *The Thrity-Seventh Asilomar Conference on Signals, Systems & Computers, 2003*, pages 1398–1402, Pacific Grove, CA, USA, 2003. IEEE. [Cited on page 8.]
- [25] Alain Hore and Djemel Ziou. Image Quality Metrics: PSNR vs. SSIM. In *2010 20th International Conference on Pattern Recognition*, pages 2366–2369, Istanbul, Turkey, August 2010. IEEE. [Cited on page 8.]
- [26] John R. Zech, Marcus A. Badgeley, Manway Liu, Anthony B. Costa, Joseph J. Titano, and Eric Karl Oermann. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. *PLOS Medicine*, 15(11):e1002683, November 2018. [Cited on page 8.]
- [27] Martin J. Willemink, Wojciech A. Koszek, Cailin Hardell, Jie Wu, Dominik Fleischmann, Hugh Harvey, Les R. Folio, Ronald M. Summers, Daniel L. Rubin, and Matthew P. Lungren. Preparing Medical Imaging Data for Machine Learning. *Radiology*, 295(1):4–15, April 2020. [Cited on page 8.]

- [28] Christopher J. Kelly, Alan Karthikesalingam, Mustafa Suleyman, Greg Corrado, and Dominic King. Key challenges for delivering clinical impact with artificial intelligence. *BMC Medicine*, 17(1):195, December 2019. [Cited on page 8.]
- [29] Eri Haneda, Nils Peters, Jiayong Zhang, Grigorios Karageorgos, Wenjun Xia, Harald Paganetti, Ge Wang, Yi Guo, Jianhua Ma, Hyoung Suk Park, Kiwan Jeon, Fuxin Fan, Mareike Thies, and Bruno De Man. AAPM CT metal artifact reduction grand challenge. *Medical Physics*, 52(10):e70050, October 2025. [Cited on page 9.]
- [30] xcist. AAPM CT Metal Artifact Reduction (CT-MAR) Grand Challenge Benchmark Tool. https://github.com/xcist/example/tree/main/AAPM_datachallenge/. [Cited on page 9.]
- [31] Nils Peters, Eri Haneda, Jiayong Zhang, Grigorios Karageorgos, Wenjun Xia, Joost Verburg, Ge Wang, Harald Paganetti, and Bruno De Man. A hybrid training database and evaluation benchmark for assessing metal artifact reduction methods for X-ray CT imaging. *Medical Physics*, 52(10):e70020, October 2025. [Cited on page 9.]
- [32] Mingye Wu, Paul FitzGerald, Jiayong Zhang, W. Paul Segars, Hengyong Yu, Yongshun Xu, and Bruno De Man. XCIST – An Open Access X-ray/CT Simulation Toolkit. *Physics in medicine and biology*, 67(19):10.1088/1361-6560/ac9174, September 2022. [Cited on page 9.]
- [33] Nir Goren, James Avery, Thomas Dowrick, Eleanor Mackle, Anna Witkowska-Wrobel, David Werring, and David Holder. Multi-frequency electrical impedance tomography and neuroimaging data in stroke patients. *Scientific Data*, 5:180112, July 2018. [Cited on pages 9 and 10.]
- [34] Nir Goren, Thomas Dowrick, James Avery, and David Holder. UCLH Stroke EIT Dataset - Radiology Data, August 2017. [Cited on page 9.]
- [35] Ke Yan, Xiaosong Wang, Le Lu, and Ronald M. Summers. DeepLesion: Automated mining of large-scale lesion annotations and universal lesion detection with deep learning. *Journal of Medical Imaging*, 5(3):036501, July 2018. [Cited on pages 9 and 10.]
- [36] Mustafa Umit Oner, Yi-Chih Cheng, Hwee Kuan Lee, and Wing-Kin Sung. Training machine learning models on patient level data segregation is crucial in practical clinical applications, April 2020. [Cited on page 10.]

- [37] Jiang Hsieh. *Computed Tomography: Principles, Design, Artifacts, and Recent Advances*. Wiley Interscience ; SPIE Press, Hoboken, N.J. : Bellingham, Wash, 2nd ed edition, 2009. [Cited on page 11.]
- [38] Saeid Asgari Taghanaki, Yefeng Zheng, S. Kevin Zhou, Bogdan Georgescu, Puneet Sharma, Daguang Xu, Dorin Comaniciu, and Ghassan Hamarneh. Combo Loss: Handling Input and Output Imbalance in Multi-Organ Segmentation, September 2021. [Cited on pages 16 and 22.]
- [39] Uğur Kesimal, Habip Eser Akkaya, Önder Polat, and Murat Sağlam. Use of Deep Learning Models in the Diagnosis of Proptosis Through Orbital Magnetic Resonance Imaging. *Medical Science Monitor*, 32, April 2026. [Cited on page 16.]
- [40] Yang Li, Bobo Yan, Jianxin Hou, Bingyang Bai, Xiaoyu Huang, Canfei Xu, and Limei Fang. UNet based on dynamic convolution decomposition and triplet attention. *Scientific Reports*, 14(1):271, January 2024. [Cited on page 17.]
- [41] Rich Caruana. Multitask Learning. [Cited on page 18.]
- [42] Sebastian Ruder. An Overview of Multi-Task Learning in Deep Neural Networks, June 2017. [Cited on page 18.]
- [43] Hang Zhao, Orazio Gallo, Iuri Frosio, and Jan Kautz. Loss Functions for Image Restoration With Neural Networks. *IEEE Transactions on Computational Imaging*, 3(1):47–57, March 2017. [Cited on page 21.]
- [44] Lin Zhang, Lei Zhang, Xuanqin Mou, and D. Zhang. FSIM: A Feature Similarity Index for Image Quality Assessment. *IEEE Transactions on Image Processing*, 20(8):2378–2386, August 2011. [Cited on page 22.]
- [45] Kenneth Clark, Bruce Vendt, Kirk Smith, John Freymann, Justin Kirby, Paul Koppel, Stephen Moore, Stanley Phillips, David Maffitt, Michael Pringle, Lawrence Tarbox, and Fred Prior. The Cancer Imaging Archive (TCIA): Maintaining and operating a public information repository. *Journal of Digital Imaging*, 26(6):1045–1057, December 2013. [Cited on page 27.]
- [46] B. Albertina, M. Watson, C. Holback, R. Jarosz, S. Kirk, Y. Lee, K. Rieger-Christ, and J. Lemmerman. The Cancer Genome Atlas Lung Adenocarcinoma Collection (TCGA-LUAD), 2016. [Cited on page 27.]

- [47] Samuel G. Armato, Geoffrey McLennan, Luc Bidaut, Michael F. McNitt-Gray, Charles R. Meyer, Anthony P. Reeves, Binsheng Zhao, Denise R. Aberle, Claudia I. Henschke, Eric A. Hoffman, Ella A. Kazerooni, Heber MacMahon, Edwin J. R. Van Beek, David Yankelevitz, Alberto M. Biancardi, Peyton H. Bland, Matthew S. Brown, Roger M. Engelmann, Gary E. Laderach, Daniel Max, Richard C. Pais, David P.-Y. Qing, Rachael Y. Roberts, Amanda R. Smith, Adam Starkey, Poonam Batra, Philip Caligiuri, Ali Farooqi, Gregory W. Gladish, C. Matilda Jude, Reginald F. Munden, Iva Petkovska, Leslie E. Quint, Lawrence H. Schwartz, Baskaran Sundaram, Lori E. Dodd, Charles Fenimore, David Gur, Nicholas Petrick, John Freymann, Justin Kirby, Brian Hughes, Alessi Vande Casteele, Sangeeta Gupte, Maha Sallam, Michael D. Heath, Michael H. Kuhn, Ekta Dharaiya, Richard Burns, David S. Fryd, Marcos Salganicoff, Vikram Anand, Uri Shreter, Stephen Vastagh, Barbara Y. Croft, and Laurence P. Clarke. Data from LIDC-IDRI, 2015. [Cited on page 27.]
- [48] The Research for Precision Oncology Program and the Applied Proteogenomics Organizational Learning and Outcomes (APOLLO) Research Network. VA Research Precision Oncology Program – APOLLO (VAREPOP-APOLLO), 2024. [Cited on page 27.]
- [49] National Cancer Institute Clinical Proteomic Tumor Analysis Consortium (CPTAC). The Clinical Proteomic Tumor Analysis Consortium Sarcomas Collection (CPTAC-SAR), 2019. [Cited on page 27.]
- [50] National Cancer Institute Clinical Proteomic Tumor Analysis Consortium (CPTAC). The Clinical Proteomic Tumor Analysis Consortium Pancreatic Ductal Adenocarcinoma Collection (CPTAC-PDA), 2018. [Cited on page 27.]

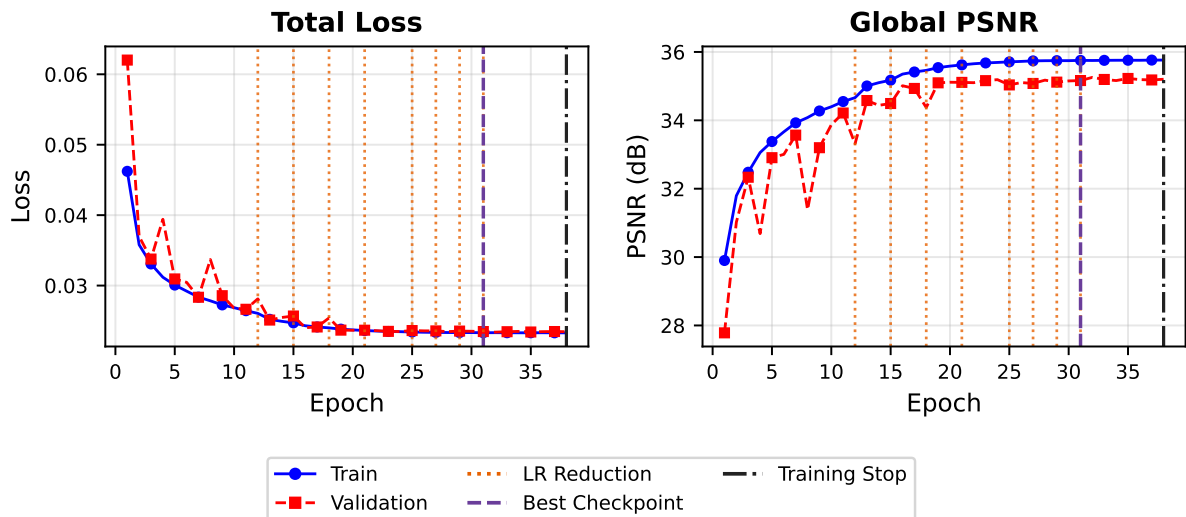
Appendix A

Supplementary Internal Evaluation

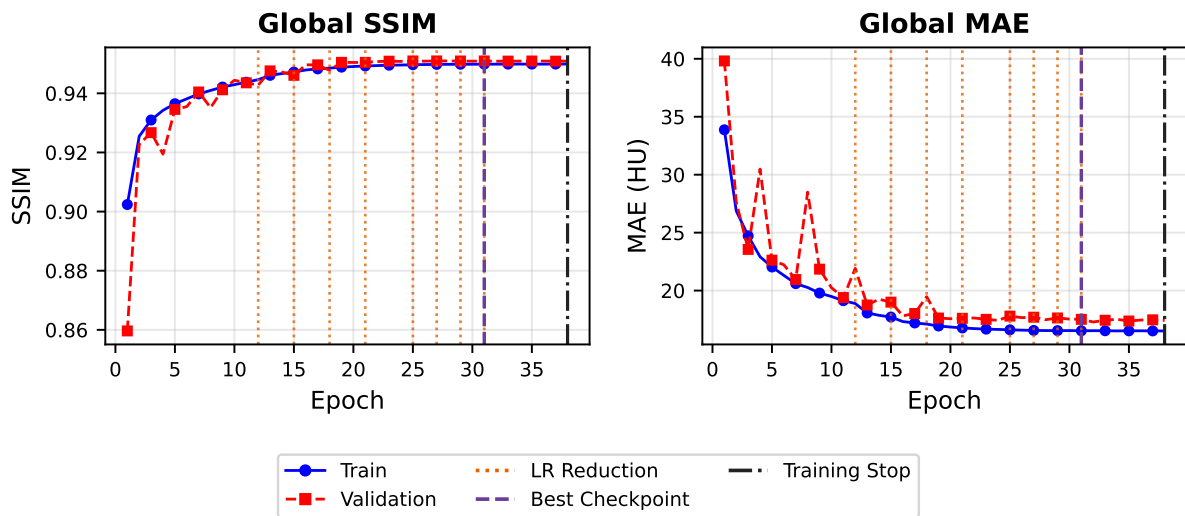
A.1 Training Convergence Across Runs

This section provides the complete convergence documentation for the six deterministic training runs. Each run’s checkpoint was selected exclusively from validation reconstruction [SSIM](#), as specified in Materials and Methods. The six frozen checkpoints were subsequently compared once on the internal test partition to select the external-evaluation candidate by Artifact-ROI Δ [SSIM](#), with a descriptive Distant-Zone preservation safeguard. The figures and tables below document per-run checkpoint selection and did not, by themselves, determine the final external-evaluation candidate.

Single Model — Training Convergence



Single Model — Training Convergence



Single Model — Training Convergence

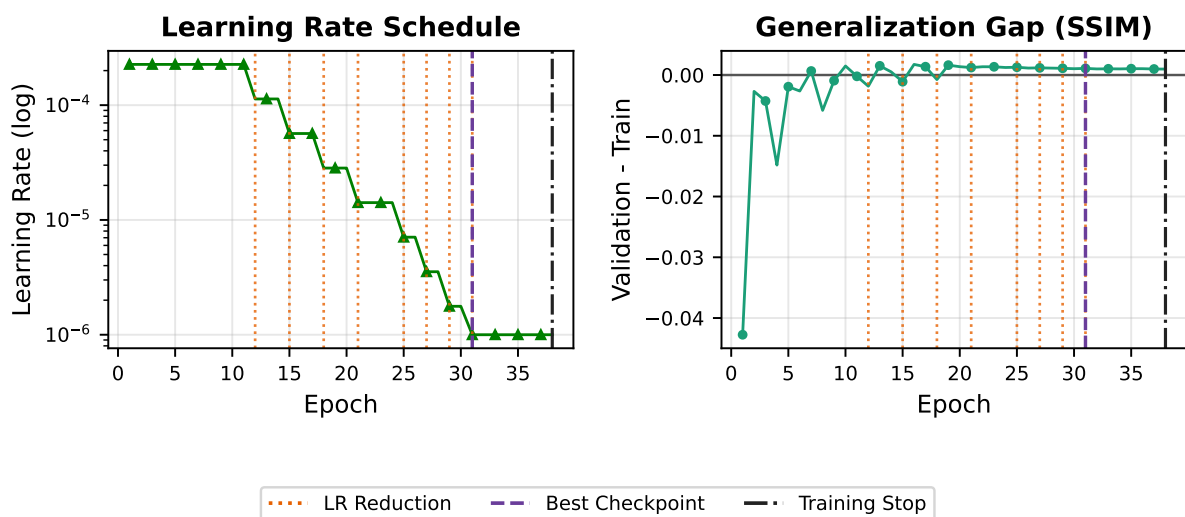
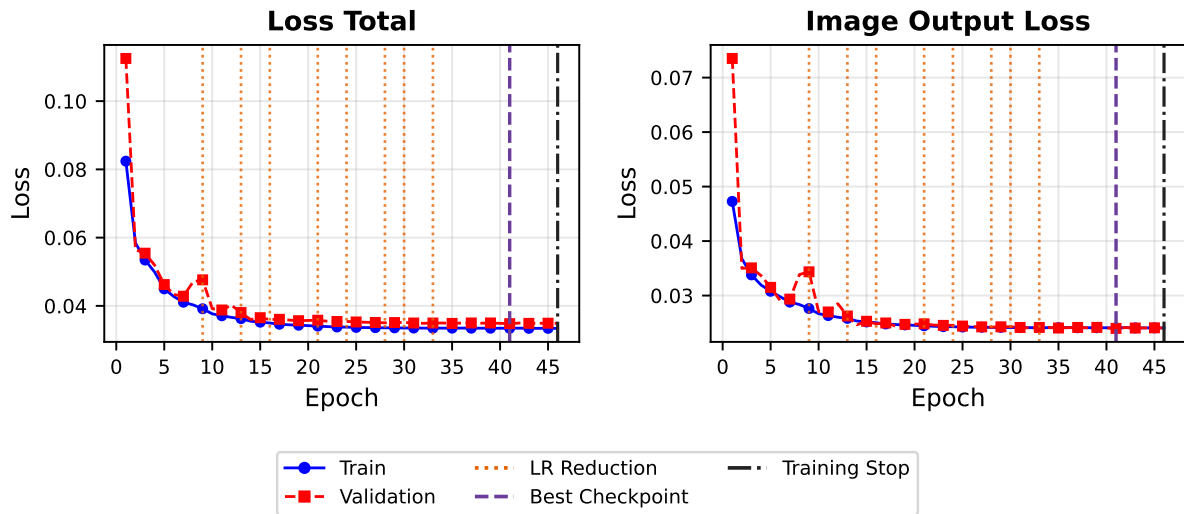
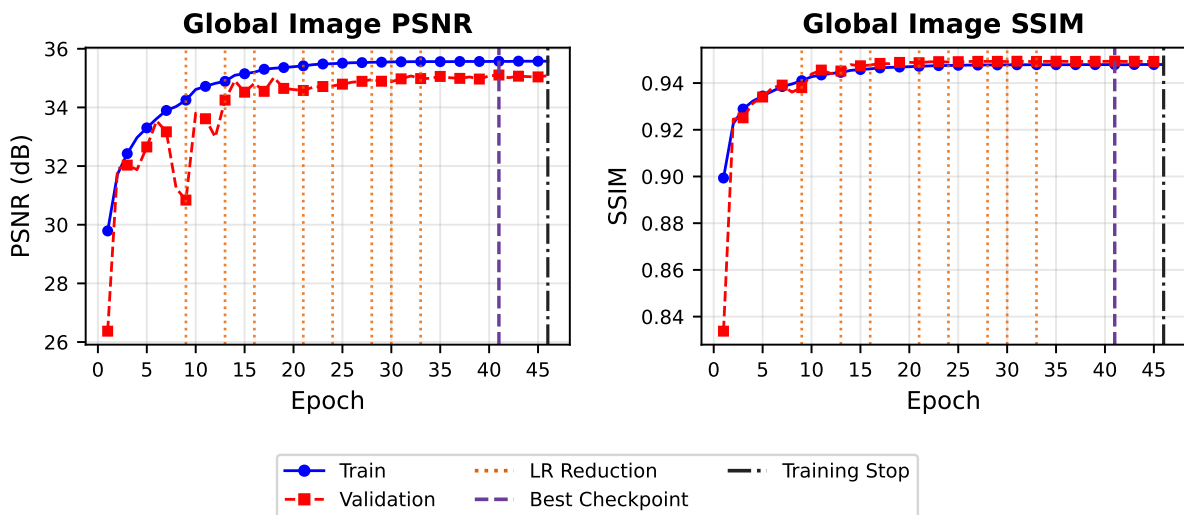


Fig.A.1 Training convergence for the Single Model, seed 42. Each chart pair repeats the shared legend so it remains interpretable if the group spans multiple pages. Orange dotted vertical lines indicate learning-rate reductions; the purple dashed line marks the epoch of the best validation checkpoint; and the black dash-dotted line marks training termination. The *generalization-gap* panel reports validation minus training SSIM and supports inspection for persistent divergence compatible with overfitting.

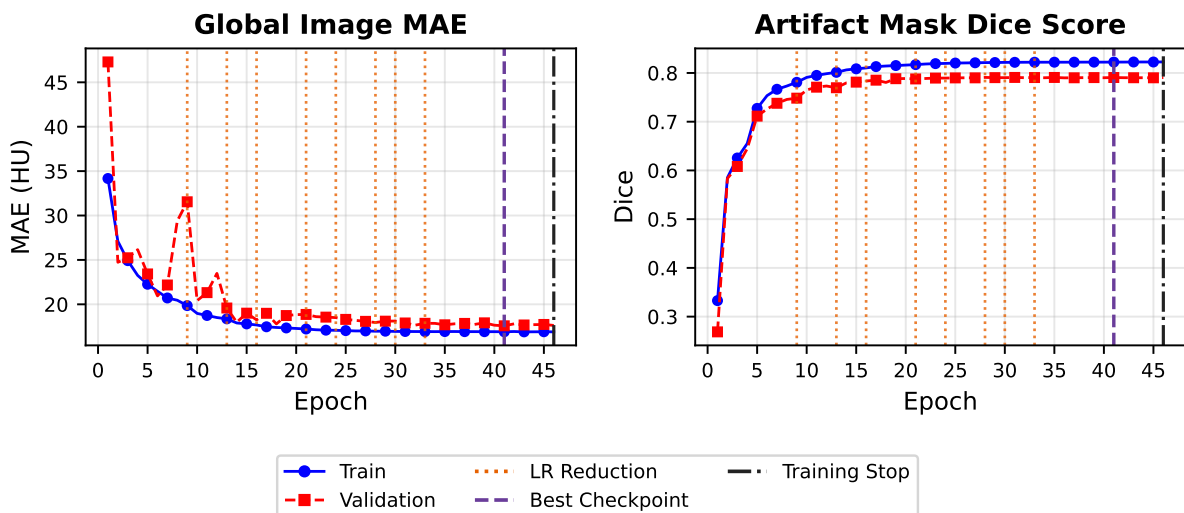
Dual Model — Training Convergence



Dual Model — Training Convergence



Dual Model — Training Convergence



Dual Model — Training Convergence

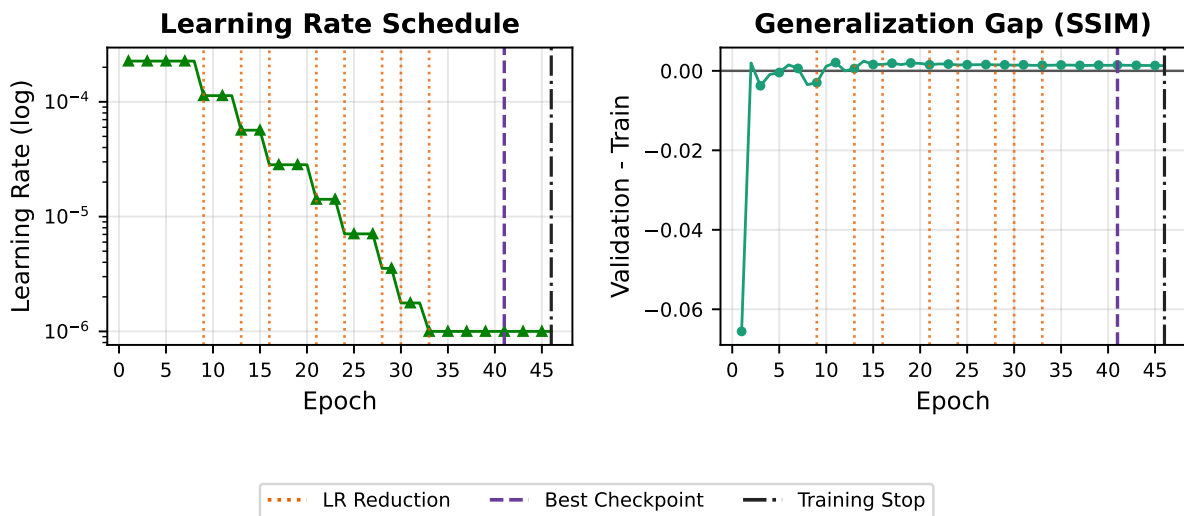
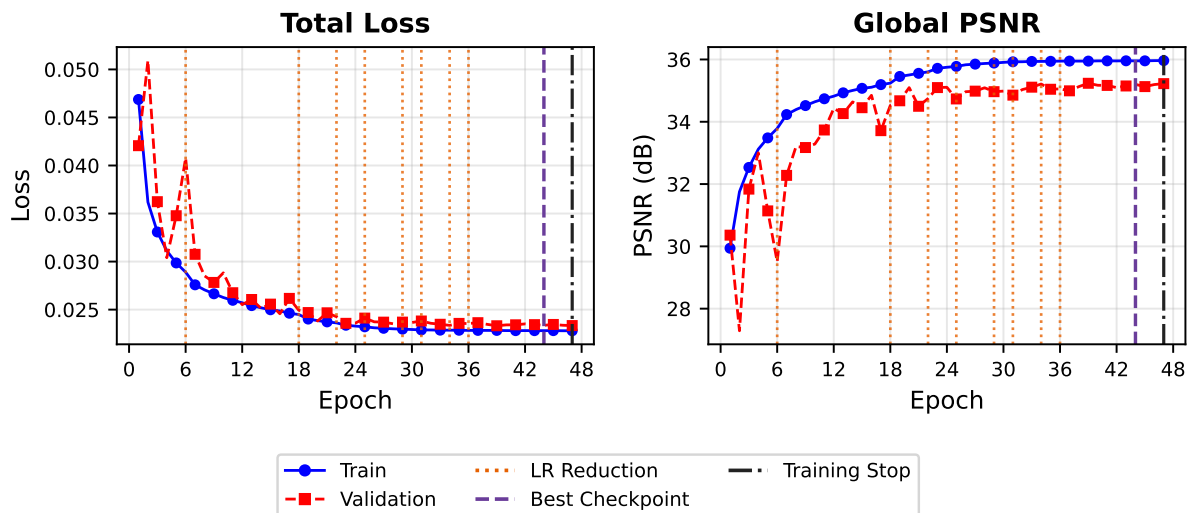
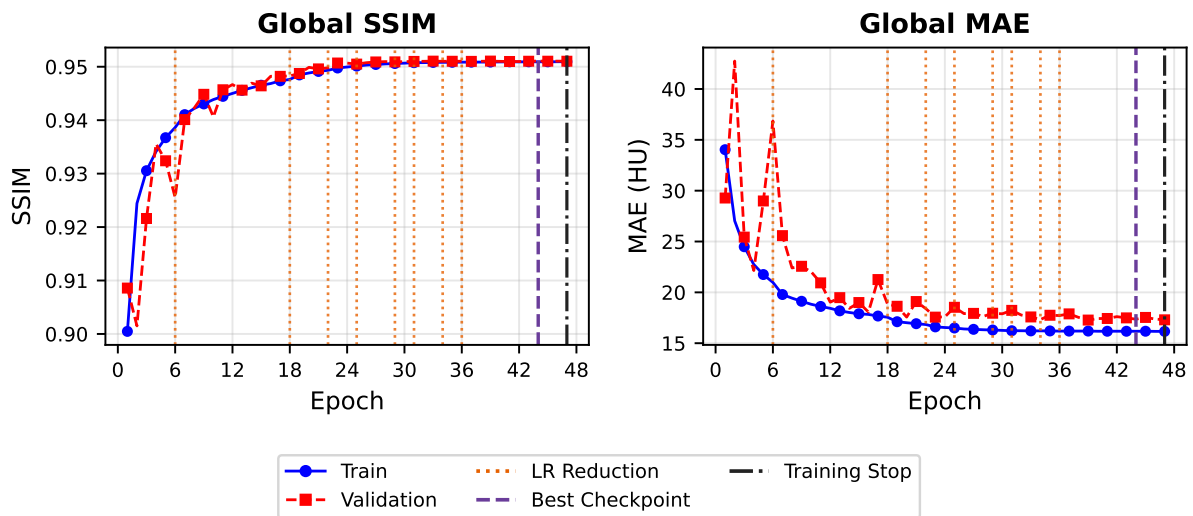


Fig.A.2 Training convergence for the Dual Model, seed 42. Each chart pair repeats the shared legend so it remains interpretable if the group spans multiple pages. Orange dotted vertical lines indicate learning-rate reductions; the purple dashed line marks the epoch of the best validation checkpoint; and the black dash-dotted line marks training termination. The *generalization-gap* panel reports validation minus training SSIM and supports inspection for persistent divergence compatible with overfitting.

Single Model — Training Convergence



Single Model — Training Convergence



Single Model — Training Convergence

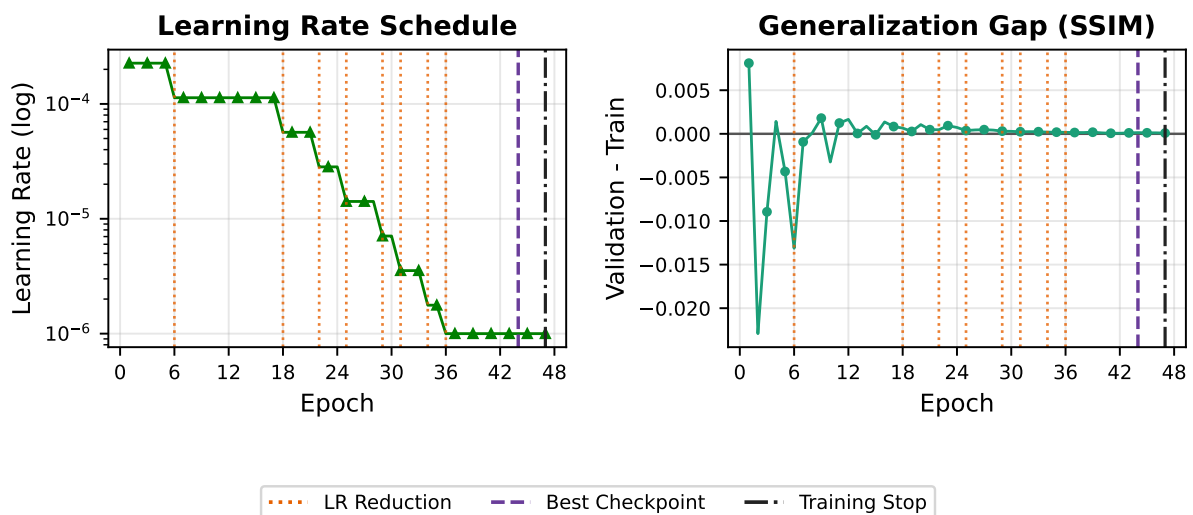
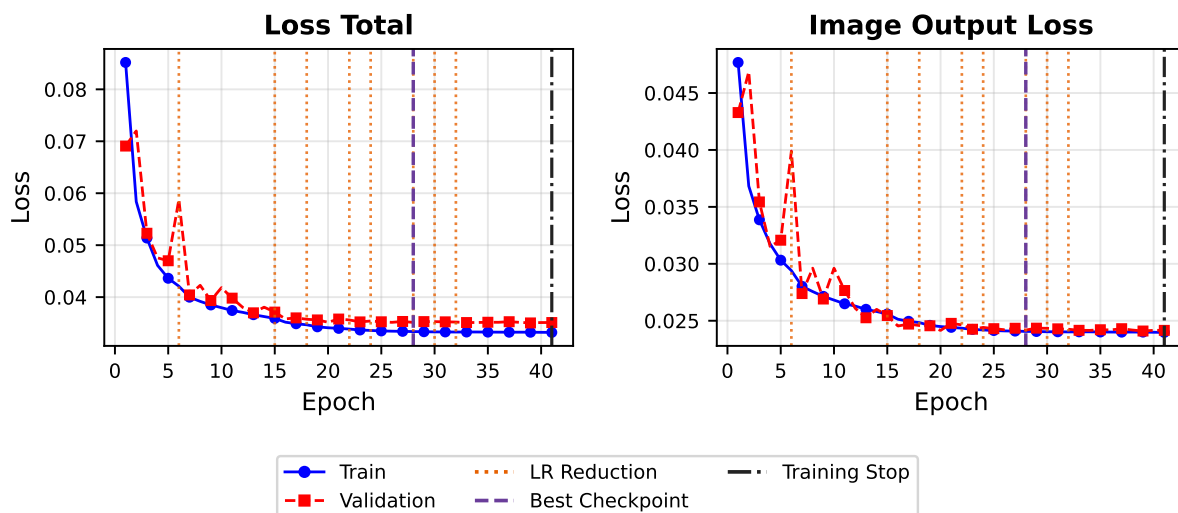
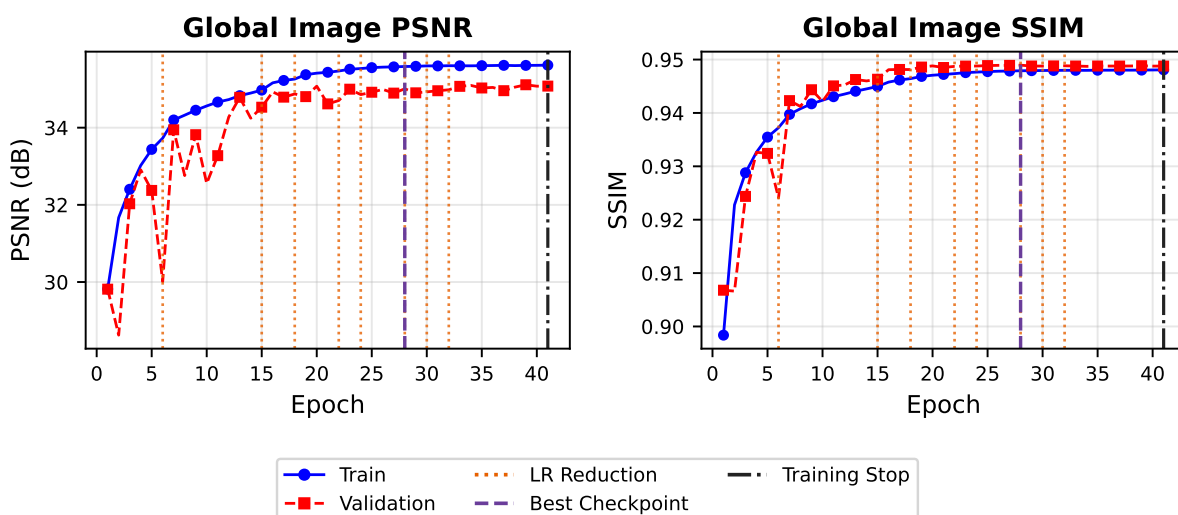


Fig.A.3 Training convergence for the Single Model, seed 1337. Each chart pair repeats the shared legend so it remains interpretable if the group spans multiple pages. Orange dotted vertical lines indicate learning-rate reductions; the purple dashed line marks the epoch of the best validation checkpoint; and the black dash-dotted line marks training termination. The *generalization-gap* panel reports validation minus training SSIM and supports inspection for persistent divergence compatible with overfitting.

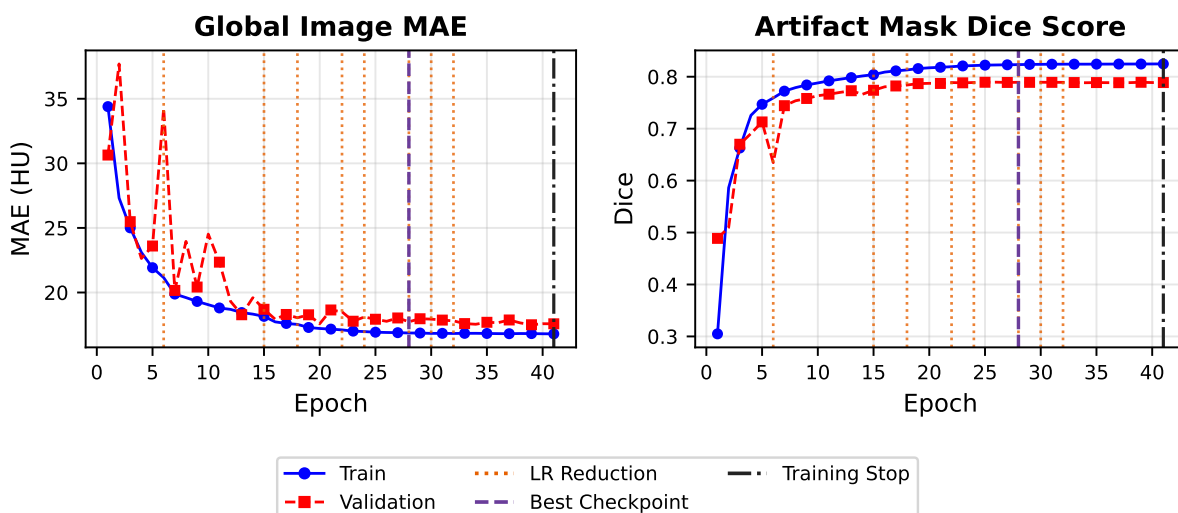
Dual Model — Training Convergence



Dual Model — Training Convergence



Dual Model — Training Convergence



Dual Model — Training Convergence

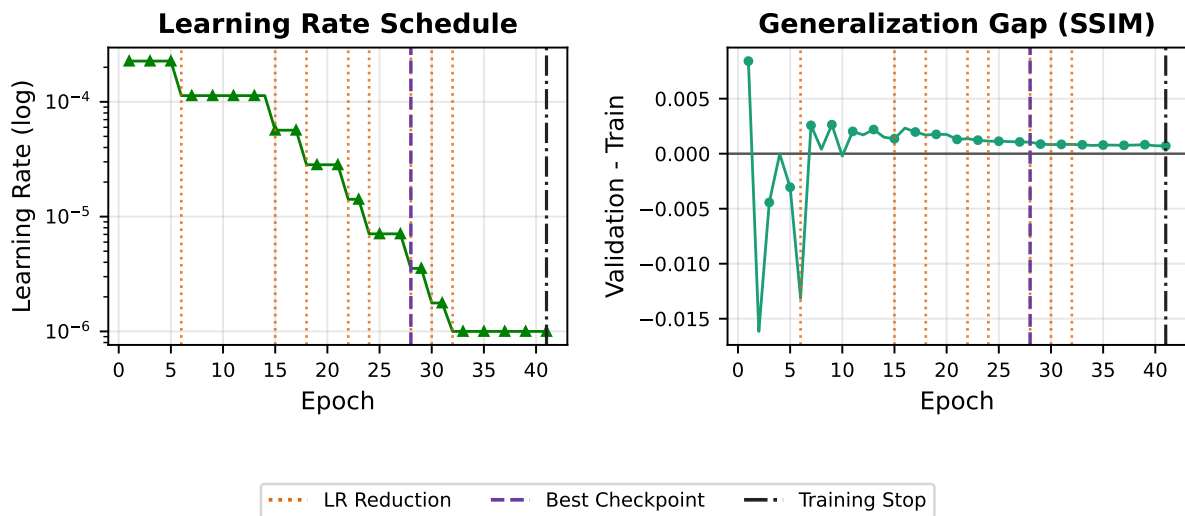
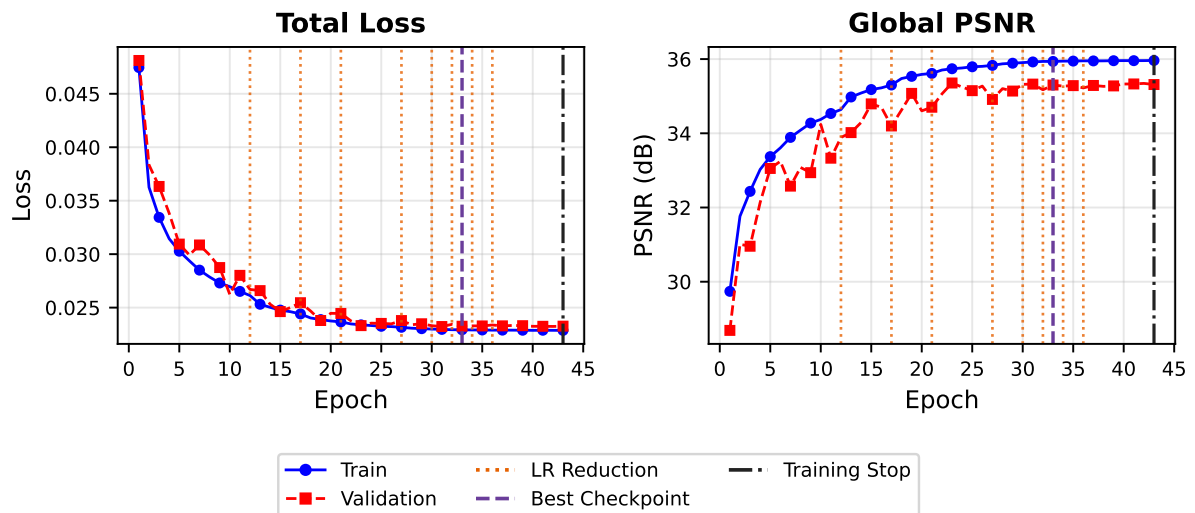
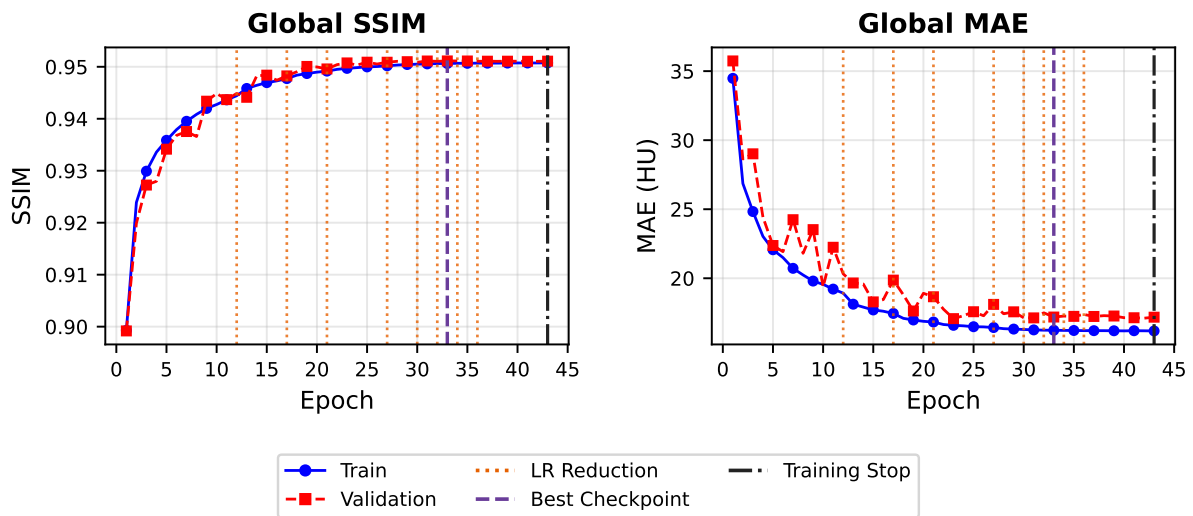


Fig.A.4 Training convergence for the Dual Model, seed 1337. Each chart pair repeats the shared legend so it remains interpretable if the group spans multiple pages. Orange dotted vertical lines indicate learning-rate reductions; the purple dashed line marks the epoch of the best validation checkpoint; and the black dash-dotted line marks training termination. The *generalization-gap* panel reports validation minus training SSIM and supports inspection for persistent divergence compatible with overfitting.

Single Model — Training Convergence



Single Model — Training Convergence



Single Model — Training Convergence

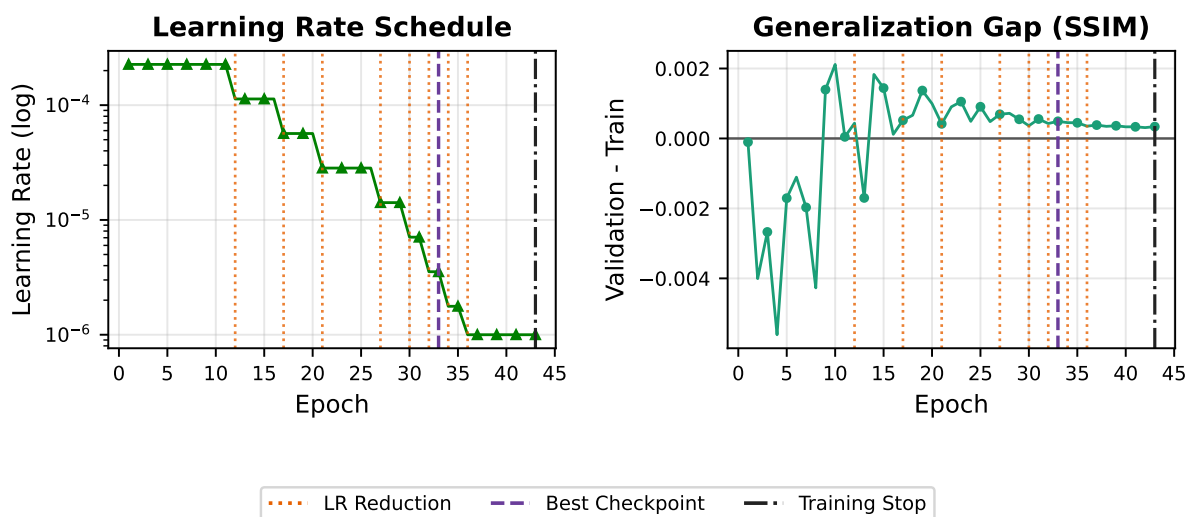
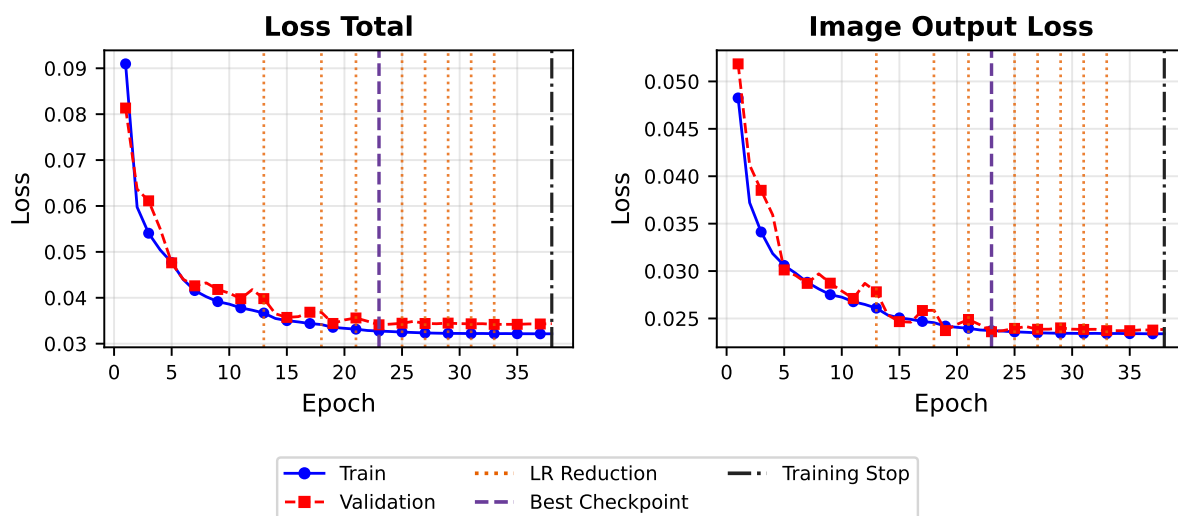
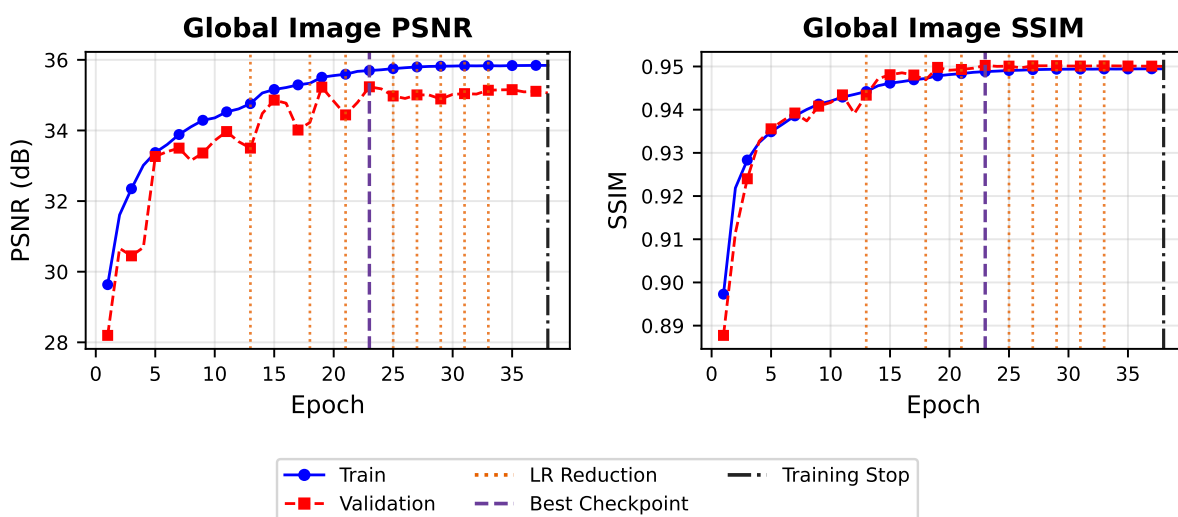


Fig.A.5 Training convergence for the Single Model, seed 2024. Each chart pair repeats the shared legend so it remains interpretable if the group spans multiple pages. Orange dotted vertical lines indicate learning-rate reductions; the purple dashed line marks the epoch of the best validation checkpoint; and the black dash-dotted line marks training termination. The *generalization-gap* panel reports validation minus training SSIM and supports inspection for persistent divergence compatible with overfitting.

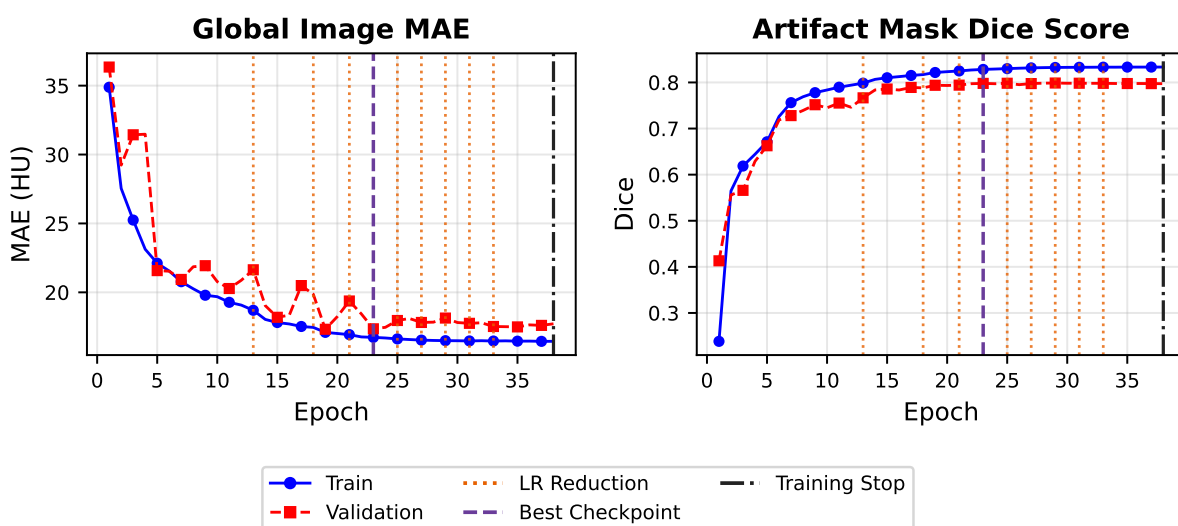
Dual Model — Training Convergence



Dual Model — Training Convergence



Dual Model — Training Convergence



Dual Model — Training Convergence

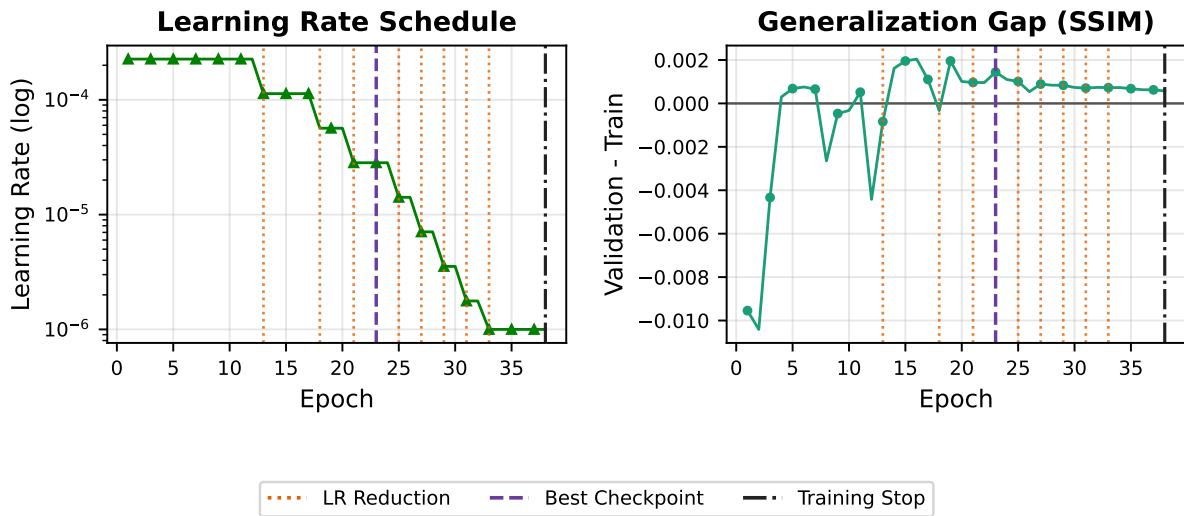


Fig.A.6 Training convergence for the Dual Model, seed 2024. Each chart pair repeats the shared legend so it remains interpretable if the group spans multiple pages. Orange dotted vertical lines indicate learning-rate reductions; the purple dashed line marks the epoch of the best validation checkpoint; and the black dash-dotted line marks training termination. The *generalization-gap* panel reports validation minus training SSIM and supports inspection for persistent divergence compatible with overfitting.

Best SSIM (HU)			
Seed	42	1337	2024
Single Model	0.95096 (31)*	0.95106 (44)	0.95113 (33)
Dual Model	0.94938 (41)	0.94895 (28)	0.95025 (23)

*Values in parentheses indicate the corresponding epoch.

Table A.1 Highest validation SSIM by model and seed, with the corresponding epoch. For the Dual Model, SSIM was calculated on the reconstructed-image output.

PSNR (dB) - Same Best SSIM Epoch			
Seed	42	1337	2024
Single Model	35.16365 (31)	35.20020 (44)	35.30099 (33)
Dual Model	35.10422 (41)	35.00705 (28)	35.23622 (23)

Table A.2 Validation PSNR at the epoch selected by the highest validation SSIM, by model and seed. For the Dual Model, both metrics were calculated on the reconstructed-image output.

MAE (HU) - Same Best SSIM Epoch			
Seed	42	1337	2024
Single Model	17.52002 (31)	17.36296 (44)	17.19180 (33)
Dual Model	17.58006 (41)	17.73584 (28)	17.35387 (23)

Table A.3 Validation MAE at the epoch selected by the highest validation SSIM, by model and seed. For the Dual Model, both metrics were calculated on the reconstructed-image output.

This appendix presents complementary quantitative checks for the internal evaluation. The following figure is included solely as a visual diagnostic of pixel-level radiometric agreement

and does not constitute inferential analysis, because sampled pixels are not independent observations.

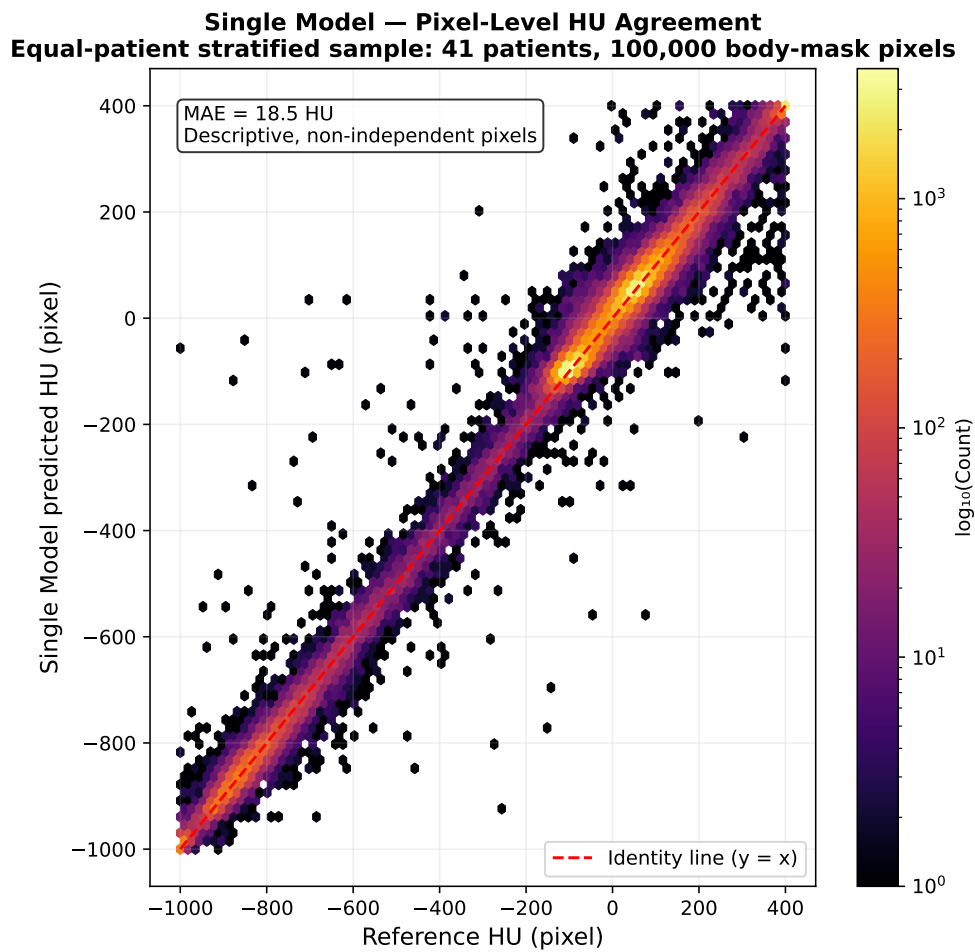


Fig.A.7 Pixel-level HU agreement between the Single Model reconstruction and the clean reference. The hexbin uses a fixed equal-patient stratified sample of 100,000 valid body-mask pixels from 41 test-set patients. It is a descriptive calibration diagnostic; no hypothesis test or confidence interval is derived from the sampled pixels.

The following plots complement the global-delta histograms presented in the main text. Each point represents one test slice; accordingly, these plots were not used for statistical inference.

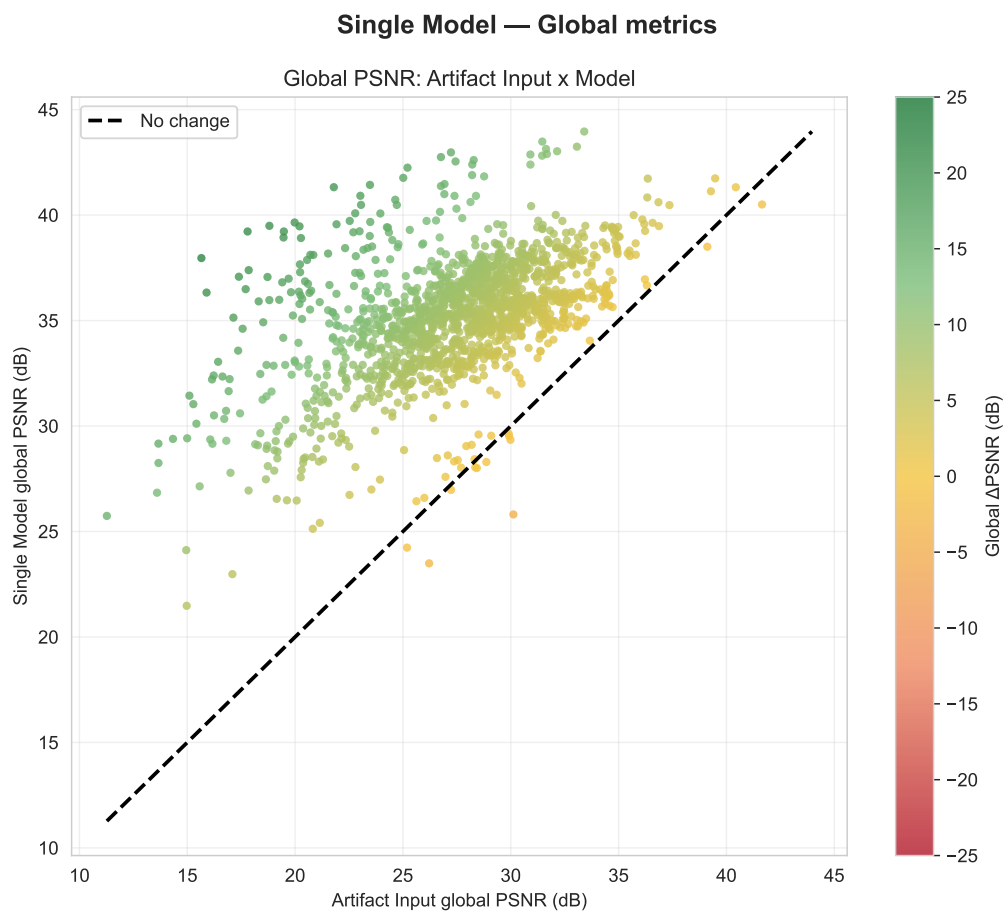
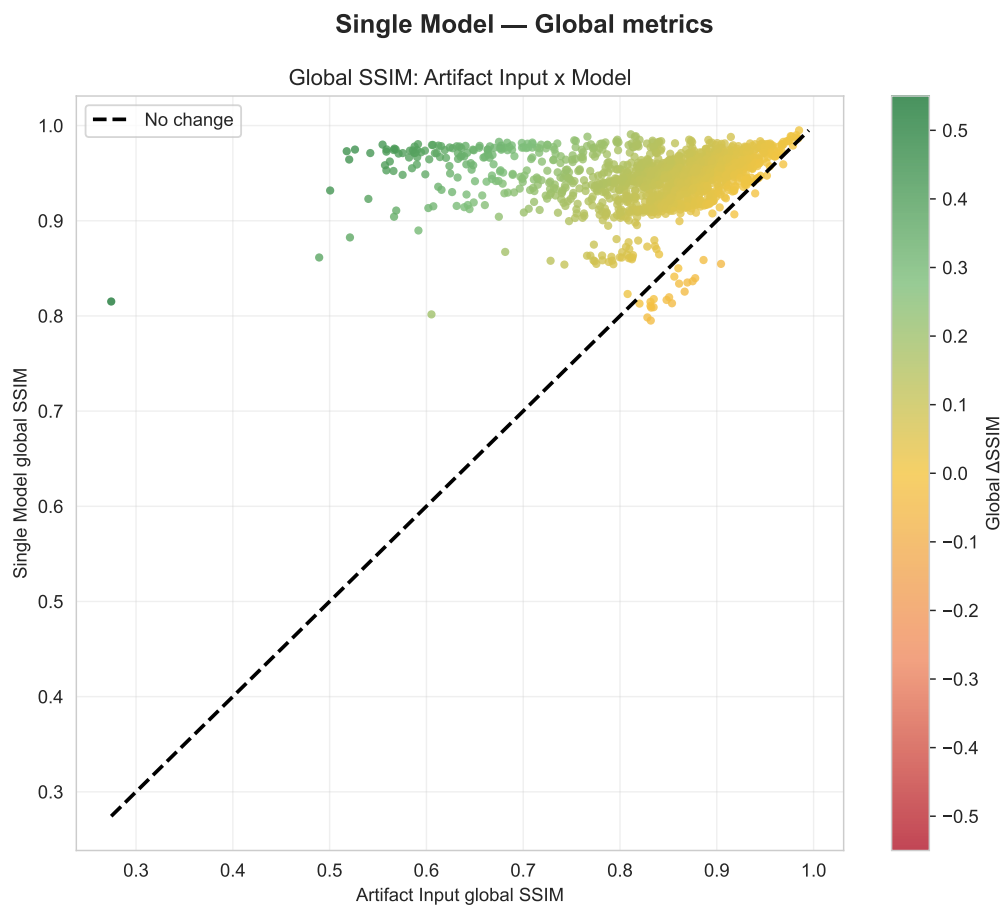
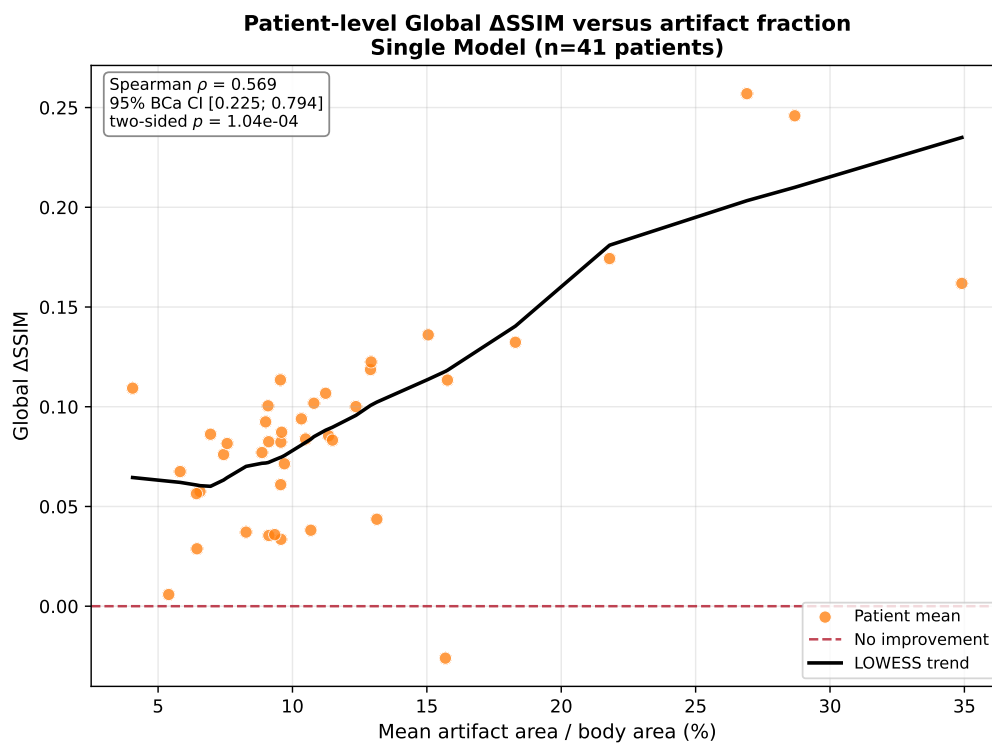


Fig.A.8 Per-image global SSIM (top) and PSNR (bottom) for the corrupted input and the Single Model reconstruction, seed 2024. The identity line denotes no change, marker color encodes the corresponding global delta. These slice-level relationships are descriptive, whereas statistical inference used patient-level aggregation.

As an exploratory analysis of heterogeneity, [Figure A.9](#) relates artifact burden to global improvement. Each point represents the mean across all slices of one patient ($n = 41$), giving each patient equal weight. Artifact burden was defined as the mean artifact-area/body-area fraction per slice; deltas correspond to the reconstruction metric minus the input metric, both calculated against the clean reference. Two-sided Spearman correlations were calculated with paired BCa bootstrap 95% confidence intervals (10 000 resamples; seed 1337), and the trend curve is a LOWESS estimate from *Statsmodels*. This analysis was not part of the primary hypothesis-testing family and did not receive Holm adjustment. A positive association was observed between artifact burden and improvement: $\rho = 0,479$ for ΔPSNR (95% CI [0,156; 0,717], $p = 0,00154$) and $\rho = 0,569$ for ΔSSIM (95% CI [0,225; 0,794], $p = 0,000104$).



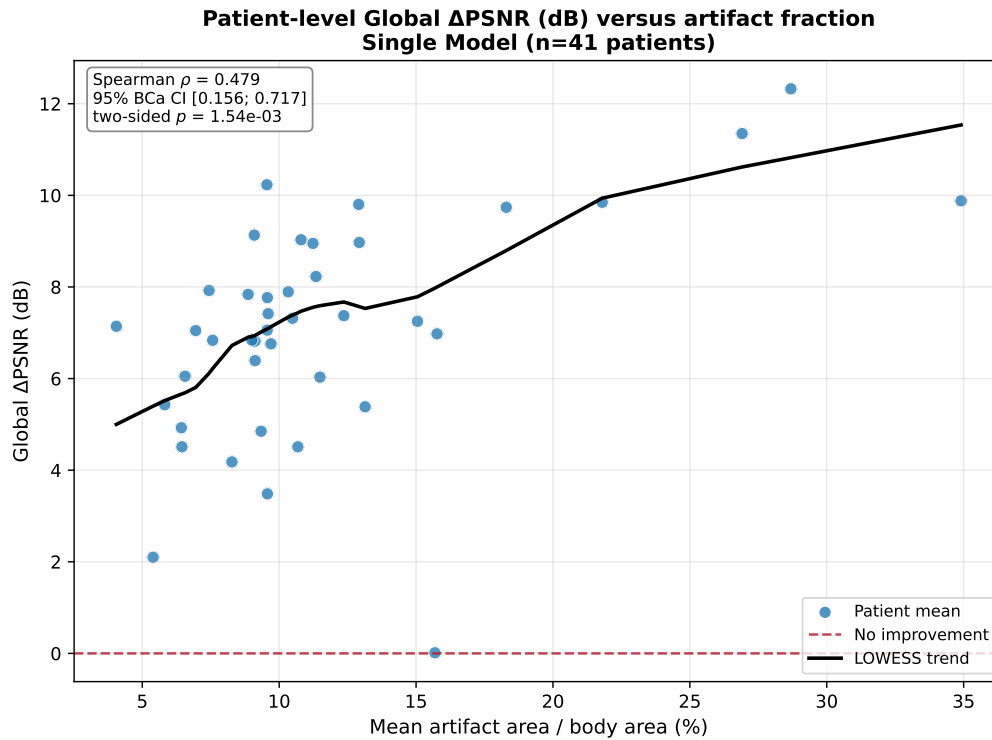


Fig.A.9 Exploratory patient-level association between mean artifact fraction and global improvement for the Single Model, seed 2024: Δ SSIM (top) and Δ PSNR (bottom). Each point is one patient's mean across its test-set slices ($n = 41$). Lines show Statsmodels LOWESS trends; annotations report two-sided Spearman correlations and paired BCa bootstrap 95% confidence intervals.

A.2 Worst Performance Cases

As a complementary characterization of lower-performing slices, two slice-level groups were identified for the Single Model, seed 2024, in the test set: first by the bottom fifth percentile of global Δ SSIM and then, complementarily, by the bottom fifth percentile of global Δ PSNR. This slice-level classification is purely descriptive: it does not constitute a hypothesis test or reproduce the patient-level candidate-selection analysis, whose primary endpoint was Artifact-ROI Δ SSIM, and it serves only to quantitatively situate the slice groups whose individual visual examples are discussed in [subsection 5.3.4](#).

Using the global Δ SSIM criterion ($N = 74/1478$ slices; threshold = 0,0146), mean Δ PSNR improvement remained positive (1,88 dB), but mean Δ SSIM improvement was slightly negative ($-0,0019$), with a mean Artifact-Zone MAE reduction of 65,2 HU ([Figure A.10](#)). Thus, the most unfavorable subset according to this descriptive global criterion included a small mean SSIM degradation, even though most remaining slices continued to improve substantially.

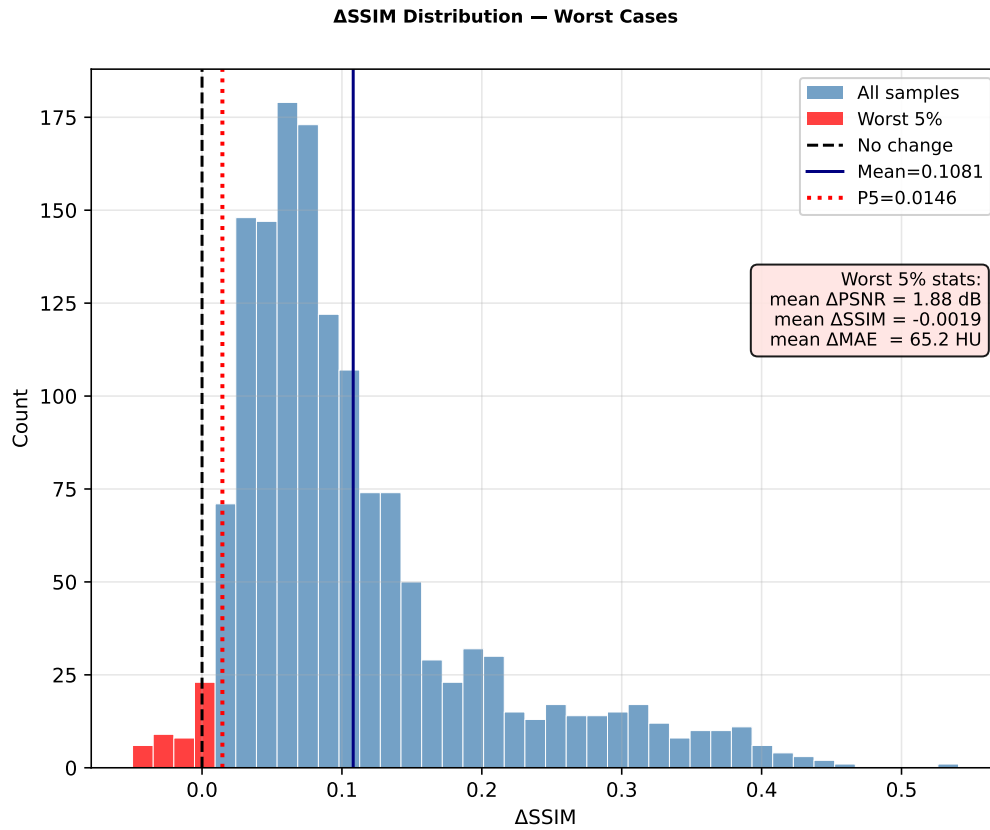


Fig.A.10 Bottom 5th-percentile slices by global Δ SSIM for the Single Model, seed 2024, test split ($N = 74/1478$, threshold = 0.0146, highlighted in red). Descriptive; no inferential test is derived from this subset.

Using the complementary Δ PSNR criterion ($N = 74/1478$ slices; threshold = 2,59 dB), this group was not identical to that defined by Δ SSIM: mean improvement remained positive for all three metrics (mean Δ PSNR = 1,23 dB; mean Δ SSIM = 0,0071; mean Artifact-Zone Δ MAE = 62,0 HU), although with substantial dispersion and some individual cases of degradation relative to the input, visible in the left tail of [Figure A.11](#).

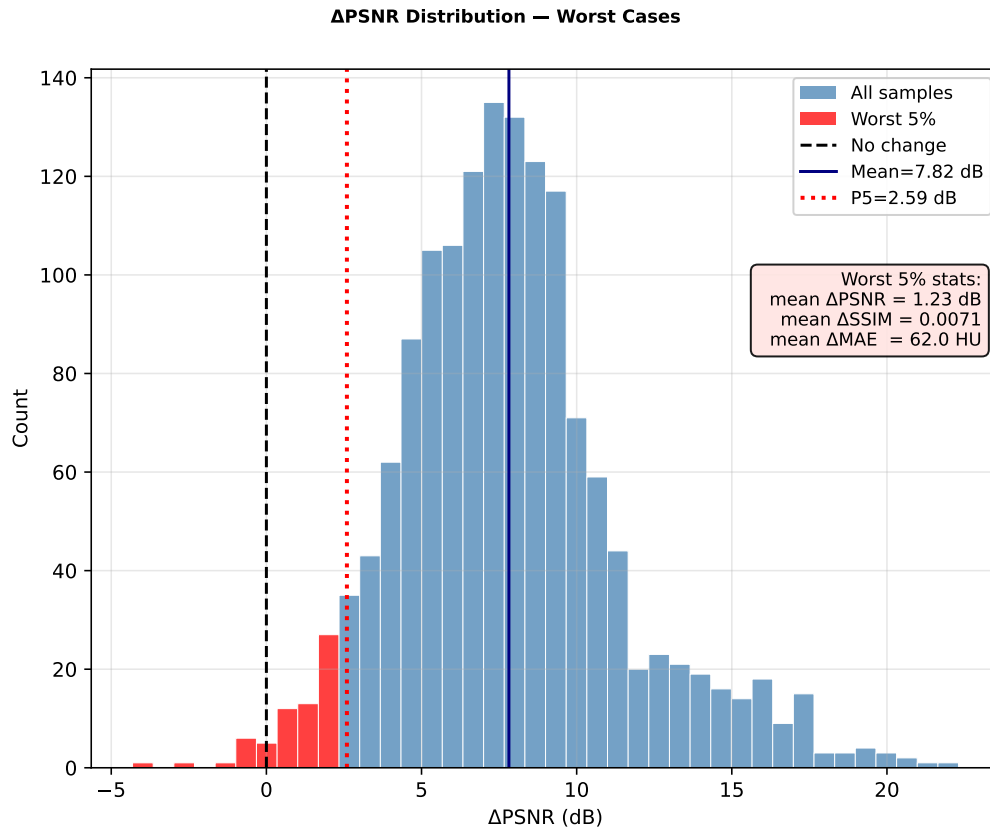


Fig.A.11 Bottom 5th-percentile slices by global Δ PSNR for the Single Model, seed 2024, test split ($N = 74/1478$, threshold = 2.59 dB, highlighted in red). Descriptive; no inferential test is derived from this subset.

Finally, [Figure A.12](#) jointly relates slice-level Δ PSNR and Δ SSIM, highlighting the lower-performance tail defined by Δ PSNR in the preceding paragraph and allowing visualization of the bivariate dispersion of unfavorable cases.

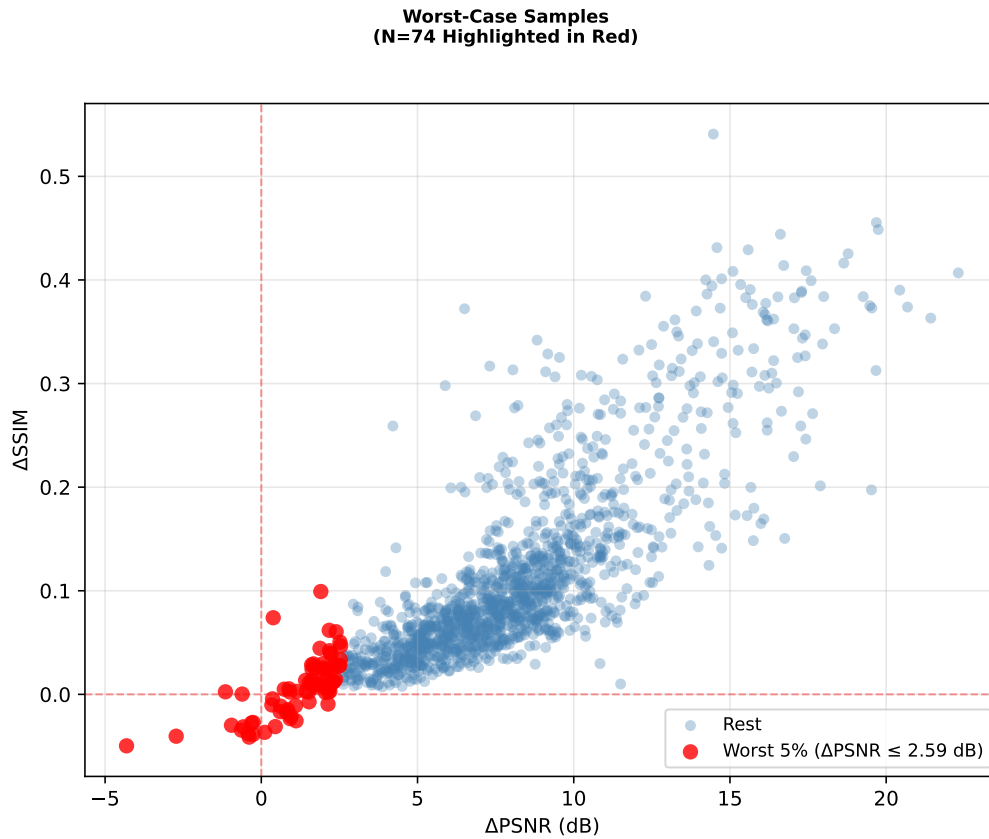


Fig.A.12 Bottom 5th-percentile slices by global ΔPSNR for the Single Model, seed 2024, test split ($N = 74/1478$, threshold = 2.59 dB, highlighted in red). ΔPSNR vs. ΔSSIM scatter. Descriptive; no inferential test is derived from this subset.

A.3 Computational Performance

Runtime performance was quantified for the checkpoint of the model selected for external evaluation, on the internal test partition ($n = 1478$ slices). The benchmark was run in the environment described in Materials and Methods, using an NVIDIA RTX A4500 GPU, batch size one, and five warm-up passes. For each slice, latency was measured from the model call to explicit synchronization of the output with the CPU. Consequently, this measurement includes inference execution and output synchronization, but excludes image loading and external preprocessing or result-presentation stages.

Mean latency was 24,13 ms per slice (standard deviation: 0,69 ms; median: 24,30 ms), corresponding to a throughput of 41,4 slices per second. The 95th and 99th percentiles were 24,66 ms and 27,53 ms per slice, respectively. The proximity of the mean, median, and 95th percentile indicates latency stability for most runs; the complete distribution and slice-level variation are shown in [Figure A.13](#). These values characterize the experimental configuration used

and do not, by themselves, represent the response time of a complete clinical workflow or of another computational environment.

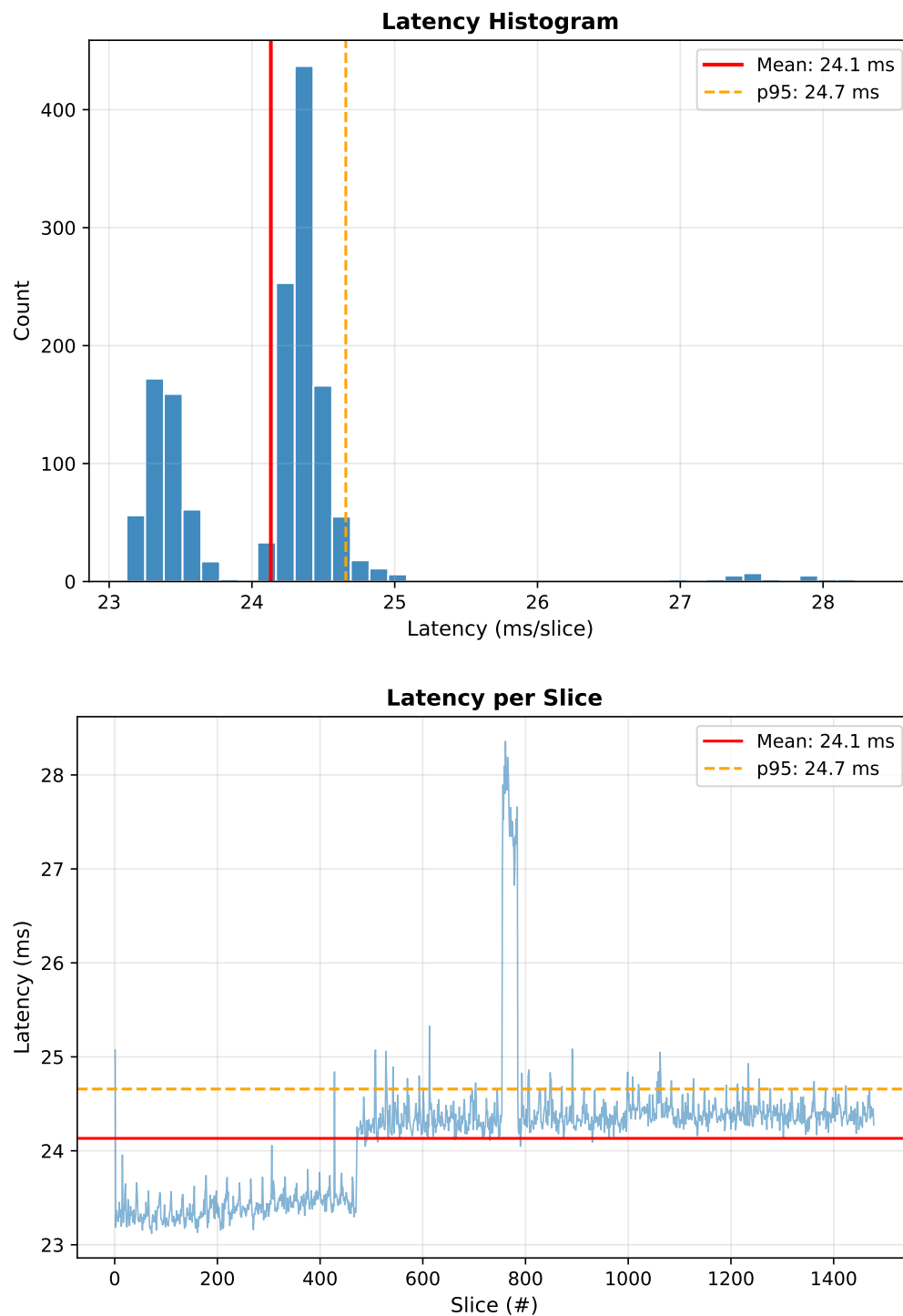


Fig.A.13 Inference benchmark: latency histogram and latency per slice.